prehension is prevalent from 1 to 2 months. Hence we see how the two cases are different: in the first association appears as constitutive of the schema itself, whereas in the second it superadds itself to already existing schemata. The concept of the associative transfer must therefore be discussed separately in each case.

Concerning the first case, the doctrine of the associative transfer is tantamount to saying that each of our schemata is constituted by virtue of a sequence of independent associations. For example, if the child has acquired the habit of sucking his tongue, then his thumb, and then seeking the breast when he is in position to nurse, this would be for the following reasons: certain sensations of lips and tongue having regularly preceded the movements of the latter, and these movements having led to agreeable sucking sensations, the first sensations (contact of the tongue and the lips, etc.) would become a sort of signal automatically starting tongue movements and leading to the desirable result. In the same way that certain sensations of empty sucking have preceded the introduction of the thumb into the mouth a sufficient number of times and this has been followed by agreeable sensations of thumb sucking, it would suffice for the child to make sucking-like movements or to have just finished his meal in order that the sensorial elements peculiar to this situation serve as a signal and set in motion by association the putting of the thumb into the mouth. Finally if these sensations peculiar to the nursing situation set in motion the search for the breast, it is because they would be associated to these movements in the capacity of a signal regularly preceding them. So also, in the realm of vision, if the glance follows objects it is because, the perception of the initial displacements having regularly preceded the movements of the muscles of the eye permitting it to rediscover the displaced object, this perception would have become a signal regulating the movements of the eye itself: hence there would be in the act of following with the glance, a chain of associative transfers. Such an interpretation thus applies to everything: it is not one of the schemata, which we have delineated, which could not be conceived of as being a combination of associative transfers.

But such a manner of speaking seems to us more convenient

than precise. The same criticisms can be applied to the associationist explanation thus renovated as can be applied to the generalization of the conditioned reflex. As the essential in every behavior pattern seems to result from an associative transfer, the essential is not the association itself; it is the fact that the association leads to a favorable or unfavorable result: without the relationship *sui generis* existing between this result and the subject himself, the association would not be consolidated in anything. When the hand is retracted when confronted by fire or the foot is raised at the step of a staircase, the precision of the sensorimotor accommodations which constitute these behavior patterns depends entirely on the meaning which the subject attributes to the flame or to the staircase. It is this active relationship between the subject and the objects that are charged with meanings which creates the association and not the association which creates this relationship. So also, when the child sucks his tongue and his thumb, seeks the breast when he is in nursing position, follows with his eyes moving objects, etc.; it goes without saying that such habits presuppose regulated associations between sensorial elements and movements, but these associative transfers were only able to be constituted and consolidated thanks to a fundamental relationship between the activity of the subject (sucking, vision, etc.) and the sensorial subject endowed with meanings because of this very activity. It may therefore be said, in a general way, that if the association of ideas presupposes judgment instead of constituting it, so also the associative transfer presupposes a relation *sui generis* between the act and its result instead of constituting it.

What, then, is this relation between the act and its objective? Here intervene the concepts of assimilation, accommodation and organization, outside of which the associative transfer seems to us meaningless. The point of departure of all individual activity is, in effect, one or several reflexes already hereditarily organized. There are no elementary habits which do not graft themselves upon reflexes; that is to say, upon an already existing organization, capable of accommodation to the environment and of assimilation of the environment to its own functioning. Now, where a habit begins—that is to say where the associative trans-

fers begin to constitute themselves—there may always be observed this relationship of combined assimilation and accommodation between the reflex activity of the subject and the new result which the nascent habit tends to attain and conserve. It is, in effect, the relation between the act and its result which alone permits the establishing of associative transfers. Now, such a relationship involves assimilation, for what makes the interest or the meaning of the new result pursued by the subject is precisely the fact that it can be assimilated to the reflex activity upon which the habit information grafts itself: hence the tongue and thumb are sucked because they serve as aliments for sucking, objects are followed by the eye because they serve as aliments for looking, etc. In short, the result of the acts, which alone gives to these acts their direction and thus "confirms" the associative transfers, sustains, with the initial reflex schemata, a functional relation of gratifying need, consequently of assimilation. Besides, and by virtue of that very fact, the assimilation of new objects to schemata preformed by the reflexes presupposes an accommodation of these schemata to the situation insomuch as it is new. Thus it is that in order to suck his tongue and his thumb the newborn child is obliged to incorporate into the movements constituting his hereditary sucking schema new movements, discovered in the course of individual experience: pulling the tongue, bringing the hand to the mouth, etc. It is precisely this incorporation of movements and of sensorial elements into the schemata which have already been constituted that is called in associationist terminology conditioned reflex or associative transfer. Only this accommodation is inseparable from assimilation and in this resides the fact that it is much more than an association. It is an insertion of new sensorimotor elements into a totality which has already been organized, which totality constitutes precisely the schema of assimilation. Hence, in sucking his tongue or his fingers, the child incorporates the new sensations he experiences into those of former sucking (sucking the breast, etc.)—therein is assimilation—and, at the same time, he inserts movements of putting out his tongue or putting his thumb into his mouth into the already organized totality of sucking movements—and this is what constitutes accommodation. It is this

progressive extension of the total schema that enriches itself while remaining organized which constitutes accommodation. That is not "association" but progressive differentiation. Thus when the child seeks the breast when he is in nursing position, it cannot simply be said that the attitudes peculiar to this position are henceforth associated to sucking. It must be said that the global schema of sucking movements has incorporated into itself these attitudes and that from this moment they form a whole with the schema itself. In short, the associative transfer is only a moment, artificially cut, in the act of accommodation, which proceeds by differentiation from an earlier schema and by the incorporation of new elements into this schema, and not by association; furthermore, this accommodation is inseparable from assimilation, since it presupposes a total schema and this schema only functions in assimilating to itself new realities. This assimilation alone can explain the satisfaction to which the act leads and which determines the so-called "associative transfers."

As far as the associations producing themselves within an identical schema are concerned, it is therefore illusory to speak of associative transfer. Only the result of an act determines its contexture, which is tantamount to saying, in associationist terminology, that sanction is necessary to consolidate training and stabilize associations. The relationship between an activity and its object cannot be dissociated from the assimilation of the objective result to this activity and of accommodation of the activity to this result. This being so, it necessarily follows that the activity proceeds by global schemata of organization and not by associations. Not only, in effect, does assimilation presuppose such schemata, but further, it unceasingly reconstitutes their unity.

If we now go on to the second possible case—that is, the coördination between two separate schemata—we do not find more associative transfers in the pure state. When the child coordinates his hearing with his vision (and tries to see what he hears) or his prehension with sucking and vision, etc., it cannot be said that this is simple association between a sensorial signal (acoustic, visual, or tactile) and the movements of eye, mouth, or hand. In effect, all the reasons previously involved with regard to the single schemata apply here. The only difference is that in the

present case there is no relationship of simple assimilation and accommodation between the activity of the subject and the object of this activity, but rather reciprocal assimilation and accommodation between two already existing schemata. Between the coördination of schemata and their internal constitution, there is therefore only a difference of degree and not of quality.

In conclusion, the first solution could not account for the facts which we have analyzed in this chapter, for reasons analogous to those which prevent simple Lamarckism from explaining the hereditary morphological reflex variations and which prevent associationism from exhausting intelligence itself. In the three realms of reflexes, sensorimotor acquisitions and intelligence, the primacy of habit or of passive association, leads to a neglect of the factor of organization, hence of combined assimilation and accommodation which is irreducible to automatism. Habit, as such, is certain only an automatization, but in order to be constituted it presupposes an activity which goes beyond simple association.

Then is it necessary to adopt the *second solution* and to consider, as do vitalism or spiritualistic intellectualism that every habit is derived from intelligence itself? The preceding remarks on the complementary relationships of assimilation and accommodation which connect the act to its result bring to mind Buytendijk's arguments concerning the intelligent finality inherent in every activity giving rise to habits, even in the animal. Must the conclusion be drawn from this that habit presupposes intelligence? For our part we would refrain from going that far. It seems indisputable, in effect, that the formation of habits precedes all truly intelligent activity. It is functionally, and not from the point of view of structure, that the behavior patterns described in this chapter can be compared to those we shall subsequently analyze as characterizing the beginnings of intelligence itself. Furthermore the workings of assimilation and accommodation do not necessitate any recourse to finalism or to "immaterial" activities. In yielding to a realism useless to psychology one deduces from the fact of psychological organization the hypothesis of a special force of organization, or one projects into assimilatory activity the structure of an implicit intelligence. Pseudopsycho-

logical realism of which one is thus the victim simply springs from the twofold illusion of philosophical common sense according to which we can grasp in ourselves our own intellectual activity as a known datum of internal experience (whence the ideas of synthetic "reason," of spiritual energy, etc., which extend the *Geist* or the "soul" itself) and according to which this given activity is structurally preformed at the most primitive stages (whence the ideas of vital force, of *a priori* reason, etc.). Quite different is the meaning which we would attribute to the concepts of organization, assimilation, and accommodation. Those are functional processes and not forces. In other words, these functions crystallize in sequential structures and never give rise to an *a priori* structure which the subject would discover directly within himself. In this respect nothing is more instructive than the comparison of the picture of the first infantile activities with the renowned studies by Maine de Biran. Probably no writer has observed better than Maine de Biran the contrast between activity and passive associations in the individual's elementary acquisitions. Concerning hearing and the voice, vision, touch and prehension and many other primordial functions, Maine de Biran always emphasizes the factors of effort and of active motive power which are contrary to the passivity of "affective sensibility" and he concludes the impossibility of an associationistic explanation. In this respect the concepts of assimilation and accommodation which we have employed could be conceived as hypotheses extending, without adding anything, Biran's doctrine of activity. But an obstacle remains, which seems to us to be the following: Biran's "effort" which is found at all levels of psychological activity and explains the "living intelligence" of the adult reflected in the formation of the first habits, is the emanation of a self which apprehends directly in the capacity of substance. It is therefore a force remaining identical to itself in the course of its history and opposing itself to the environmental forces which it comes to know by their resistance. Quite different is active adaptation such as the analysis of assimilation and accommodation obliges us to conceive it. Neither assimilation nor accommodation is a force which presents itself just as it is to consciousness and which furnishes in the capacity of im-

mediate data the experience of a "self" and of an external world. On the contrary, through the very fact that assimilation and accommodation are always on a par, neither the external world nor the self is ever known independently of the other. The environment is assimilated to the activity of the subject at the same time as the latter accommodates itself to the former. In other words, it is through a progressive construction that the concepts of the physical world and of the internal self will become elaborated as a function of each other, and the processes of assimilation and accommodation are only instruments of this construction without ever representing the actual result of it. Regarding this result, it is always relative to the construction as such, and besides there does not exist, at any level, direct experience either of the self or of the external environment. There only exist "interpreted" experiences and this is true precisely thanks to this double play of correlative assimilation and accommodation. In short, the organization peculiar to intellectual becoming is not a faculty which would form intelligence itself nor a force which would form the "self": it is only an operation of which the sequential structural crystallizations never realize intelligence as such. All the more reason why it is unlikely that the most elementary acquisitions—that is to say the subject's first habits which are under discussion, derive from the higher intellectual processes as spiritualism would have it.

But if habit does not derive from intelligence, without adding anything, it cannot be said, as in the *third* and the *fourth* solution that it does not have, or almost does not have, any relationships to intellectual activity. If association and habit are considered not in their automatized form, but to the extent that they become organized at the level at which we have considered them in analyzing the facts, it seems indisputable that they reveal close functional analogies to intelligence. The same is true, in effect, of habit as of imitation. Its automatic form is not the primitive form and the primitive form presupposes a more complex activity than the evolved forms. In the case of habit, this elementary activity is that of the sensorimotor organizations whose schemata function in the same way as intelligence itself, through complementary assimilations and accommodations. We

shall see in what follows that there are present all the transitions between these schemata and those of intelligence. Besides, it is too early to show now in which respect the Gestalt theory has exaggerated the contrast between the higher structures and the more fluctuating behavior of the elementary stages, and in which respect the schema of assimilation is to be conceived of as a system of relations less rigid than a Gestalt and itself involving an organizing activity of which it is only the expression. Let us limit ourselves to recalling that schemata such as those of sucking thumb or tongue, grasping seen objects, the coördination of hearing and of vision, etc., never arise *ex abrupto* but constitute the point of arrival of a long effort of gradual assimilation and accommodation. It is this effort which foretells intelligence. Besides, when Delacroix tells us that intellectual generalization is in a sense the opposite of habit, that is true of habit which has been formed and is degenerating in passivity, but that is not certain concerning the assimilation which is at the point of departure of this habit. There is, as we have seen, a generalizing assimilation which works in the same way as intelligence, through a sequence of choices and corrections. Even groping, which Claparède regards as the characteristic of nascent intelligence is therefore not excluded from the formation of habits which does not mean that the latter are already intelligent, but rather that there is a continuous organizing activity connecting organic adaptation to intellectual adaptation through the intermediary of the most elementary sensorimotor schemata.

It is therefore the *fifth solution* which we shall adopt. Association and habit form the automatization of an activity which functionally prepares intelligence while yet differing from it by a more elementary structure. Let us try to state these assertions precisely and to do so let us first recall the general characteristics of the stage under consideration, contrasting them to those of the preceding stage and those of the following one.

In general it may be said that the behavior patterns studied in §§1–4 consist in searchings which prolong reflex activity and which are as yet devoid of intention but which lead to new results of which the mere discovery is fortuitous and whose conservation is due to a mechanism adapted from combined sensori-

motor assimilation and accommodation. These behavior patterns prolong those of the first stage in that the needs connected with the reflex (sucking, looking, listening, crying, grasping, etc.) are still their only motive power without there yet being needs connected with derived and deferred aims (grasping in order to throw, in order to swing to and fro, etc.). But, contrary to purely reflex searching, the searching peculiar to the present stage is displayed in gropings which lead to new results. Contrary to the subsequent stage, these results are not pursued intentionally. They are therefore the product of chance, but, resembling intelligent behavior patterns, the behavior patterns of which we are speaking tend, as soon as the result is obtained, to conserve it by correlative assimilation and accommodation.

This conservation of advantageous results obtained by chance is, of course, what Baldwin has called "circular reaction." This concept of which we have made use in describing the facts seems to us exactly to define the position of the present stage. Circular reaction involves the discovery and the conservation of the new, and in that it differs from the pure reflex; but it is anterior to intention, and in that it precedes intelligence. Only, such a concept requires interpretation. If one limits oneself, as is frequently done, to explaining repetition by the "reaction of excess" and tracing, one returns to automatism to account for what is, on the contrary, preëminently active searching. If the child tends to rediscover an advantageous result, this is not because that is the course of least effort, but on the contrary, it is because the result is assimilated to an earlier schema and the question is to accommodate this schema to the new result. The "circular reaction" is, accordingly, only a global concept, embracing in reality two distinct processes. Let us try, in conclusion, to summarize what we know about these processes.

First there is *accommodation*. The great novelty inherent in circular reaction and habit compared to the reflex, is that accommodation begins to be differentiated from assimilation. At the heart of the reflex, in effect, accommodation mingles with assimilation. The use of the reflex is at once pure repetition (that is to say, assimilation of the object to an already constructed schema) and accurate accommodation to its object. On the other

hand, from the moment when the sensorimotor schema is applied to new situations and thus dilates to embrace a larger realm, accommodation and assimilation tend to be differentiated. Take, for example, thumb sucking. During the reflex stage this behavior pattern consisted in a simple, occasional animated application of the sucking schema to a new object but without having this circumstance transform the schema in any way whatever. The new object was assimilated to the old schema and this generalizing assimilation had no other effect than to exercise the reflex in general; at most it permitted it to discriminate in future breast sucking from what was not breast sucking. During the present stage, on the contrary, application of the sucking schema to a new object such as the thumb or tongue transforms the schema itself. This transformation constitutes an accommodation and this accommodation is consequently distinct from pure assimilation. In a general way, contact of some schema with a new reality results, during the present stage, in a special behavior pattern, intermediate between that of the reflex and that of intelligence. In the reflex the new is entirely assimilated to the old and accommodation thus mingles with assimilation; in intelligence, there is interest in the new as such and accommodation is consequently definitely differentiated from assimilation; in the behavior patterns of the intermediary stage, the new is still only of interest to the extent that it can be assimilated to the old, but it already cracks the old frameworks and thus forces them to an accommodation partly distinct from assimilation.

How does this accommodation work? We have seen in the above: not through association, but through differentiation of an existing schema and insertion of new sensorimotor elements among those which already form it. In effect, with the reflex activity a series of already constructed schemata are hereditarily given and their assimilatory function thus represents an activity directed toward performance dating from the very beginning of life and prior to all association. When these schemata are differentiated by accommodation—in other words, in physiological terminology, when a reflex connection is subordinated to a cortical connection and with it forms a new totality—it cannot be said that a given reaction is simply associated to new signals or

to new movements. It must be said that an activity which has already been organized from the beginnings is applied to new situations and that the sensorimotor elements connected with these new situations were comprised in the primitive schema in thus differentiating it. There is no subordination of the reflex schema to new associations nor inverse subordination. There is continuity of a single activity with complementary differentiation and interpretation.

Accommodation, then, presupposes *assimilation* as, in reflective intelligence, empirical association presupposes judgment. It is this factor of functional assimilation which forms the organizing and totalizing activity insuring the continuity between the schema considered before accommodation and the same schema after the insertion of new elements due to this accommodation. What, then, is assimilation?

Assimilation is first of all purely functional—that is to say, cumulative repetition and assimilation of the object to the function: sucking for the sake of sucking, looking for the sake of looking, etc. As such psychological assimilation prolongs without adding anything to organic functional assimilation and does not require any special explanation. Then, to the extent that assimilation of the object to the function extends to include increasingly varied objects, assimilation becomes "generalizing"; that is to say (as regards the present stage), combines with multiple accommodations. Finally, and through the very fact of this differentiation, assimilation becomes "recognitory"; that is to say, perception of objects—or more precisely, of sensorial images—as a function of the multiple activities delineated by the generalizing assimilation. Therein resides a first principle of exteriorization which combines with the exteriorization due to the coördination between heterogeneous schemata.

To describe this assimilation clearly one can do so either from the point of view of consciousness or from that of behavior. What can the consciousness of the infant be concerning the thumb he sucks, the object he looks at, the object he will grasp after having perceived it, the sounds he makes, etc.? Stern[22] as-

[22] W. Stern, *Psychologie der frühen Kindheit bis zum sechsten Lebensjahre*, Leipzig, Quelle & Meyer, 1927.

serts that an impression is only individualized if it is connected with a movement experienced as being active or at least connected with the context of the activity itself. One could, at first, object to this way of seeing the 2-month-old baby's attention to things and people (Lucienne, at 0;1 (28) looks at the trees above her, laughs when people move about in front of her). But, in order to look, there exists accommodation of the eyes and of the head and this accommodation is probably experienced as a real activity much more by the nursling than by us. Mimicry denotes unceasing effort, tension, expectation, satisfaction or disappointment, etc. Besides, perception is already extended into imitation, as we shall see later. We therefore fully concede Stern's remark.[23] Now, the following, it seems to us, results from this, from the point of view of the states of consciousness concomitant to assimilation. During the elementary stages of consciousness things are much less apprehended in their own form than is the case with the adult or the child who talks. There is not a thumb, a hand, a ribbon about to be grasped, etc. There is a variety of tactile, visual, gustatory, etc., images which are not contemplated but rather operated—that is to say, produced and reproduced, impregnated, so to speak, with the need to be supported or rediscovered. Hence this conclusion which must always be borne in mind in order to avoid the associationist error constantly reappearing under cover of the law of transfer: the new objects which are introduced to consciousness have no peculiar and separate qualities. Either they are at once assimilated to an already existing schema: a thing to suck, to look at, to grasp, etc. Or else they are vague, nebulous, because of being unassimilable and then they create an uneasiness whence comes sooner or later a new differentiation of the schemata of assimilation.

From the point of view of behavior, assimilation is revealed in the form of cycles of movements or of acts bringing each other along and closing up again on themselves. This is clearly true of the reflex whose various forms of use we have studied. This is again true of circular reaction: the performed act leaves a vacuum

23 Ch. Bühler, *op. cit.*, adds that the child's interest in a situation culminates at the moment that actual activity begins to triumph over his difficulties.

which, in order to be filled, leads to the repetition of the same act. Hence there exists the form of an ensemble or cycle of organized movements, to the extent that the act gratifies a real need. Each activity forms a whole. To be sure, the ensemble is not perfect from the outset. There is groping in its performance and it is in the course of this groping that it is easy to dissociate the sequential movements in order to describe them in terms of associative transfer. But the so-called signal which would determine the movements constitutes more of an indication in comparison with an activity which tries to gain satisfaction for itself than a trigger starting up movements. The true cause of movement is need; that is to say, the total act of assimilation. This is not to say yet that the movement is intentional. Need is nothing other, for the moment, than the vacuum created by the preceding performance of the act and, at the beginning, by the chance discovery of an interesting result, interesting because directly assimilable.

In short, the uniting of accommodation and assimilation presupposes an *organization*. Organization exists within each schema of assimilation since (we have just recalled it) each one constitutes a real whole, bestowing on each element a meaning relating to this totality. But there is above all total organization; that is to say, coördination among the various schemata of assimilation. Now, as we have seen, this coördination is not formed differently from the simple schemata, except only that each one comprises the other, in a reciprocal assimilation. At the point of departure we are in the presence of needs which attain satisfaction separately. The child looks for the sake of looking, grasps in order to grasp, etc. Then there is an accidental coördination between one schema and another (the child looks by chance at the hand which grasps, etc.) and finally, fixation. How does this fixation work? It seems at first that this might be by association. Contact of the hands with an object or of an object with the lips seems to be the signal which sets in motion the movement of the object to the lips and sucking. But the opposite procedure is also possible. The need to suck sets in motion the movement of the hand to the mouth, etc. The possibility of the two complementary actions shows sufficiently that they form but one unit. All

the more reason why this is so when the coördination of the schemata is reciprocal, when, for instance, the child grasps what he sees and brings before his eyes that which he grasps. In short, the conjunction of two cycles or of two schemata is to be conceived as a new totality, self-enclosed. There is neither association between two groups of images nor even association between two needs, but rather the formation of a new need and the organization of earlier needs as a function of this new unity.

It is then, let us remember, that assimilation becomes objectified and perception is externalized. A sensorial image which is at the point of intersection of several currents of assimilation is, through that very fact, solidified and projected into a universe where coherence makes its first appearance.

In conclusion it may be seen to what extent the activity of this stage, activity from which the first sensorimotor habits proceed, is identical, from the functional point of view, with that of intelligence, while differing from it in structure. Functionally speaking, the accommodation, assimilation and organization of the first acquired schemata are altogether comparable to those of the mobile schemata of which sensorimotor intelligence will make use and even to those of the concepts and relationships which reflective intelligence will employ. But, from the structural point of view, the first circular reactions lack intention. As long as action is entirely determined by directly perceived sensorial images there can be no question of intention. Even when the child grasps an object in order to suck or look at it, one cannot infer that there is a conscious purpose. The goal of the action is one with the point of departure only by virtue of the fact of the unity of the schema of coördination. It is with the appearance of secondary and mobile schemata and of deferred reactions that the purpose of the action, ceasing to be in some way directly perceived, presupposes a continuity in searching, and consequently a beginning of intention. But, to be sure, all gradations exist between these evolved forms of activity and the primitive forms of which we have spoken hitherto.

PART II

The Intentional Sensorimotor Adaptations

THE INTENTIONAL SENSORIMOTOR ADAPTATIONS

The coördination of vision and prehension, which we have studied in Chapter II, inaugurates a new series of behavior patterns: the intentional adaptations. Unfortunately, nothing is more difficult to define than intention. Shall it be said, as is frequently done, that an act is intentional when it is determined by representation, contrary to the elementary associations in which the act is controlled by an external stimulus? But if representation is taken in the strict sense of the word, there would not then be intentional acts prior to language—that is to say, before the faculty of thinking of reality by means of signs making up the deficiency of action. Now intelligence presupposes intention. If, on the other hand, one extends the term representation so that it comprises all consciousness of meanings, intention would exist ever since the simplest associations and almost since the beginning of reflex use. Shall it be said that intention is connected with the power of evoking images and that searching for the fruit in a closed box, for instance, is an intentional act to the extent that it is determined by the representation of the fruit in the box? But, as we shall see, it appears according to all probabilities that even this kind of representations, by images and individual symbols, makes a tardy appearance. The mental image is a product of the internalization of the acts of intelligence and not a datum preliminary to these acts. Since then we see only one method of distinguishing intentional adaptation from the simple circular reactions peculiar to sensorimotor habit: this is to invoke the number of intermediaries coming between the stimulus of the act and its result. When a 2-month-old baby sucks his thumb this cannot be called an intentional act because the coördination of the hand and of sucking is simple and direct. It therefore suffices for the child to maintain, by circular reaction,

the favorable movements which satisfy his need, in order that this behavior become habitual. On the other hand, when an 8-month-old child sets aside an obstacle in order to attain an objective, it is possible to call this intention, because the need set in motion by the stimulus of the act (by the object to be grasped) is only satisfied after a more or less lengthy series of intermediary acts (the obstacles to be set aside). Intention is thus determined by consciousness of desire, or of the direction of the act, this awareness being itself a function of the number of intermediary actions necessitated by the principal act. In a sense, there is therefore only a difference of degree between the elementary adaptations and the intentional adaptations. The intentional act is only a more complex totality subsuming the secondary values under the essential values and subordinating the intermediary movements or *means* to the principal steps which assign an end to the action. But, in another sense, intention involves a reversing in the data of consciousness. There is henceforth the influence of recurrent consciousness of direction impressed on the action or no longer only on its result. Consciousness arises from dis-adaptation and thus proceeds from the periphery to the center.

In practice, we can acknowledge—provided we bear in mind that this division is artificial and that all the transitions connect the acts of the second stage to those of the third—that intentional adaptation begins as soon as the child transcends the level of simple corporal activities (sucking, listening and making sounds, looking and grasping) and acts upon things and uses the interrelationships of objects. In effect, to the extent that the subject is limited to sucking, looking, listening, grasping, etc., he satisfies in a more or less direct way his immediate needs, and, if he acts upon things, it is simply in order to perform his own functions. In such a case it is hardly possible to speak of ends and means. The schemata serving as means become mingled with those which assign an end to the action and there is no occasion for this influence of consciousness *sui generis* which determines intention. On the contrary, as soon as the subject, possessing the coördinated schemata of prehension, vision, etc., utilizes them in order to assimilate to himself the totality of his universe, the multiple combinations which then present themselves (by gen-

eralizing assimilation and accommodation, combined) bring with them the momentary hierarchies of ends and means; that is to say, there is the influence of consciousness of the direction of the act or of its intention.

From the theoretical point of view, intention therefore denotes the extension of the totalities and relationships acquired during the preceding stage and, by the fact of their extension, their greater dissociation into real totalities and ideal totalities in relationships of fact and relationships of value. As soon as there is intention, in effect, there is a goal to reach and means to use, consequently the influence of consciousness of values (the value or the interest of the intermediary acts serving as *means* is subordinated to that of the goal) and of the ideal (the act to be accomplished is part of an ideal totality or *goal,* in relation to the real totality of the acts already organized). Thus it may be seen that the functional categories related to the function of organization will henceforth become more precise, from the time of the global schemata of the preceding stage. Concerning the functions of assimilation and accommodation, intentional adaptation also brings with it a more pronounced differentiation of their respective categories, ever since the relatively undifferentiated state of the first stages. Assimilation, after having proceeded as hitherto, by nearly rigid schemata (the sensorimotor schemata of sucking, prehension, etc.) will henceforth engender more mobile schemata, capable of various involvements and in which we shall find the functional equivalent of the qualitative concepts and of the quantitative relationships peculiar to reflective intelligence. With regard to accommodation, by clasping more tightly the external universe, it will clarify the space-time relationships as well as those of substance and causality, hitherto enveloped in the subject's psycho-organic activity.

In other words, we now arrive at the problem of intelligence which we shall study with regard to stages III to VI. Hitherto we have stayed on this side of actual intelligence. During the first stage this was self-evident, since pure reflexes were involved. Concerning the second stage it was not known how, despite the functional resemblances, to identify habit and intelligent adaptation, since it is precisely intention that separates them. This is

not the place to define this structural difference which analyzing the facts alone will permit us to fathom and which we shall take up again at the conclusion of this book. Let us say only that the sequence of our stages corresponds in the main to the system outlined by Claparède in a remarkable article on intelligence published in 1917.[1] To Claparède intelligence is an adaptation to new situations as opposed to reflexes and habitual associations which also constitute adaptations, either hereditary or due to personal experience, but adaptations to situations which repeat themselves. Now these new situations to which the child will have to adapt himself appear precisely when the habitual schemata, elaborated during the second stage, will be applied for the first time to the external environment in its complexity.

Furthermore, there may be distinguished, among the intentional acts which constitute intelligence, two relatively opposite types, corresponding in the main to what Claparède calls empirical intelligence and systematic intelligence. The first consists in operations controlled by the things themselves and not by deduction alone. The second consists in operations controlled from within by the consciousness of relationships and thus marks the beginning of deduction. We shall consider the first of these behavior patterns as characteristic of the stages III to V and shall make the appearance of the second behavior patterns the criterion of a sixth stage.

On the other hand, the concept of "empirical intelligence" remains a little vague as long as one does not put into effect, in the sequence of facts, some divisions intended, not to make discontinuous an actual continuity, but to permit analysis of the increasing complication of the behavior patterns. This is why we shall distinguish three stages between the beginnings of the action upon things and those of systematic intelligence: stages III to IV.

The third stage appearing with the prehension of visual objectives is characterized by the appearance of a behavior pattern which is already almost intentional, in the sense indicated before, which also foretells empirical intelligence but which nevertheless remains intermediary between the acquired association belonging to the second stage and the true act of intelligence. This is the

[1] Republished in *Education fonctionnelle, op. cit.*

"secondary circular reaction," that is to say, the behavior which consists in rediscovering the gestures which by chance exercised an advantageous action upon things. Such a behavior pattern, in effect, goes beyond acquired association to the extent that almost intentional searching is necessary to reproduce the movements until then performed fortuitously. But it does not yet constitute a typical act of intelligence since this searching simply consists in rediscovering that which has just been done and not in inventing again or in applying the known to new circumstances: the "means" are hardly yet differentiated from the "ends" or at least they are only differentiated after the event, at the time the act is repeated.

A fourth stage begins at around 8 to 9 months and lasts until the end of the first year. It is characterized by the appearance of certain behavior patterns which are superimposed on the preceding ones and their essence is "the application of known means to new situations." Such behavior patterns differ from the preceding ones both in their functional meaning and in their structural mechanism. From the functional point of view for the first time they fully correspond to the current definition of intelligence: adaptation to new circumstances. Given a habitual goal temporarily thwarted by unforeseen obstacles, the problem is to surmount these difficulties. The simplest procedure consists in trying out different known schemata and in adjusting them to the goal pursued: in this consist the present behavior patterns. From the structural point of view they therefore constitute a combination of schemata among themselves, so that some are subordinated to others in the capacity of "means"; hence two results: a greater mobility of the schemata and a more accurate accommodation to external conditions. If this stage is to be distinguished from the preceding one with respect to the functioning of intelligence, it is to be distinguished still more with regard to the structure of objects, space and causality: it marks the beginnings of the permanence of things, of "objective" spatial "groups" and of spatial and objectified causality.

At the beginning of the second year a fifth stage makes itself manifest, characterized by the first real experimentations; hence the possibility of a "discovery of new means through active

experimentation." This is the impetus of the instrumental behavior patterns and the acme of empirical intelligence.

Finally this totality of the behavior patterns, the application of which determines the beginning of the sixth stage, will be crowned by the "invention of new means through mental combination."

CHAPTER III

THE THIRD STAGE:

The "Secondary Circular Reactions" and the Procedures Destined to Make Interesting Sights Last

From the simple reflex to the most systematic intelligence, the same method of operation seems to us to continue through all the stages, thus establishing complete continuity between increasingly complex structures. But this functional continuity in no way excludes a transformation of the structures being on an equal footing with an actual reversal of perspectives in the subject's consciousness. At the beginning of intellectual evolution, in effect, the act is set in motion all at once and by an external stimulus and the individual's initiative consisted merely in being able to reproduce his action when confronted with stimuli analogous to the normal stimulus, or by simple empty repetition. At the end of the evolution, on the other hand, every action involves an organization versatile in making dissociations and unlimited regroupings, the subject thus being able to assign to himself goals which are increasingly independent of instigation by the immediate environment.

How does such a reversal work? Due to the progressive complication of the schemata: by constantly renewing his acts through reproductive and generalizing assimilation, the child surpasses simple reflex use, discovers circular reaction and thus forms his first habits. Such a process is obviously capable of unlimited extension. After applying it to his own body, the subject will utilize it sooner or later in order to adapt himself to the unforeseen phenomena of the external world, whence the be-

havior patterns of exploration, experimentation, etc. Wherefore the possibility, subsequently, of decomposing and recomposing the same schemata: gradually as the schemata apply themselves to more varied external situations the subject is, in effect, led to dissociate their elements and to consider them as means or as ends, while at the same time regrouping them among themselves in all sorts of ways. It is this distinction of means and ends which sets intention free and so reverses the act's direction. Instead of being turned toward the past—that is to say, toward repetition—the action is directed toward new combinations and actual invention.

Now the stage which we are about to describe forms exactly the transition between the behavior patterns of the first type and those of the second. The "secondary circular reactions" prolong, in effect, without adding anything to, the circular reactions under examination hitherto; that is to say, they essentially tend toward repetition. After reproducing the interesting results discovered by chance on his own body, the child tries sooner or later to conserve also those which he obtains when his action bears on the external environment. It is this very simple transition which determines the appearance of the "secondary" reactions; accordingly it may be seen how they are related to the "primary" reactions. But it is necessary immediately to add that, the more the effort of reproduction bears upon results removed from those of reflex activity, the clearer becomes the distinction between means and ends. To the extent that it is a hereditary assembly the reflex schema forms an indissociable totality. The repetition belonging to "reflex use" would only know how to make the machine go by activating it completely without distinction between the transitive terms and the final terms. In the case of the first organic habits (thumb sucking, for example) the complexity of the schema augments since an acquired element is inserted among the reflex movements; repeating the interesting result therefore will involve a coördination between terms not necessarily united with each other. But, as their union, although an acquired one, was in a way imposed by the conformation of the body itself and sanctioned by a strengthening of reflex activity, it is still easy for the child to rediscover through simple repetition

the result obtained without distinguishing the transitive terms and the final term of the act. On the other hand, as soon as the result to be reproduced is connected with the external environment—that is to say, with independent objects (even if their mutual relationships and their permanence are still unknown to the child)—the effort to rediscover a propitious gesture will lead the subject, after the event, to distinguish in his action the transitive terms or "means" and a final term or "end." It is from this time on that it is possible really to speak of "intention" and of a reversal in acquiring consciousness of the act. But this reversal will only be definitive when the different terms will be sufficiently dissociated so that they may recombine among themselves in various ways; that is to say, when there will be a possibility of applying known means to new ends or, in a word, when an intercoördination of the schemata will exist (fourth stage). The "secondary circular reaction" is not yet at this level; it tends simply to reproduce every advantageous result obtained in relation to the external environment, without the child's yet dissociating or regrouping the schemata thus obtained. The goal is therefore not set ahead of time, but only when the act is repeated.—Therein the present stage forms the exact transition between preintelligent operations and truly intentional acts. The behavior patterns which characterize it still owe a great deal to repetition while being superior to it from the point of view of complexity and they already possess intelligent coördination while remaining inferior to it from the point of view of dissociation of means and ends.

This intermediary characteristic will be found again, as we shall see in Volume II, in all the behavior patterns of the same stage, whether it is a matter of the content of intelligence or of real categories (object and space, causality and time) as well as of its form (which we shall study).

With regard to the object, for example, the child at this stage arrives at a behavior pattern exactly intermediary between those of nonpermanence, belonging to the lower stages, and the new behavior relating to objects which have disappeared. On the one hand, in effect, the child knows henceforth how to grasp the objects he sees, bring to sight those he touches, etc., and this

coördination between vision and prehension indicates notable progress in the solidification of the external world: acting upon things, he considered them unyielding and permanent to the extent that they prolong his action or thwart it. On the other hand, to the extent that objects leave the perceptual field and consequently, the child's direct action, the child no longer reacts and does not apply himself to active searching to find them again, as he will do during the next stage. If there is permanence of the object it is still relative to the action in progress and not a fact in itself.

Concerning space, the actions brought to bear on things by the child at the third stage result in forming a perception of "groups"; that is to say, of schemata of displacements capable of returning to their point of departure. In this sense, the behavior patterns of this stage mark important progress in relation to the preceding ones in fixing securely the intercoördination of the various practical spaces (visual, tactile, buccal spaces, etc.). But the "groups" thus formed remain "subjective" for, beyond the immediate action, the child does not yet take into consideration the spatial interrelations of objects. Causality, too, takes form to the extent that the child acts upon the external environment: henceforth it unites certain separate phenomena to the acts which correspond to them. But, precisely because the schemata belonging to this stage are not yet dissociated in their elements, the child only gets a confused and global feeling of the causal connection and does not know how to objectify or spatialize causality.

The same is true *a fortiori* of the temporal series which will link the different phases of the act but not yet the different events produced in an environment independent of the self.

In short, during the first two stages—that is to say, so long as the child's activity consists in mere repetitions without intention—the universe is not yet at all dissociated from the action itself and the categories remain subjective. As soon as the schemata become, on the other hand, capable of intentional decompositions and recombinations—that is to say, of really intelligent activity—the consciousness of the relations thus implicated by distinguishing means and ends will necessarily bring

with it the elaboration of a world independent of the self. From this point of view of the contents of intelligence, the third stage, therefore, also marks a turning point. Its particular reactions remain midway between the solipsist universe of the beginning and the objective universe belonging to intelligence. Without being requisite to the description of the facts which follows, such thoughts nevertheless clarify many aspects of them.

§1. THE "SECONDARY CIRCULAR REACTIONS"—I. THE FACTS AND REPRODUCTIVE ASSIMILATION.—We can call the circular reactions of the second stage "primary." Their character consists in simple organic movements centered on themselves (with or without intercoördination) and not destined to maintain a result produced in the external environment. So it is that the child grasps for the sake of grasping, sucking, or looking, but not yet in order to swing to and fro, to rub, or to reproduce sounds. Moreover the external objects upon which the subject acts are one with his action which is simple, the means being confused with the end. On the other hand, in the circular reactions which we shall call "secondary" and which characterize the present stage, the movements are centered on a result produced in the external environment and the sole aim of the action is to maintain this result; furthermore it is more complex, the means beginning to be differentiated from the end, at least after the event.

Of course, all the intermediaries are produced between the primary circular reactions and the secondary reactions. It is by convention that we choose, as criterion of the appearance of the latter, the action exerted upon the external environment. Henceforth if they make their appearance, for the most part, after the definitive acquisition of prehension, it is nevertheless possible to find some examples of this phenomenon prior to that.

First here are some examples of circular reactions relating to the movements the child gives to his bassinet and to the hanging objects:

Observation 94.—At 0;3 (5) Lucienne shakes her bassinet by moving her legs violently (bending and unbending them, etc.), which makes the cloth dolls swing from the hood. Lucienne looks at them, smiling, and

recommences at once. These movements are simply the concomitants of joy. When she experiences great pleasure Lucienne externalizes it in a total reaction including leg movements. As she often smiles at her knick-knacks she caused them to swing. But does she keep this up through consciously coördinated circular reaction or is it pleasure constantly springing up again that explains her behavior?

That evening, when Lucienne is quiet, I gently swing her dolls. The morning's reaction starts up again, but both interpretations remain possible.

The next day, at 0;3 (6) I present the dolls: Lucienne immediately moves, shakes her legs, but this time without smiling. Her interest is intense and sustained and there also seems to be an intentional circular reaction.

At 0;3 (8) I again find Lucienne swinging her dolls. An hour later I make them move slightly: Lucienne looks at them, smiles, stirs a little, then resumes looking at her hands as she was doing shortly before. A chance movement disturbs the dolls: Lucienne again looks at them and this time shakes herself with regularity. She stares at the dolls, barely smiles and moves her legs vigorously and thoroughly. At each moment she is distracted by her hands which pass again into the visual field: she examines them for a moment and then returns to the dolls. This time there is definite circular reaction.

At 0;3 (13) Lucienne looks at her hand with more coördination than usually (see Obs. 67). In her joy at seeing her hand come and go between her face and the pillow, she shakes herself in front of this hand as when faced by the dolls. Now this reaction of shaking reminds her of the dolls which she looks at immediately after as though she foresaw their movement. She also looks at the bassinet hood which also moves. At certain times her glance oscillates between her hand, the hood, and the dolls. Then her attention attaches itself to the dolls which she then shakes with regularity.

At 0;3 (16) as soon as I suspend the dolls she immediately shakes them, without smiling, with precise and rhythmical movements with quite an interval between shakes, as though she were studying the phenomenon. Success gradually causes her to smile. This time the circular reaction is indisputable. Same reaction at 0;3 (24). Same observations during the succeeding months and until 0;6 (10) and 0;7 (27) at sight of a puppet and at 0;6 (13) with a celluloid bird, etc.

Observation 94 repeated.—At 0;3 (9) Lucienne is in her bassinet without the dolls. I shake the bassinet two or three times without her seeing me. She looks very interested and serious and begins again, for a long stretch of time, rough and definitely intentional shaking.—That evening I rediscover Lucienne in the act of shaking her hood spontaneously. She laughs at this sight.

Here is involved, therefore, the schema described in the foregoing

observation, but applied to a new object. Same observation on the following days.

At 0;4 (4) in a new bassinet, she moves her loins violently in order to shake the hood. At 0;4 (13) she moves her legs very rapidly while looking at the festoons on the bassinet hood; as soon as she sees them again, after a pause, she begins once more. Same reaction with regard to the hood in general. At 0;4 (19) she recommences by examining each part of the hood in detail. At 0;4 (21) she does the same in her carriage (and no longer in the bassinet): she studies the result of her shaking most attentively. Same observations at 0;5 (5) etc., until 0;7 (20) and later.

Observation 95.—Lucienne, at 0;4 (27) is lying in her bassinet. I hang a doll over her feet which immediately sets in motion the schema of shakes (see the foregoing observations). But her feet reach the doll right away and give it a violent movement which Lucienne surveys with delight. Afterward she looks at her motionless foot for a second, then recommences. There is no visual control of the foot, for the movements are the same when Lucienne only looks at the doll or when I place the doll over her head. On the other hand, the tactile control of the foot is apparent: after the first shakes, Lucienne makes slow foot movements as though to grasp and explore. For instance, when she tries to kick the doll and misses her aim, she begins again very slowly until she succeeds (without seeing her feet). In the same way I cover Lucienne's face or distract her attention for a moment in another direction: she nevertheless continues to hit the doll and control its movements.

At 0;4 (28), as soon as Lucienne sees the doll she moves her feet. When I move the doll toward her face she increases her movements and thus resumes the behavior described in the preceding observations. So also at 0;5 (0) she oscillates between the global reaction and specific foot movements, but at 0;5 (1) she resumes the latter movements only and even seems to regulate them (without seeing them) when I raise the doll a little. A moment later she gropes until she has felt contact between her naked foot and the doll: she then increases her movements. Same reaction at 0;5 (7) and the days following.

At 0;5 (18) I place the doll at different heights, sometimes to the left, sometimes to the right: Lucienne first tries to reach it with her feet, and then, when she has succeeded, shakes it. The schema is therefore definitely acquired and begins to be differentiated through accommodation to various situations.

Observation 96.—At 0;5 (8) Jacqueline looks at a doll attached to a string which is stretched from the hood to the handle of the bassinet. The doll is at approximately the same level as the child's feet. Jacqueline moves her feet and finally strikes the doll whose movement she immediately notices. A circular reaction ensues comparable to that in the

foregoing observation but less coördinated in view of Jacqueline's retarded development as she was born in winter and had less physical exercise than Lucienne. The feet move, first without conscious coordination, then certainly by circular reaction. The activity of the feet grows increasingly regular whereas Jacqueline's eyes are fixed on the doll. Moreover, when I remove the doll Jacqueline occupies herself quite differently, and when I replace it, after a moment, she immediately recommences to move her legs. But, contrary to Lucienne, Jacqueline does not understand the necessity for contact between feet and doll. She limits herself to noting the connection between the movement of the object and the total activity of her own body. This is why, as soon as she sees the doll, she places herself in the situation of total movement in which she has seen the doll swing. She moves her arms, her torso and her legs in a global reaction without paying particular attention to her feet. The counterproof is easy to furnish. I place the doll above Jacqueline's face, outside the range of any possible contact: Jacqueline recommences to wriggle her arms, torso, and feet, exactly as before, while staring only at the doll (and not at her feet). Jacqueline accordingly establishes a connection between her movements in general and those of the object and not between her feet and the object. I do not observe any tactile control either.

The objection might then be raised that Jacqueline establishes no connection and limits herself to manifesting her joy at the doll's movements without attributing them to her own activity. The child's wriggling would thus be only an attitude accompanying pleasure and not a circular reaction tending toward an objective result. But, without having proofs in the particular case, we can conclude that there is an intentional connection by analogy with the foregoing and following observations in which the child's much more precocious reactions permitted us to make quite a different interpretation.

Observation 97.—Laurent, from the middle of the third month, revealed global reactions of pleasure, while looking at the toys hanging from the hood of his bassinet, or at the hood itself, etc. He babbles, arches himself, beats the air with his arms, moves his legs, etc. He thus moves the bassinet and recommences more vigorously. But it is not yet possible to speak of circular reaction: there is no connection felt between the movements of his limbs and the spectacle seen, but only an attitude of joy and of physical exertion. Again, at 0;2 (17) I observe that when his movements induce those of the toys, he stops to contemplate them, far from grasping that it is he who produces them; when the toys are motionless, he resumes, and so on. On the other hand, at 0;2 (24) I made the following experiment which set in motion a beginning of secondary circular reaction. As Laurent was striking his chest and shaking his hands which were bandaged and held by strings attached to the handle of the bassinet (to prevent him from sucking), I had the

idea of using the thing and I attached the strings to the celluloid balls hanging from the hood. Laurent naturally shook the balls by chance and looked at them at once (the rattle made a noise inside them). As the shaking was repeated more and more frequently Laurent arched himself, waved his arms and legs—in short, he revealed increasing pleasure and through this maintained the interesting result. But nothing yet authorizes us to speak of circular reaction; this could still be a simple attitude of pleasure and not a conscious connection.

The next day, at 0;2 (25) I connect his right hand to the celluloid balls but leave the string a little slack in order to necessitate ampler movements of the right arm and thus limit the effect of chance. The left hand is free. At first the arm movements are inadequate and the rattle does not move. Then the movements become more extensive, more regular, and the rattle moves periodically while the child's glance is directed at this sight. There seems to be conscious coördination but both arms move equally and it is not yet possible to be sure that this is not a mere pleasure reaction. The next day, same reactions.

At 0;2 (27), on the other hand, conscious coördination seems definite, for the following four reasons: (1) Laurent was surprised and frightened by the first shake of the rattle which was unexpected. On the other hand, since the second or third shake, he swung his right arm (connected to the rattle) with regularity, whereas the left remained almost motionless. Now the right could easily move freely without moving the rattle, the string being loose enough to permit Laurent to suck his thumb, for instance, without pulling at the balls. It therefore seems that the swinging was intentional. (2) Laurent's eye blinks beforehand, as soon as his hand moves and before the rattle moves, as though the child knew he was going to shake it. (3) When Laurent temporarily gives up the game and joins his hands for a moment, the right hand (connected to the rattle) alone resumes the movement while the left stays motionless. (4) The regular shakes that Laurent gives the rattle reveal a certain skill; the movement is regular and the child must stretch his arm backward sufficiently to make the rattle sound.—The reaction is the same on the following days: the right arm connected to the rattle is always more active than the left. Moreover the interest is growing and Laurent swings his right arm as soon as he has heard the rattle (while I fasten the string) without waiting to shake it by chance.

At 0;3 (0) I attached the string to the left arm after six days of experiments with the right. The first shake is given by chance: fright, curiosity, etc. Then, at once, there is coördinated circular reaction: this time the right arm is outstretched and barely mobile while the left swings. Now Laurent has plenty of room to do something else with his left arm than shake the rattle, but he does not try at all to free his hand from the string and his glance is directed at the result.—This time

it is therefore possible to speak definitely of secondary circular reaction, even though Laurent only learned a week later to coördinate his prehension with his vision. The fact is all the more certain because at 0;2 (29) I observed the following. Putting my middle finger in his left hand I made his arm swing in a movement analogous to that required to set the rattle in motion: when I stopped, Laurent continued this movement by himself and directed my finger. Such a movement therefore lends itself to intentional coördination beginning at this age.

At 0;3 (10), after Laurent has learned to grasp what he sees, I place the string, which is attached to the rattle, in his right hand, merely unrolling it a little so that he may grasp it better. For a moment nothing happens but, at the first shake due to chance movements of his hand, the reaction is immediate: Laurent starts when looking at the rattle and then violently strikes his right hand alone, as if he felt the resistance and the effect. The operation lasts fully a quarter of an hour during which Laurent emits peals of laughter. The phenomenon is all the more clear because, the string being slack, the child must stretch his arm sufficiently and put the right amount of effort into it.

Observation 98.—Subsequently, at 0;3 (12) Laurent is subjected to the following experiment. I attach to the rattles (hanging from the bassinet hood) my watch chain and let it hang vertically until it almost reaches his face in order to see whether he will grasp it and thus shake the celluloid balls. The result is completely negative: when I put the chain in his hands and he shakes it by chance and hears the noise, he immediately waves his hand (as in the foregoing observation) but lets the chain go without understanding that he must grasp it in order to shake the rattle. The following day, however, he discovers the procedure. At first, when I place the chain in his hand (I only do so in order to start the experiment as this act of prehension would in any case be produced, sooner or later and fortuitously), Laurent waves his hand, then lets the chain go, while looking at the balls. Then he strikes great blows at random which shake the chain (and the rattle) without his grasping it. Then, without looking, he takes hold of the sheet in front of him (doubtless to suck it, as he does a part of the day) and at the same time grasps the chain without recognizing it. The chain then moves the rattle and Laurent again is interested in this sight. Little by little, Laurent thus arrives at discriminating tactilely the chain itself: his hand searches for it and as soon as the outer side of his fingers strikes it, he lets go the sheet or coverlet in order to hold the chain alone. Then he immediately swings his arm while looking at the rattle. He therefore seems to have understood that it is the chain, and not his body movements in general, that shakes the rattle. At a certain moment he looks at his hand which holds the chain; then he looks at the chain from top to bottom.

That evening, as soon as he hears the sound of the rattle and sees the chain hanging, he tries to grasp it without looking either at his hand or at the lower end of the chain (he only looks at the rattle). It happened exactly as follows: while looking at the rattle, Laurent let go with his right hand the sheet which he was sucking (he keeps it in his mouth with his left hand) and searched for the chain, his right hand open and the thumb opposed; as soon as he made contact with the chain he grasped and shook it. After a few moments of this, he resumes sucking his fingers. But when the chain touches him lightly he at once removes his right hand from his mouth, grasps the chain, pulls it very slowly while looking at the toys and apparently expecting a noise: after a few seconds during which he still pulls the chain very gently, he shakes much harder and succeeds. He then bursts into peals of laughter, babbles and swings the chain as much as possible.

At 0;3 (14) Laurent looks at the rattle at the moment I hang up the chain. He remains immobile for a second. Then he tries to grasp the chain (without looking at it), brushes it with the back of his hand, grasps it but continues to look at the rattle without moving his arms. Then he shakes the chain gently while studying the effect. Afterward he shakes it more and more vigorously. A smile and expression of delight.

But, a moment later, Laurent lets go the chain, unawares. He then keeps his left hand (which thitherto held the chain) tightly closed whereas his right hand is open and immobile and he continues to shake his left arm as though he held the chain, while looking at the rattle. He continues thus for at least five minutes.—This last observation shows that, although Laurent knows how to coördinate his grasp movements and the shaking of the rattle, he little understands the mechanism of these connections.

The following days, Laurent grasps and shakes the chain as soon as I hang it up and thus shakes the rattle, but he does not look at the chain before grasping it: he limits himself to searching for it with his hand (right or left, as the case may be) and to grasping it when he touches it. At 0;3 (18) on the other hand, he looks first at the rattle, then at the chain which he grasps after having seen it. The chain has therefore acquired a visual and no longer only a tactile meaning: Laurent knows henceforth that this visual obstacle is simultaneously a thing to be grasped and a means of shaking the rattle. But this tactile-visual coördination relating to the chain in no way implies that Laurent has understood the particulars of the mechanism. There is simply an efficient connection between the grasping of the chain, followed by the adaptation of the arm, and the movements of the rattle. The rest of the observation (see below, Obs. 112) will show us, in effect, to what extent this schema remains phenomenalistic: the chain is not conceived as the extension of the rattle, it is simply a thing to be

grasped and shaken when one desires to see and hear the rattle in motion.

Observation 99.—After having thus discovered the use of the chain hanging from the toys, Laurent generalizes this behavior pattern by applying it to everything that hangs from the hood of his bassinet.

For example, at 0;3 (23) he takes hold of the string attaching a rubber doll to the hood and shakes it immediately. This gesture, simple assimilation of the perceived string to the habitual schema, has of course the effect of shaking the hood and the toys attached to it. Laurent, who did not seem to expect this result, considers it with growing interest and his vigor increases, this time apparently in order to make the spectacle last. After an interruption, I myself shake the hood (from behind): Laurent then looks for the string, grasps and shakes it. He then succeeds in grasping and shaking the doll itself.

The following evening: identical reactions. I observe that Laurent, when he pulls the string, looks at it from top to bottom: he therefore is expecting the result of his act. He also looks at it before grasping it, but not in general: he does not need to do so, since he knows the visual meaning of this object and how to guide his arm by using his kinesthetic sense.

At 0;4 (3) he pulls at will the chain or the string in order to shake the rattle and make it sound: the intention is clear. I now attach a paper knife to the string. The same day I attach a new toy half as high as the string (instead of the paper knife): Laurent begins by shaking himself while looking at it, then waves his arms in the air and finally takes hold of the rubber doll which he shakes while looking at the toy. The coördination is clearly intentional.

At 0;4 (30) Laurent, seeing the doll hang from the rattles at the hood at once looks at these and then shakes the doll only: it is clearly in order to shake the rattles that he grasps the doll.

At 0;5 (25) the same reactions on seeing the string. Furthermore, if I shake the hood (from behind, without being seen) this suffices to make Laurent pull the string in order to make this movement continue.

Observation 100.—At 0;7 (16) Jacqueline reveals a circular reaction analogous to that in Observation 99, but with the delay explained by the three-month retardation which separate her from Laurent from the point of view of prehension of seen objects. She is presented with a doll suspended from the string which connects the hood to the handle of the bassinet. In grasping this doll she shakes the bassinet hood; she immediately notices this effect and begins again at least twenty times in succession, more and more violently and while watching the shaken hood laughingly.

At 0;7 (23) Jacqueline looks at the hood of her bassinet which I shake several times while I remain unseen. As soon as I have finished

she grasps and pulls a cord hanging where the doll had been. My movement of course reminds her of the familiar schema and she pulls the string at the usual place without having necessarily understood the mechanism in detail. Same reaction, but entirely spontaneous, at 0;8 (8), 0;8 (9), 0;8 (13), 0;8 (16), etc.

Observation 100 repeated.—Lucienne, similarly, at 0;6 (5) pulls a doll hanging from the hood in order to make it move; she even looks at the hood ahead of time while grasping the doll, thus revealing accurate foresight. Same observation at 0;6 (10), 0;8 (10), etc.

Observation 101.—Finally two other procedures were employed by Jacqueline, Lucienne, and Laurent in order to shake their bassinet or the objects hanging from the hood. At 0;7 (20) Lucienne looks at the hood and the hanging ribbons; her arms are outstretched and slightly straightened, at an equal distance from her face. She gently opens and closes her hands, then more and more rapidly with involuntary arm movements which thus gently shake the hood. Lucienne then repeats these movements with increasing speed. Same reaction at 0;7 (27), etc. I again observe the phenomenon at 0;10 (27): she moves her bassinet while waving her hands.

At 0;8 (5), Lucienne shakes her head from side to side in order to shake her bassinet, the hood, ribbons, fringes, etc.

Jacqueline, in the same way, shakes her bassinet, at 0;8 (19) while swinging her arms. She even succeeds in differentiating her movements in order to conserve certain effects obtained by chance. She waves her right arm in a certain way (obliquely to her trunk) in order to make the bassinet grate while shaking it all over. In case of failure she corrects herself and gropes, places her arms perpendicularly to her trunk, then places them more and more obliquely until she succeeds. At 0;11 (16) she shakes at a distance (at the end of her bassinet) a jack-in-the-box, while swinging her arms.

At the end of the fourth month Laurent discovered these same two circular reactions which shows that they are general. Thus at 0;3 (23) I find him shaking his head spontaneously (a lateral movement) when confronted by the hanging toys, before grasping the cord which enables him to shake them. Actually this head movement sufficed to shake the whole cradle slightly.

With regard to the arm movements, they are partly the result of the reactions learned in Observations 97 and 98, but partly also of the movements of the whole body which the child sometimes performs in order to shake his bassinet. At 0;3 (25), for example, and at 0;4 (6) he begins by shaking himself all over when confronted by hanging objects, then he shakes his right arm in space. The reaction becomes general during the following days.

Here are some observations concerning secondary circular reactions relating to objects, usually not hanging, that the child grasps in order to move, swing, shake, rub them against others, cause them to make a noise, etc.

Observation 102.—The simplest example is doubtless that of the objects the child simply shakes as soon as he has grasped them. From this elementary schema, which is almost "primary," the following is immediately derived: if the objects brandished produce a sound this is enough to make the child attempt to reproduce it.

As early as 0;2 (26) Laurent, in whose right hand I have put the handle of a rattle, shakes it by chance, hears the noise and laughs at the result. But he does not see the rattle and looks for it in the direction of the hood, at the place from which such a sound usually comes. When he finally sees the rattle he does not understand that this is the object which is making the noise nor that he himself makes it move. He nevertheless continues his activity.

At 0;3 (6), during the fourth stage of prehension, he grasps the rattle after having seen his hand in the same visual field, then brings it to his mouth. But the sound thus produced arouses the schema of the hanging rattle: Laurent shakes himself all over, especially moving his arm, and finally only moves the latter, astonished and slightly worried by the increasing noise.

From 0;3 (15)—that is to say, the present stage—it suffices for Laurent to grasp an object to make him shake it and that he observe the rattle with the handle to cause him to grasp and shake it properly. But subsequently the reaction becomes complicated through the fact that Laurent tries rather to strike it with one hand while holding it with the other, to rub it against the edges of the bassinet, etc. We shall return to this in connection with these latter schemata.

At 0;4 (15) Lucienne grasps the handle of a rattle in the shape of a celluloid ball. The movements of the hand in grasping the rattle result in shaking it and producing a sudden and violent noise. Lucienne at once moves her whole body, and especially her feet, to make the noise last. She has a demented expression of mingled fear and pleasure, but she continues. Hitherto the reaction is comparable to that of Observations 94–95 repeated, and the movement of the hands is not yet maintained for itself, independently of the reaction of the whole body. This reaction lasts a few days but then Lucienne, when she is in possession of the rattle, limits herself to shaking it with the hand that holds it. But—a curious thing—at 0;5 (10) and again at 0;5 (12) she accompanies this movement of the hands with shakes of the feet analogous to those she makes to shake a hanging object (see Obs. 95).

Jacqueline, too, at 0;9 (5) shakes while holding a celluloid rattle in the form of a parrot which she has just been given. She smiles when

the noise is slight, is anxious when it is too loud and knows very well how to gradate the phenomenon. She progressively increases the noise until she is too frightened and then returns to the soft sounds. Furthermore, when the rattle is stuck at one of the ends, she shakes the parrot by turning it in another direction and thus knows how to reëstablish the noise.

Observation 103.—A second classic schema is that of "striking." At 0;4 (28) Lucienne tries to grasp the rattle in Observation 102 when it is attached to the bassinet hood and hangs in front of her face. During an unlucky attempt she strikes it violently: fright, then a vague smile. She brings her hand back with a doubtless intentional suddenness: a new blow. The phenomenon then becomes systematic: Lucienne hits the rattle with regularity a very great number of times.

At 0;5 (0) the same happens to her hanging dolls which she strikes violently.

At 0;6 (2) she looks at a wooden Pierrot which I have hung before her and with which she has rarely played. Lucienne at first tries to grasp it. But the movement she makes in holding out her hand shakes the Pierrot before she has touched it: Lucienne at once shakes her legs and feet in a regular and rapid rhythm in order to maintain the swinging of the object (see Obs. 94). Then she grasps and pulls it. The Pierrot again escapes her and swings; Lucienne reacts by shaking her legs again. Finally she rediscovers the schema of 0;4 (28) and 0;5 (0): she strikes the toy more and more vigorously, without trying to grasp it, and bursts out laughing at the Pierrot's antics. Same reactions at 0;6 (3). At 0;6 (10) she begins by striking the puppet that I hang up, and so makes it swing, then she maintains the movement by shakes of the legs. At 0;6 (19) she strikes the hanging dolls in order to make them swing.

Jacqueline in the same way strikes her toys, from 0;7 (15); at 0;7 (28) strikes her duck, at 0;8 (5) a doll, at 0;9 (5) her cushions, from 0;8 (5) to 0;9 (0) her parrot, etc.

With regard to Laurent the schema of striking arose in the following way. At 0;4 (7) Laurent looks at a paper knife attached to the strings of a hanging doll. He tries to grasp the doll or the paper knife, but, at each attempt, the clumsiness of his movements results in causing him to knock these objects. He then looks at them with interest and recommences.

The next day, at 0;4 (8) same reaction. Laurent still does not strike intentionally but, trying to grasp the paper knife, and noting that he fails each time, he only outlines the gesture of grasping and thus limits himself to knocking one of the extremities of the object.

At 0;4 (9), the next day, Laurent tries to grasp the doll hanging in front of him; but he only succeeds in swinging it without holding it. He then shakes it altogether, while waving his arms (see Obs. 101). But

he thus happens to strike the doll: he then begins again intentionally a series of times. A quarter of an hour later, as soon as he is confronted by the same doll in the same circumstances he begins to hit it again.

At 0;4 (15), faced by another hanging doll, Laurent tries to grasp it, then he shakes himself in order to make it swing, happens to knock it and then tries to strike it. The schema is therefore almost differentiated from the preceding ones but it does not yet constitute a principal and independent behavior pattern.

At 0;4 (18) Laurent strikes my hands without trying to grasp them, but he has begun by simply waving his arms in the air and only "hit" subsequently.

At 0;4 (19), at last, Laurent directly strikes a hanging doll. The schema is therefore completely differentiated. At 0;4 (21) he strikes the hanging toys in the same way and thus swings them as much as possible. Same reaction on the following days.

From 0;5 (2) Laurent strikes the objects with one hand while holding them with the other. Thus he holds a rubber doll in the left hand and strikes it with his right. At 0;5 (6) he grasps a rattle with a handle and strikes it immediately. At 0;5 (7) I hand him different objects which are new to him (a wooden penguin, etc.): he hardly looks at them but strikes them systematically.

It may thus be seen how the schema of striking hanging objects became differentiated little by little from simpler schemata and gave rise to the schema of hitting objects held in the hands. It is noteworthy, however, that though the 4- to 7-month-old child thus learns to swing hanging objects by hitting them as hard as possible, he does not attempt simply to start their swinging to observe it, but often achieves this by chance. It is only at about 0;8 (10) that I observed the latter behavior pattern in Lucienne and Jacqueline and about 0;8 (30) in Laurent. But it definitely differs from the preceding one both from the point of view of causality and that of the intellectual mechanism. The child who strikes in order to swing is, in effect, active himself, whereas he who limits himself to starting the swinging transfers this activity to the object as such. Therefore that is no longer a simple secondary circular reaction but an exploration and almost a sort of experimentation. For this reason we shall not speak of this behavior here but shall study it in connection with the next stage.

Observation 104.—A final noteworthy example is the behavior pattern consisting in rubbing objects against hard surfaces such as the wicker of the bassinet. Lucienne, from 0;5 (12), and Jacqueline a little later, about 0;7 (20) used the toys they held in their hands to rub the surfaces of the bassinet. Laurent discovered this at 0;4 (6) in circumstances which it is worthwhile to analyze.

At 0;3 (29) Laurent grasps a paper knife which he sees for the first time; he looks at it a moment and then swings it while holding it in

his right hand. During these movements the object happens to rub against the wicker of the bassinet: Laurent then waves his arm vigorously and obviously tries to reproduce the sound he has heard, but without understanding the necessity of contact between the paper knife and the wicker and, consequently, without achieving this contact otherwise than by chance.

At 0;4 (3) same reactions, but Laurent looks at the object at the time when it happens to rub against the wicker of the bassinet. The same still occurs at 0;4 (5) but there is slight progress toward systematization.

Finally, at 0;4 (6) the movement becomes intentional: as soon as the child has the object in his hand he rubs it with regularity against the wicker of the bassinet. He does the same, subsequently, with his dolls and rattles (see Obs. 102), etc.

These few examples of secondary circular reactions thus constitute the first behavior patterns involving an action brought to bear upon things and no longer a utilization, in some way organic, of reality. Such a question again raises the whole problem of mental assimilation.

When the nursling takes the breast for the first time and then immediately begins to suck and swallow or, even earlier, when he impulsively moves his lips and then continues empty sucking, one might suppose that this reproductive assimilation as well as the recognitions and generalizations which prolong it are themselves dependent for their conditioning upon an earlier need: the organic need to take nourishment and to suck. So also, when the child learns to look, listen, or grasp, it could be asserted that this functional activity is only assimilatory because it constitutes first of all a satisfaction of physiological needs. If such were the case, it would not be possible to understand how the child's activity can become centered beginning at 4 to 6 months on results such as those of the secondary circular reactions which do not correspond, in their externality, to any internal, definite and particular need.

But, as we have seen (Chapter I, §3) the retention in consciousness of a physiological need is not a simple fact nor an immediate condition, and it is fitting to distinguish two distinct series in the humblest act of repetition, by which reflex use or acquired association begin: the organic series and the psychic se-

ries. From the physiological point of view it is certain that need explains repetition: it is because sucking corresponds to a need that the nursling does not stop sucking, and it is because of the connection established between thumb sucking and satisfying this need that the 1- to 2-month-old child puts his thumb back into his mouth as soon as he is capable of this coördination. But it is noteworthy, from this strictly physiological point of view, that all needs depend, either immediately or remotely, upon a fundamental need which is that of the organism's development; that is to say precisely, upon assimilation: it is due to the subordination of the organs to this chief tendency—which defines life itself—that the functioning of each one gives rise to a particular need. Now, from the psychological point of view, it proceeds in exactly the same way. The need sets in motion the act and its functioning, but this functioning itself engenders a greater need which from the very first goes beyond the pure satisfaction of the initial need. It is therefore fruitless to ask if it is need that explains repetition or the reverse: together they form an inseparable unity. Hence the primary fact is neither the need anterior to the act nor repetition, the source of satisfaction, but it is the total relation of the need to the satisfaction. From the point of view of behavior, this relation is none other than the operation by which an already organized mechanism becomes established by functioning and functions by utilizing a condition external to itself: it is therefore functional assimilation. From the point of view of consciousness, this relation is also of an operative nature, and this is why one cannot seek the fundamental truth of psychology either in a state of simple consciousness or in an isolated tendency. Need and satisfaction are, in effect, vicarious, and oscillate between the purely organic and the functional; moreover, they are experienced relatively to one another. Consequently they are both connected to a fundamental operation—of which they are only the influence of variable and approximative consciousness—by which the behavior pattern puts its own functioning into relationship with the conditions of the environment. The relation of need and satisfaction thus reveals a relation anterior to assimilation, according to which the subject only apprehends the object relatively to his own activity. Conse-

quently, just as all physiological needs depend on a main tendency—that of the organism's development through assimilation to the environment—so also every elementary psychic function, however subordinated it may seem to be to the satisfaction of a precise physiological need, involves an activity which will gradually integrate the whole of the behavior patterns: the assimilation of the object to the subject in general.

When these principles are remembered it is easy to understand how needs which were at first chiefly organic can be subordinated little by little to functional needs and how the latter can give rise to operations concerning the interrelations of things and no longer the relations of things to the organs of the body. For example, how does it happen that the child, instead of merely grasping the doll hanging from the hood of his bassinet, makes use of it in order to shake the hood (Obs. 100)? Until then, in effect, the doll was an object to be looked at, grasped, sucked, heard, etc., but not at all a thing to produce extrinsic results such as the shaking of the hood. The transition from the first state to the second must therefore be explained. With regard to the movements of the hood, either they are perceived for the first time, and then it must be understood why from the very first they give rise to an attempt at repetition, or else they have already been a thing to look at, to hear, etc., and it must be understood how they are transformed into a result to be maintained by new means.

The question is simplified as soon as one sees the essential fact that, among the unknown phenomena observed by the child only those which are experienced as dependent on the activity itself give rise to a secondary circular reaction. But let us note that this is not as natural as it might seem. One might very well think that the child, confronted by any new spectacle at all, even one independent of him from the observer's point of view, tries from the very first to reproduce it or make it continue. This is exactly what is subsequently revealed when, accustomed to repeat everything through circular reaction, the child generalizes this behavior pattern and tries to discover "procedures to make interesting sights last" (see Obs. 112–118). But observation shows that that is derived behavior and that, before practicing the sec-

ondary circular reaction, the child, in order to assimilate new sights, limits himself to utilizing the primary reactions. For instance, when he sees the hanging toys move without yet knowing that it is he who sets them going, or when he sees the rattle with a handle without yet realizing that he is the cause of the effect produced, Laurent is already interested in these phenomena; that is to say, he tries to assimilate them but he tries to conserve them only by looking at or listening to them without yet attempting to reproduce them by means of hand or arm movements. This, however, does not mean that these phenomena are conceived by him as "objective" and independent of his activity in general. It is very possible, on the contrary, that when looking at an object or turning his head to listen to it, etc., the subject may have the impression of participating in the repetition or continuation of the sensorial image. But, to be precise, such a relationship must be experienced in order that the attempt at repetition which constitutes the secondary circular reaction may begin.

Hence it cannot be said that the present behavior pattern consists in repeating everything which happens to appear in the child's visual field. The secondary circular reaction only begins when a chance effect of the action itself is understood to be the result of this activity. Thereafter it is easy to grasp the continuity which exists between the primary and secondary reactions. Just as, in the first, the objective is aliment for sucking, vision or prehension, so also, in the second, it becomes aliment for any movement produced by differentiation of prehension and the movements of the forearm. True, the difference is great between the somewhat centripetal interest of sucking, or even of sight, and the centrifugal interest of the present level, an interest directed at the external result of the acts. But this contrast is modified if it is recalled that a sensorial image is the more objectified and externalized because it coördinates within itself more schemata and because all the intermediaries between the primary and secondary reactions therefore exist. A visual objective, for example, is much nearer the actual "object" if it is simultaneously a thing to see, to hear and to touch than if it is only an image to contemplate. Consequently the movement of the hood or the sound

of a stick against the wicker of the bassinet will give rise to an externalization all the greater because they are simultaneously to be seen, heard, and reproduced by means of hand movements. Through a paradox analogous to that of the development of the sciences, it happens that reality is objectified in proportion as it is elaborated by the thinking and working of the subject's schemata, whereas the phenomenalism of immediate perception is only subjectivism. Furthermore, while incorporating in his activity results so remote from himself, the child introduces into his proceedings a series of intermediaries. For instance, when he shakes the hood of his bassinet by grasping a hanging doll he is obliged, even without any understanding of the relations which exist between the two states, to see in the hood's movement the prolongation of the act of grasping the doll. The assimilation of the movements of the hood to the schema of prehension thus presupposes putting these movements into relationship with those of the doll. Such a process explains why every reproductive assimilation of a remote spectacle brings with it an active elaboration of relationships. The action ceases to be simple in order to introduce a beginning of differentiation between means and ends, and the assimilation of things to the self becomes construction of relationships between things.

The assimilation characteristic of secondary circular reaction is, in short, only the development of assimilation at work in the primary reactions. Just as everything, in the child's primitive universe, is for sucking, looking, listening, touching, and grasping, everything gradually becomes something to be shaken, swung, rubbed, etc., according to differentiations of the manual and visual schemata. But before discerning the mechanism according to which these progressive accommodations operate, there remains to be explained how a remote spectacle can thus be conceived as produced by the action itself (which is, as we have just noted, the condition for the advent of the secondary reaction). This question can be answered in one word: this discovery is made through reciprocal assimilation of the present schemata. We recall, in this connection, how coördination such as that of vision or hearing is established: by seeking to see that which he hears and to hear that which he sees, the child perceives little

by little that a same given object is simultaneously the source of sounds and visual image. It is in an analogous way that the coordination of vision and prehension subsequently is brought about. After having looked at his hands and at the objects grasped, the child tries to move the visual image which he also sees; in that way he discovers that one may grasp what one sees as well as look at what one grasps. Now in the case of the beginnings of secondary circular reaction a phenomenon of the same kind ensues. For instance when Laurent unwittingly starts a movement of the toys by pulling a chain or rubs a paper knife against the wicker of his bassinet he is looking at, listening to, etc., the effect thus produced without trying to conserve it by other means. But, precisely because he is in the act of shaking the chain or the paper knife while he looks at or listens to the result of these movements, the two kinds of schemata sooner or later end by being reciprocally assimilated. The child then applies himself to moving with his hand the image that he looks at, just as formerly he was led intentionally to move the visual image of his own limbs. This does not yet mean that he tries to reproduce the objective phenomenon as such (which will constitute the secondary circular reaction), but simply that his visual and manual schemata, being simultaneously active, tend to assimilate each other according to a general law. But as soon as this reciprocal assimilation has been formed the child understands that the external result which he has observed (the movements of the toys or the sound of the paper knife against the wicker) depends upon his manual as well as his visual or auditory activity, and this understanding thereafter gives rise to an immediate circular reaction; that is to say, to an act of reproductive assimilation. From the point of view of assimilation itself, secondary circular reaction thus prolongs the primary reaction, and the child's interest only becomes externalized on the interrelations of things as a function of the increasing coördination of the present schemata (the primary schemata).

§2. THE SECONDARY CIRCULAR REACTIONS—II. ACCOMMODATION AND ORGANIZATION OF THE SCHEMATA.—Until the present behavior patterns—that is, dur-

ing the entire stage of the pure primary reactions—accommodation remained relatively subordinated to assimilation. Sucking, looking, grasping, consisted in incorporating observed objects into corresponding schemata of assimilation, free to accommodate these schemata to a variety of things. So it is that the movements and positions of hands, eyes, and mouth vary as a function of the objectives, in a continuous accommodation concomitant to assimilation, although of opposite direction. At the other extreme of sensorimotor behavior patterns—that is to say, in the tertiary circular reactions, we shall see—on the contrary, that accommodation precedes assimilation, in a sense. Confronted by new objects the child intentionally seeks to find out in what way they are new and so experiments upon them before assimilating them to a schema constructed on their effect. Hence accommodation evolves from the simple differentiation of schemata, peculiar to the primary reactions, to the search for the new, peculiar to the tertiary reactions. What of the secondary circular reaction?

In its point of departure the latter reveals no accommodation other than that of the primary reactions: a simple differentiation of schemata as function of the object. So it is that Laurent discovers the possibility of hitting a hanging doll, simply while trying to grasp it (Obs. 103), and that Lucienne and Laurent learn to rub a toy against the side of the bassinet simply while swinging it (Obs. 104, etc.). But, contrary to that which occurs in the primary reactions, this initial differentiation of the schema does not, without adding something, lead to its application to new objects, precisely since Laurent does not succeed in grasping the doll nor in moving the rattle as he intends to, but discovers an unforeseen phenomenon due to this very defeat: the doll swings when one strikes it and the rattle rubs the wood of the bassinet. It is then that the specific accommodation of the secondary circular reaction is produced: the child tries to rediscover the movements which lead to the result observed. As we have previously demonstrated, the child begins, in effect, by trying to assimilate this new result while limiting himself to looking at it, etc. (primary schemata). Then, as soon as he has discovered, through reciprocal assimilation of the schemata, that this result depends on his manual activity, he tries to reproduce it by as-

similation to this activity. But, as it is just in differentiating the latter that the subject has by chance obtained the new result, the question is to establish this differentiation intentionally and it is in this that the accommodation peculiar to the secondary reactions consists: rediscovering the movements which have given rise to the result observed. This accommodation, without preceding assimilation as is the case in the tertiary reaction, or simply doubling it, as is the case in the primary reaction, consists, then, in completing it at the moment when the new schema is formed. Therefore accommodation is no longer an almost automatic differentiation of the schemata, nor yet an intentional search for novelty as such, but it is a voluntary and systematic fixation of the differentiations imposed by new realities which arise by chance. A concrete example will enable us to understand:

Observation 105.—Laurent, from 0;4 (19) as has been seen (Obs. 103) knows how to strike hanging objects intentionally with his hand. At 0;4 (22) he holds a stick; he does not know what to do with it and slowly passes it from hand to hand. The stick then happens to strike a toy hanging from the bassinet hood. Laurent, immediately interested by this unexpected result, keeps the stick raised in the same position, then brings it noticeably nearer to the toy. He strikes it a second time. Then he draws the stick back but moving it as little as possible as though trying to conserve the favorable position, then he brings it nearer to the toy, and so on, more and more rapidly.

The dual character of this accommodation may be seen. On the one hand, the new phenomenon makes its appearance by simple fortuitous insertion in an already formed schema and hence differentiates it. But, on the other hand, the child, intentionally and systematically, applies himself to rediscovering the conditions which led him to this unexpected result.

It goes without saying that the use of the stick described in this example was only episodical: it has nothing to do with the "behavior pattern of the stick" which we shall describe in connection with the fifth stage.

This analysis of the accommodation peculiar to the secondary circular reactions makes it possible to understand why the child's activity, which hitherto seemed to us essentially conservative, henceforth appears to be indefinitely diversified.

That activity should be conservative at the reflex stage is

only natural. The schemata belonging to the reflexes having already been hereditarily elaborated, reflex behavior simply consists in assimilating the datum to these schemata and in accommodating them to reality by simple use without transforming them. With regard to the primary circular reactions and the habits which derive from them the same is true fundamentally, despite the obvious acquisitions characteristic of these behavior patterns. When the child learns to grasp, to look, to listen, to suck for the sake of sucking (and no longer only in order to eat), he assimilates to his reflex schemata an increasing number of realities and, if acquired accommodation to these realities exists, they remain nevertheless simple aliments for the conservation of the schemata. With regard to acquisitions through the coördination of schemata, the question is, as we have seen, only one of reciprocal assimilation, that is to say, again of conservation. This assimilation, consequently, does not exclude enrichment and cannot be reduced to identification pure and simple—this goes without saying—but it nevertheless remains essentially conservative.

How, then, can it be explained that at a given moment the circle of conservation seems to break and the reproduction of new results prolongs the primary reaction by thus creating multiple relations between the things themselves? Is it reality alone which makes the framework of assimilation crack by confining the child's activity to a progressive diversification, or can this diversification be considered as a function of assimilation and as always depending on conservation?

There is no doubt that both are present. On the one hand, reality forces the child to make indefinite accommodations. As soon as the child knows how to grasp that which he sees, the objects he manipulates put him brutally in the presence of the most varied experiences. The rattles which swing while producing disturbing sounds, the bassinet which shakes while producing the movement of the hanging toys, the boxes which offer opposition by reason of their weight and shapes, the coverlets or strings held back or attached in an unpredictable way, everything is an opportunity for new experiments, and the content

of these experiments cannot give rise to assimilation without a continuous accommodation which opposes it in a way.

But, on the other hand, this accommodation is never pure, and secondary circular reaction would not be explainable if the child's behavior did not remain basically assimilatory and conservative. As we have just seen, each of the secondary circular reactions which appear in the child is derived by differentiation from a primary circular reaction or a secondary reaction grafted upon a primary reaction. Everything thus goes back to movements of legs or feet, arms or hands, and it is these "circular" movements of prehension which become differentiated in movements directed at pulling, shaking, swinging, displacing, rubbing, etc. When Lucienne, at 3 to 4 months, shakes her bassinet and her dolls (Obs. 94–95) she limits herself to moving feet and legs, in conformity to a primary schema. When Laurent at 0;2 (24)–0;3 (0) shakes a rattle attached to his arm (Observation 97) before knowing how to grasp, he is only prolonging the spontaneous circular movements of that arm. And when at 0;3 (13) he learns to shake the rattle by means of a chain, this is simply because he uses his nascent schema of prehension (Obs. 98). The same applies to all the secondary circular reactions: each is the prolongation of an already existing schema. With regard to "procedures to make interesting sights last" of which we shall speak later on, they in turn prolong these circular reactions. The only difference between the secondary reactions and the primary reactions is, therefore, that henceforth the interest is centered on the external result and no longer only on the activity as such. But that is not contradictory to the conservative character of this function: in effect, the external result, arising suddenly at the very center of the child's activity, interests him at one and the same time inasmuch as it is related to his essential schemata and inasmuch as it is unforeseen and baffling. If it were only new it would merely deserve momentary attention; but on the contrary, it appears to the subject as being connected with his most familiar acts or with his schemata in actual use. Moreover this unforeseen result leads astray all that of which these schemata habitually admit. The attention is, therefore, perforce centered on the exterior and no longer only on the function. In short, the

secondary circular reactions are essentially conservative and assimilatory since they prolong without adding anything to the primary reactions and, if the child's interest is displaced and externalized on the material result of the acts, it is simply because this result is function of an increasingly rich assimilatory activity.

What do these acquisitions mean now from the point of view of organization?

Organization, we recall, is the internal aspect of the functioning of the schemata to which assimilation tends to reduce the external environment. It is therefore, one might say, an internal adaptation of which accommodation and assimilation combined form the external expression. Actually each schema, or each ensemble of schemata, consists in a "totality" independently of which no assimilation would be possible and which in turn rests upon a number of interdependent elements (see Introduction, §2). Furthermore, to the extent that these totalities are not entirely realized but are in process of elaboration, they involve a differentiation between "means" and "ends," between "values" subordinated to the formation of the whole, this whole not complete for an "ideal' totality. This fundamental mechanism of organization accompanies internally the external manifestations of adaptation. How then does it function, during this stage, and in what form is it made manifest in the child's behavior?

It is not difficult to see, in effect, that the secondary schemata, once they have been elaborated through complementary assimilation and accommodation, consist in organized schemata: in the capacity of a practical concept in which the schema thus constitutes a "totality," whereas the "relations" on which it rests determine the reciprocal relationships which constitute this totality.

With regard to the interorganization of the schemata—that is to say, the coördination of the secondary schemata—it only manifests itself during the following stage. We shall therefore speak of it again in connection with that fourth stage. But it is apparent that the different schemata of this stage, without intercoördination in intentional series conscious of their unity, have already reached balance among themselves and constitute a sys-

tem of unconsciously interdependent relationships. Without this total subjacent organization it would be impossible to explain how any object which is presented to a child is immediately classified, that is, assimilated through a simultaneously reproductive and recognitory act of assimilation suitable to this object and no other.

The totalities in process of being constituted or reconstituted remain to be examined. An original totality is constituted, in effect, every time that a new schema is elaborated by contact with things, and this contact is reconstituted every time the subject again finds himself confronted by suitable objects and assimilates them to the schema in question. The organization of these totalities marks progress over that of the "primary" schemata in this sense that, for the first time, and to the extent that the "relations" of which we shall presently speak are formed, the "means" begin to be distinguished from the "ends": consequently the movements made and the objects used henceforth assume different "values" subordinated to an "ideal" totality, that is to say, not yet realized. For instance when, in Observation 98, Laurent discovers that the hanging chain can be used to shake the rattle to which it is attached, it is certain that the action of pulling the chain is conceived as a "means" toward the "end" of reproducing the interesting result, although the means have been supplied at the same time as the end in the initial action reproduced by circular reaction. It is after the event, and when the subject seeks the result for himself, that he distinguishes between means and ends. Now, such a distinction is certainly new to the child's consciousness. It is true that one could in the same way analyze any primary schema such as that of thumb sucking. The action of introducing the thumb into the mouth could be conceived as a means at the service of the end which consists in sucking. But it is clear that such a description does not correspond to anything from the point of view of the subject himself, since the thumb is not known independently of the act which consists in sucking it; on the contrary, the chain serving to shake the rattle has been perceived and manipulated before being conceived as a "means" and does not cease to be regarded as distinct from the rattle. With regard to the coördinations among primary

schemata (grasping in order to suck, etc.) they make manifest, it is true, the present distinction between means and ends, precisely since "secondary circular reaction" is only made possible by this kind of coördination (that of prehension and vision and, in the elementary cases, that of foot movements with vision). But, as we have seen, they consist in simple reciprocal assimilations, resulting in the formation of new global entities in which, consequently, the difference of which we speak is immediately obliterated.

But, if the distinction between means and ends is only established in the course of the elaboration of secondary schemata, we must not allow ourselves to believe that it is thus achieved and to identify it with that which it will become during the next stage—that is to say, at the time of the coördination of the same schemata. In effect, we have just seen that, during the present stage, the secondary schemata are not yet intercoördinated; each one constitutes a more or less self-enclosed totality instead of arranging itself in series analogous to that which in reflective thought is reason or the implication of concepts. From the fourth stage, on the contrary, these schemata will become intercoördinated when it will be a question of adapting themselves to unforeseen circumstances, thus giving rise to behavior patterns which we shall call the "application of known schemata to new situations." It is only in this connection that the "means" will be definitely dissociated from the "ends": the same schema being capable of serving as "means" to different ends will therefore assume an instrumental value which is much clearer than that which can be offered, during the present stage, by a movement (such as pulling the chain) always connected to the same end (shaking the rattle) and whose function of "means" has been discovered fortuitously.

In conclusion, it can be said that the secondary circular reactions make intelligent adaptation manifest, without, however, constituting true acts of intelligence. If compared to primary circular reactions they reveal intelligence because they elaborate an ensemble of almost intentional relations between things and the subject's activity. In effect these relations with the environment being at the outset complex, they give rise, as we have just

seen, to a beginning of differentiation between means and ends and thereby to a rudiment of intention. When the child pulls the chain in order to shake a rattle, he carries out a much higher behavior pattern than simply grasping an object which he sees.

But, furthermore, the secondary circular reactions do not yet constitute complete acts of intelligence for two reasons. The first is that the relations utilized by the child (shaking himself in order to move the bassinet, pulling a chain in order to shake a rattle, etc.) were discovered fortuitously and not for the purpose of resolving a problem or satisfying a need. The need arises from the discovery and not the discovery from the need. On the contrary, in the true act of intelligence there is pursuit of an end and only subsequently discovery of means. The second reason, closely connected with the first, is that the only need involved, in the secondary circular reactions, is a need of repetition. For the child, the question is simply to conserve and reproduce the beneficial result discovered by chance. It is need that sets the act in motion at each new turn of the circle of circular reaction and it can surely be said, in this sense, that need is anterior to the act; in any case it is this fact which makes it possible to speak of intention and of intelligence. But, this need being only a desire for repetition, the means put to work in order to reproduce the desired result are already found. They are entirely contained in the fortuitous action which is at the point of departure of the ensemble of the reaction and which it is simply a question of repeating. The role of intelligence involved in such behavior patterns therefore consists simply in rediscovering the series of movements which have given rise to the beneficial result and the intention of these behavior patterns merely consists in trying to reproduce this result. Therein, let us repeat, is an outline of an intelligent act, but not a complete act. In effect, in a true act of intelligence, the need which serves as motive power not only consists in repeating, but in adapting, that is to say, in assimilating a new situation to old schemata and in accommodating these schemata to new circumstances. It is to this that secondary circular reaction will lead by extension; but, in this respect, it is not yet there.

All the more reason why it is impossible to attribute to such

behavior patterns the capacity to engender or to utilize representations. There could be no question, at first, of a representation of the means employed: the child does not know in advance that he will perform a certain movement, since he simply tries to rediscover the motor combination which succeeded and since he subsequently limits himself to repeating his acts. With regard to the goal, does the child, for example, keep the memory of the shaken rattle in the form of visual or auditory images and does he try to reproduce something which conforms to this representation? There is no need for so complicated a mechanism to account for such behavior patterns. It is enough that the sight of the rattle created a sufficiently powerful interest for this interest to orient the activity in the direction already followed an instant before. In other words, when the rattle stops moving, there ensues a vacuum which the child immediately tries to fill and he does so by utilizing the movements which have just been performed. When these movements lead to a result which resembles the earlier spectacle, there is certainly recognition, but recognition does not presuppose the existence of representation. Recognition simply requires that the new result embrace entirely the structure of the assimilatory schema outlined at the beginning of circular reaction. Of course, if this mechanism repeats itself indefinitely, there can be a beginning of representation but, without being able to determine precisely when this appears, it can be said that it is not pristine and that it is useless to the formation of the present behavior patterns.

On the other hand, the secondary schemata constitute the first outline of what will become "classes" or concepts in reflective intelligence: perceiving an object as being something "to shake," "to rub," etc. This is, in effect, the functional equivalent of the operation of classification peculiar to conceptual thought. We shall return to this, in connection with the fourth stage, when the secondary schemata will have become more "mobile," but the observation arises now.

Furthermore, just as the logic of classes is correlative to that of "relations," so also the secondary schemata involve putting things into conscious interrelationships. Herein resides, as we have seen, their chief novelty in relation to the primary sche-

mata. What are these relationships? It is apparent, since they are established within the same schema and not due to coördinations between separate secondary schemata, that they remain essentially practical and, consequently, global and phenomenalistic, without yet involving the elaboration of really "objective" substantial, spatial, or causal structures. When, in the example already commented upon, the child pulls a chain in order to shake a rattle, the relation he establishes between the chain and the rattle is not yet a spatial, causal and temporal relation between two "objects," it is a simple practical relationship between the act of pulling and the result observed. It is during the fourth stage, with the coördination of the secondary schemata and the implications which result, that these relations begin to become objectified, with the sole disadvantage of not being really objectified until the fifth stage.

But, however empirical these relations remain, they nevertheless constitute, from the formal point of view, the beginning of a system separate from that of "classes" which will later become increasingly differentiated. Furthermore, this elementary elaboration of relations, like the "logic of relations" of reflective intelligence, leads, at the outset, to the discovery of quantitative relations as distinguished from simple qualitative comparisons inherent in the classification as such.

We know, in effect, that if the concepts or "classes" only structure reality as a function of the qualitative resemblances or differences of the beings so classed, the "relations," on the other hand, involve quantity and lead to the elaboration of mathematical series. Even the relations with qualitative content, such as "darker" or "brother of" constitute, in effect, a "seriation" of a kind other than the relations of appurtenance or of inherence and hence presuppose either the concepts of "more" and "less" which are frankly quantitative, or a discrimination and ordering of individuals, which relationships envelop number.

Now this is precisely what happens on the sensorimotor plane as soon as the first relations are elaborated. For instance, the relations established by the child between the act of pulling the chain and the movements of the rattle (Obs. 98) at the outset leads the subject to the discovery of a quantitative connection

immanent in this relation: the more the chain is shaken the more violently the rattles move.

Observation 106.—In the evening of 0;3 (13) Laurent by chance strikes the chain while sucking his fingers (Obs. 98): he grasps it and slowly displaces it while looking at the rattles. He then begins to swing it very gently which produces a slight movement of the hanging rattles and an as yet faint sound inside them. Laurent then definitely increases by degrees his own movements: he shakes the chain more and more vigorously and laughs uproariously at the result obtained.—On seeing the child's expression it is impossible not to deem this gradation intentional.

At 0;4 (21) as well, when he strikes with his hand the toys hanging from his bassinet hood (Obs. 103) he visibly gradates his movements as function of the result: at first he strikes gently and then continues more and more strongly, etc.

These gradations are found in nearly all the preceding observations as well as in the use of "procedures to make interesting sights last" (see below, Obs. 112–118).

Thus it may be seen how the secondary schemata constitutes not only a kind of concept or practical "class" but also a system of relationships enveloping the quantity itself.

§3. RECOGNITORY ASSIMILATION AND THE SYSTEM OF MEANINGS.—The facts hitherto studied constitute essentially phenomena of reproductive assimilation: through repetition rediscovering a fortuitous result. Before seeing how this behavior is extended into generalizing assimilation and thus gives rise to "procedures to make interesting sights last," let us once more emphasize a group of facts, which no longer constitute circular reactions in themselves but which are derived from secondary reactions, in the capacity of recognitory assimilations. What happens, in effect, is that the child, confronted by objects or sights which habitually set in motion his secondary circular reactions, limits himself to outlining the customary movements instead of actually performing them. Everything takes place as though the child were satisfied to recognize these objects or sights and to make a note of this recognition, but could not recognize them except by working, rather than thinking, the schema helpful to recognition. Now this schema is none other than that of

the secondary circular reaction corresponding to the object in question.

Here are some examples:

Observation 107.—At 0;5 (3) Lucienne tries to grasp spools suspended above her by means of elastic bands. She usually uses them in order to suck them, but sometimes she swings them while shaking herself when they are there (see Obs. 94 and 94 repeated). She manages to touch but not yet to grasp them. Having shaken them fortuitously, she then breaks off to shake herself a moment while looking at them (shakes of the legs and trunk), then she resumes her attempts at grasping.

Why has she broken off in order to shake herself a few seconds? It was not in order to shake the spools because she did not persevere and was busy with something else at the moment when she made this movement: neither was it in order to facilitate her attempts at grasping. Is it a question of a purely mechanical movement started by the sight of their chance swinging? It would seem so, but the rest of the observation shows that this behavior pattern was renewed too often to be automatic: it therefore certainly has a meaning. Neither is it a question of a sort of ritual analogous to those we shall study in connection with the beginnings of play because the child, far from seeming to amuse herself, was perfectly serious. Everything transpires as though the subject, endowed for a moment with reflection and internal language, had said to himself something like this: "Yes, I see that this object could be swung, but it is not what I am looking for." But, lacking language, it is by working the schema that Lucienne would have thought that, before resuming his attempts to grasp. In this hypothesis, the short interlude of swinging would thus be equivalent to a sort of motor recognition.

Such an interpretation would remain completely hazardous when confronted by a single fact. But its probability increases along with the following observations. For instance at 0;5 (10) Lucienne again relapses into states identical to those vis-à-vis a rattle. So also, at 0;6 (5) she shakes herself several times in succession, very briefly each time, as soon as she has caught sight of her hand (which comes out of her mouth or by chance enters the visual field, etc.). One cannot see what this movement might mean if not that it is the outline of some action suggested by this sight.

At 0;6 (12) Lucienne perceives from a distance two celluloid parrots attached to a chandelier and which she had sometimes had in her bassinet. As soon as she sees them, she definitely but briefly shakes her legs without trying to act upon them from a distance. This can only be a matter of motor recognition. So too, at 0;6 (19) it suffices that she catches sight of her dolls from a distance for her to outline the movement of swinging them with her hand.

From 0;7 (27) certain too familiar situations no longer set in motion

secondary circular reactions, but simply outlines of schemata. Thus when seeing a doll which she actually swung many times, Lucienne limits herself to opening and closing her hands or shaking her legs, but very briefly and without real effort. At 0;10 (28) she is sitting in her bassinet. With my hand I slightly shake the whole apparatus by touching the handle. Lucienne laughs and responds by gently shaking her hand, but this is not an attempt to make me continue; it is only a sort of acknowledgment.

Observation 107 repeated.—Laurent, too, at 0;4 (21) has an object in his hands when, in order to distract him, I shake the hanging rattles which he is in the habit of striking. He then looks at the rattles without relinquishing his toy and outlines with his right hand the movement of "striking." From 0;5 I often note such outlines of acts when confronted by familiar objectives; they are similar to Lucienne's.

It may be seen how such behavior patterns constitute a separate class. It is no longer a question of a simple secondary circular reaction, since the child reveals no effort to arrive at a result. It is true that there might be a simple automatization of earlier reactions. But, on the one hand, the child's expression does not give the impression that he acts mechanically and, on the other hand, we do not see at all why an automatic reproduction of useless acts would last so long (we have only chosen one or two examples from among innumerable ones). In the second place, these behavior patterns cannot be identified with the "procedures to make an interesting spectacle last," of which we shall speak presently. These "procedures . . ." appear at the moment when a sight contemplated by the child is interrupted, and their purpose is to act upon the things themselves, while the present behavior patterns arise at simple contact with an object, regardless of whether this is immobile or mobile, and without an attempt at acting upon it. In the third place, neither is it possible to reduce these behavior patterns to "explorations" and "tertiary circular reactions" of which we shall speak subsequently. The latter relate to new objects whereas the present behavior patterns are set in motion by familiar objects.

We therefore only see one interpretation for Observations 107–107 (repeated): they are acts of recognitory assimilation. Confronted by a familiar object or event, but whose sudden appearance was not foreseen by the child, the latter needs to adapt him-

self to the unexpected. This is what occurs, for example, when Lucienne sees a spool swing at the moment she wishes to grasp it, or perceives her hand, the parrots, etc., at a moment and in a place she did not expect them, etc. To adapt oneself means, in such cases, simply noting the event, in so far as it is known and of no use at present: it is then, without adding anything, a matter of recognizing and classing the thing. The subject will subsequently do this in enunciated words or in internal language but, due to his present lack of such symbolic instruments, the child is limited to outlining the gestures of the corresponding schema, used thus in the capacity of a recognitory schema. In other words, instead of saying: "Oh! the spool is swinging," or: "There is my hand. . . . There is the parrot. . . There is the bassinet which is moving," the child assimilates these facts by means of motor concepts, not yet verbal, and, by shaking his own legs or hands, so indicates to himself that he understands what he perceives.

The existence of this recognitory assimilation might seem doubtful if it had not been prepared by all the reproductive assimilation of the secondary circular reaction. Two circumstances show that reproductive assimilation brings with it at the outset the formation of a sensorimotor recognition. In the first place, the very fact of rediscovering an interesting result—that is, the definition of secondary circular reaction—entails an increasingly accurate recognition. In the second place, once the schema has been constituted, it is reactivated by each new contact with the objects due to which it arose. Each time, for example, the child sees the doll hanging he is in the habit of swinging by shaking himself or striking it, etc., of his own accord he resumes shaking himself, striking, etc. This activation of the schema by immediate assimilation of the object to its function is simultaneously a recognitory and reproductive fact of assimilation, these two aspects of the assimilatory process being as yet undifferentiated during this initial phase. It is therefore very natural that simply recognitory assimilation should dissociate itself at a given moment from reproductive or purely active assimilation. At first, as revealed by the beginning of Observation 107, it can happen that the child finds himself incited by external facts to activate a schema at the exact moment when his interest is elsewhere and

is already acting there according to a different schema. In this case the schema which interferes with the main action will simply be outlined whereas the activity in progress will be pursued normally. Then, it can happen, as revealed by the end of the same Observation 107, that the schema excited by the external events is too familiar to give rise to a real action and so is again limited to a short and simple indication of it. In both cases the outline of activity, replacing real activity, is consequently equivalent to a step which is more contemplative than active; in other words, to an act of simple recognition or simple classification rather than to effective action. So it may be seen that recognitory assimilation, at first involved in reproductive assimilation, detaches itself from it little by little, to remain in the half-active, half-verifying state which is the state nearest to the pure judgment of verification of which the sensorimotor intelligence is capable.

These remarks lead us to the analysis of "meanings" and to the study of the signals or signs characteristic of this third stage. To understand the nature of the following facts it behooves us first to remind ourselves briefly how the problem of meaning arises.

To assimilate a sensorial image or an object, whether through simple assimilation, recognition, or generalizing extension, is to insert it in a system of schemata, in other words, to give it a "meaning." Regardless of whether these schemata are global and vague or, as in the recognition of an individual factor, they are circumscribed and precise, consciousness does not know any state except in reference to a more or less organized totality. Ever since then it is necessary to distinguish, in every mental element, two indissolubly bound aspects whose relationship constitutes meaning: the signifier and the signified. With regard to "meanings" of a higher order, which are also collective meanings, the distinction is clear: the signifier is the verbal expression, that is, a certain articulated sound to which one has agreed to attribute a definite meaning, and the signified is the concept in which the meaning of the verbal sign consists. But with regard to elementary meanings (significations) such as that of the perceived object, or even, in the small child and prior to the formation of

substantial objects, that of the simply "presented" sensory images, the same applies. The "signified" of objective perceptions such as that of the mountain I see from my window or the inkwell on my table is the objects themselves, definable not only by a system of sensorimotor and practical schemata (climbing a mountain, dipping pen in ink) or by a system of general concepts (an inkwell is a container which . . ., etc.), but also by their individual characteristics: position in space, dimensions, solidity and resistance, color in different lights, etc. Now the latter characteristics, although perceived in the object itself, presuppose an extremely complex intellectual elaboration; for example, in order to attribute real dimensions to the little spots which I perceive to be a mountain or an inkwell, I must place them in a substantial and causal universe, in an organized space, etc., and accordingly construct them intellectually. The signified of a perception —that is to say, the object itself—is therefore essentially intellectual. No one has ever "seen" a mountain or even an inkwell from all sides at once in a simultaneous view of their different aspects from above and below, from East and West, from within and without, etc. In order to perceive these individual realities as real objects it is essential to complete what one sees by what one knows. Concerning the "signifier," it is nothing other than the few perceptible qualities recorded simultaneously and at the present time by my sensory organs, qualities by which I recognize a mountain and an inkwell. Common sense, which prolongs in each of us the habits of infantile realism, certainly considers the signifier as being the object itself and as being more "real" than any intellectual construction. But when one has understood that every concrete object is the product of geometric, kinematic, causal, etc., elaborations, in short, the product of a series of acts of intelligence, there no longer remains any doubt that the true signified of perception is the object in the capacity of intellectual reality and that the apprehensible elements considered at a fixed moment of perception serve only as signs, consequently as "signifiers."

With regard to the simplest sensory images which are assimilated by the nursling and which are anterior to the permanent and substantial object, the same distinctions can be made, though

to a lesser degree. Thus when the baby gets ready to grasp the rattle which he sees, the visual appearance of this toy is only a "signifier" in relation to the "signified" which the other qualities of the same object constitute and which are not given simultaneously but are collected by the mind in a unique bundle (in particular its quality of object to be grasped). Here again the signifier refers to a system of schemata (of vision, prehension, hearing, sucking, etc.) and only has meaning, even with regard to the precise image given through perception, in relation to the whole of the system.

But if we interpret the idea of signification in this way, including the complementary ideas of "signifier" and "signified" it is necessary at once to distinguish between three kinds of signifiers, which we shall call the "indication," the "symbol" and the "sign" so as to place in their true perspective the facts of comprehension of significations that we shall presently describe.

The "symbol" and the "sign" are the signifiers of abstract meanings, such as those which involve representation. A "symbol" is an image evoked mentally or a material object intentionally chosen to designate a class of actions or objects. So it is that the mental image of a tree symbolizes in the mind trees in general, a particular tree which the individual remembers, or a certain action pertaining to trees, etc. Hence the symbol presupposes representation. We shall see it made manifest during the child's second year at the time of the appearance of the symbolic game (the game of make-believe) or when the progress, intelligence and use of practical deduction will really evoke absent objects. The "sign," moreover, is a collective symbol, and consequently "arbitrary." It also makes its appearance in the second year, with the beginning of language and doubtless in synchronism with the formation of the symbol. Symbol and sign are only the two poles, individual and social, of the same elaboration of meanings.

Concerning the "indication," this is the concrete signifier connected with direct perception and not with representation. In a general way we shall call indication every sensory impression or directly perceived quality whose signification (the "signified") is an object or a sensorimotor schema. In the strict and limited

sense of the word, an indication is a perceptible fact which announces the presence of an object or the imminence of an event (the door which opens and announces a person). But as we have just seen, the concept of indication could be extended to include every sensorimotor assimilation. What I see of an inkwell or of a mountain is an indication of the existence of these objects; the rattle which the baby looks at is an indication of virtual prehension; the nipple which the nursling's lips touch is an indication of possible sucking, etc. The facts belinging to the present stage thus belong in the class of concrete significations of which the signifier is "indication."

But, in order to understand the true nature of these facts, it is fitting first to divide into different types the different varieties of indications and, to do this, to recapitulate the whole of the "significations" hitherto under study.

In the first place, we have been able to speak of recognitory assimilation since the very beginning of the reflex (Chapter I). When the child is hungry and is not limited to sucking for the sake of sucking (reproductive assimilation) nor to sucking the first object that reaches his lips (generalizing assimilation), he well knows how to seek the nipple and discern it in relation to the surrounding teguments. What does this mean if not that the nipple has a meaning for him, in contrast and in relation to other significations (that of empty sucking, etc.)? This first type of signification is the simplest possible. In such a case the signifier is the elementary sensory impression accompanying the play of the reflex (whence the impression which serves as "excitant" to sucking) and the signified is the sucking schema. The proof of such an interpretation has in it nothing artificial; it is that this schema involves, as we have just recalled, a certain number of differentiated subschemata: contact with the nipple entails sucking with swallowing, whereas contact with the surrounding teguments or with an object only entails sucking for the sake of sucking, the erethism of the buccal apparatus entails empty sucking, etc. Each of these sensory impressions is therefore already classed and corresponds to a fixed subschema. At the very least, when a child is hungry and seeks the nipple, it can be said that the impression peculiar to this contact is subject to recognitory

assimilation and consequently that it comprises a precise "signified."

In the second place come the significations peculiar to the first habits and to assimilation through acquired schemata (primary). But, as we have seen, the recognition characteristic of this level presupposes as "signifiers" in addition, simple sensory impressions identical to those of the preceding level which it has been agreed to call "signals." The signal is an as yet elementary indication; it consists in a sensory impression simply associated to the reaction and to the perceptual images characteristic of any schema; thereafter it announces these images and sets in motion these reactions to the extent that it is assimilated to the schema under consideration. For example, the consciousness of a certain attitude in the position for nursing sets the sucking schema in motion. What does this mean if not that this consciousness is a signal or a signifier for the signified which the feeding constitutes? Such a signifier is surely more complex than that of the first type (direct sensory contact with the nipple or the surrounding teguments), since it presupposes an acquired extension of the schema of assimilation, but the signification which it permits remains elementary. The consciousness of the position for nursing does not signify anything more, from the subject's point of view, than the awaiting and the beginning of the sensory images connected with sucking. It is therefore necessary to avoid comparing, as is sometimes done, the signal to the "arbitrary" sign. Doubtless any signal at all can serve to set any reaction in motion: training operates in this way, in animals, establishing the most varied associations. But, as we have seen, association only becomes "fixed" if the signal is incorporated in a schema of assimilation and thus receives its meaning from the single act connecting the effort with its result. Thereafter, to the subject's consciousness, the signal is an indication and not a sign; an indication, that is to say, a given objective aspect of external reality, as the track of paws is to the hunter the indication of the passing of the game. The signal is therefore no more "arbitrary" in the sense that linguistic scholars give to this word than the association of sound and sight in perceiving a clock in movement, is arbitrary.

The latter example evokes a particular variety of this second

type: the signals founded on the coördination of heterogeneous schemata. As we have established in analyzing the different coordinations of sight and hearing, of sight and sucking, of grasping with sucking and sight, etc., the objects which give rise to such coördinations through this very fact acquire a complex signification: they begin to assume a certain solid and permanent contexture. In looking at a feeding bottle or a rattle, the child understands that it is a thing to suck or a thing to grasp; in listening to a noise the child understands that the thing heard is to be looked at, etc. An active search then ensues which comprises progress in foresight. Upon hearing a certain sound the child prepares himself to see a certain image, etc. But in such significations the signifier is always constituted by sensory impressions or signals, simply more varied than before, and the signified still consists in coördinated practical schemata.

Finally there comes the third type of significations which we shall now emphasize—that of the indication belonging to the secondary circular reactions.

Whether there is a secondary circular reaction such as pulling a chain or a string in order to shake objects hanging from the hood (Obs. 99 and 100) or a procedure to make interesting spectacles last, such as pulling the same string in order to swing these objects from a distance (Obs. 113), it is apparent that the significations involved in such cases are much more complex than the preceding ones while being derived from them by differentiation. In effect, the significations of the second type remain essentially functional and related to the subject's own activity. That which the sensory signals announce is that a certain thing is to be seen, heard, grasped, etc. On the contrary, the significations of this third type comprise from the beginning an element of foresight related to the things themselves: the string hanging from the bassinet hood is not only to be seen, grasped, and pulled, it serves to swing the objects from a distance, etc. There is, accordingly, in the signification of the string a content related to the foresight of events. Without yet understanding, of course, the details of this connection, the child knows that the gesture of pulling the string brings with it the movement of other objects. But this foresight is not always independent of the action. The string is

still a signal whose signification is the schema of "pulling in order to shake the hood." The foresight is therefore not yet pure; it is comprised in a motor schema. But, in relation to the significations of the second type, there is certainly progress and, in addition to the merely active "signal," one already anticipates the "indication" in the strict sense of the word: the string is indication of a series of possible movements.

This characteristic of transition between the "signal" belonging to the preceding stages and the "indication" belonging to the fourth stage which will set free foresight of the context of the action in progress is found again in a series of signs which are made manifest between 0;4 and 0;8, independently of the circular reactions under study hitherto.

Observation 108.—From 0;4 (12) to approximately 0;4 (30) Laurent cried with rage when, after his feedings, a handkerchief or napkin was placed under his chin: they announced a few spoonsful of a beverage he disliked.

At 0;7 (10) he cries in the morning as soon as he hears his mother's bed creak. Until then, although awake, he did not show his hunger. But, at the slightest creak, he moans and thus demands his bottle.—The same applies, for a stronger reason, to the noises of the door, but he remains insensible to external sounds (noises in the hall or neighboring rooms).

From 0;7 (15), also in the morning, when I play with him and his mother appears he immediately cries with hunger.

The same applies at 0;9 (20) when a maid and not his mother gives him his morning bottle: at sight of the maid he loses all interest in what is going on, even when he is in his mother's bed.

Observation 109.—At 0;8 (3) Jacqueline smiles and says *aa* as soon as the door to her room opens, before seeing the person who enters. She therefore understands by this sign that someone will appear. At 0;8 (10) she cries from hunger as soon as her mother enters the room; she does not do this for her father. Same reaction in negative form at 0;9 (9): she moans at sight of her mother (due to lack of appetite[1]) when she had been laughing and enjoying herself.

At 0;8 (13) she raises her hand to grasp her mother's face when the latter whispered in her ear from behind her. Without seeing anything, Jacqueline understands that there is someone behind her. Likewise, at 0;9 (27) she laughs and turns when I blow down her neck, even though she has neither seen me nor heard me arrive.

1 At that time she suffered from anorexia.

At 0;8 (18) she is still not hungry and cries when her bib is put on, knowing that a meal awaits her. Furthermore, she opens her mouth as soon as her forehead is touched by her sponge (which she has not seen) because every day she amuses herself by biting it.

Such recognitions of indications at first appear to be sufficiently detached from the action to give rise to truly objective foresight as will be the case during the fourth stage. But in reality the signs in question here are not yet "mobile" in the sense in which we shall interpret this term in connection with the fourth stage; that is to say, they do not give rise to foresight related to the activities of the objects themselves independently to the subjects. The indications described in Observations 108–109 all make up a part of the global schema: either that of the meal, in which the child is certainly active, or else that of an "interesting spectacle" (such as having his neck or hands blown on, etc.) comparable to those "interesting spectacles" which the child maintains due to procedures which are still "circular" and which we shall study in the next paragraph. If such indications already announce objective foresight, it cannot then be said that they are entirely detached from secondary circular reaction. They are simply inserted in the preëstablished schemata and only acquire meaning as a function of the latter. Like the indications and significations which we have recalled, they merely form a transition between the primary "signals" and the actual indications of the fourth stage.

§4. GENERALIZING ASSIMILATION AND THE CONSTITUTION OF "PROCEDURES TO MAKE INTERESTING SPECTACLES LAST."—The generalization of secondary schemata is produced when the child is confronted by new objects. In such cases the child from the outset makes use of his usual behavior patterns and assimilates the unfamiliar to their schemata, without adding anything. It is a remarkable thing that the younger the child, the less novelties seem new to him. Unfortunately, it is impossible to compare in this respect secondary with primary reactions in the presence of unfamiliar objects for there is no appreciable common gauge for them. But if the re-

actions of the present stage are compared to those of the following one and above all to the "tertiary circular reactions" of the fifth stage, the difference is all the more striking as the situations become more homogeneous. In the face of a new phenomenon, the child in the fifth stage is capable of adopting the attitude of experimentation (this does not mean that he necessarily adopts it, but he is apt to do so). He seeks novelty and varies the conditions of the phenomenon in order to examine all of its modalities. The child in the fourth stage, without reaching these true "experiments to see," is also interested in the new object in itself. But, in order to "understand" it, he tries to apply to it in turn the whole of the known schemata in order to find which one in particular will be most suitable to it. On the other hand, the child at the present stage, while sometimes feeling surprise in the presence of an unknown object, nevertheless from the outset treats it as a familiar object and employs it in the use of his habitual schemata. Thereafter one has the impression that the child, far from still being interested in the thing in itself and far from appreciating its novelty as such, merely tries to use his secondary schemata by pure functional assimilation, as he did hitherto by means of the primary schemata. Consequently there exists a simple generalization of secondary schemata.

Here are examples of this elementary generalizing assimilation:

Observation 110.—At 0;3 (29) for the first time Laurent sees the paper knife which figured in Observation 104. He grasps and looks at it, but only for a moment. Afterward he immediately swings it with his right hand as he does all the objects grasped (see the schema of Obs. 102). He then rubs it by chance against the wicker of the bassinet and tries to reproduce the sound heard, as though it were a rattle (see Obs. 102). It then suffices that I place the object in his left hand for him to shake it in the same way. He ends by sucking it. The novelty of the object has therefore in no way interested the child, except for the brief glance at the beginning: the paper knife from the outset was used as aliment for the habitual schemata.

At 0;4 (8) I place a large rubber monkey in front of Laurent; the mobile limbs and tail as well as its expressive head constitute an absolutely new entity for him. Laurent reveals, in effect, lively astonishment and even a certain fright. But he at once calms down and applies to the monkey some of the schemata which he uses to swing hanging objects;

he shakes himself, strikes with his hands, etc., gradating his effort according to the result obtained.

Likewise, at 0;5 (25) and the days following, Laurent looks at an unfolded newspaper which I place on the hood of his bassinet. He immediately begins to pull the strings hanging from the hood, to shake himself or his feet and arms. He bursts out laughing on seeing the movements of the newspaper just as he frequently does when the rattles shake.

At 0;6 (0) Laurent at once grasps a big box of lozenges which is unfamiliar to him. He hardly looks at it but immediately uses it to rub against the sides of the bassinet, then he passes it from one hand to the other and rubs the object against the opposite side of the bassinet.

At 0;6 (1) he grasps a new rattle made of three parts: the handle, a middle ball of medium size and the end ball, a large one. Laurent looks at the object quite a long time while passing it from one hand to the other and even seems to palpate the surface which foretells the behavior patterns of the following stage. But he quickly desists in order to move the object in the air, at first slowly, then more and more rapidly, and finally he shakes it, rubs it against the sides of the bassinet, etc.

At 0;6 (7) I offer him various new objects to see if he will resume his attempts at spatial exploration which seemed to appear in connection with the last object. This does not occur; the child utilizes the new object as aliment for his habitual schemata. So it is that a penguin with long feet and a wagging head is only looked at briefly: at first Laurent strikes it, then rubs it against the side of the bassinet, etc., without paying attention to the end by which he grasped it. Several knick-knacks receive the same treatment: he grasps them with one hand and strikes them with the other.

At 0;6 (14) he takes hold of a new doll, looks at it for a moment without investigating either its shape or clothing: he strikes it, rubs it against the wicker, shakes it in the air, etc.

At 0;6 (18) a pipe holds his attention more but is subsequently utilized in the same way. At 0;6 (16) a new swan, encircled by a ring and with a handle is looked at with curiosity and immediately struck, shaken, rubbed, etc. At 0;6 (26) a series of unfamiliar objects (a rattle with a bell, a bear, a lamb, etc.) are barely examined before being struck, shaken, etc.

At 0;7 (2) he only looks a little at an unfamiliar bird of complicated shape mounted on a plank with wheels. He limits himself to shaking and striking it, and rubbing it against the side of the bassinet.

Observation 111.—At 0;5 (3) Lucienne only has one schema at her disposition which she employs in the course of her circular reactions and attempts to make interesting spectacles last: that of shaking her foot or entire body to cause swinging (see Obs. 116). Furthermore, of course, she

knows how to grasp, suck, etc. When a new object is presented to her there ensues the curious result that she tries in turn the schemata of prehension and of shaking the feet, but applying the first chiefly to immobile and near objects and the second mainly to objects in motion or hanging before her. Here is the series of attempts:

First of all, before a cross of Malta which hangs above her, Lucienne immediately moves her feet only. Then she slows down her movements and begins empty sucking while looking at the object; after this she grasps it and brings it in front of her eyes in order to examine it.

A pipe, motionless: attempts at prehension, sucking at a distance and foot movements, all simultaneous.

An eraser: surprise, sucking at a distance and prehension. Once the eraser has been grasped, Lucienne looks at it briefly, in her hand, then immediately begins to move her feet.

Again the cross of Malta: immediate and sustained foot movements. Then Lucienne's hand having knocked against the object, there is an attempt at prehension, but this second reaction is obviously due to a fortuitous cause.

A hanging puppet: she grasps it and pulls but, not succeeding in drawing it to her, she periodically desists in order to give hard shakes of the feet. She then resumes, grasping, then moves her legs again: there is constant alternation between these two activities.

A slide rule: exclusive attempts at prehension. No movement of the feet.

A strap which I swing slowly: shakes of the feet, then attempts at prehension.

A stick of sealing wax; only prehension.

A watch placed very near her face: first prehension, then when I raise it too high, shakes of the feet.

This observation consequently shows us how much the new object is immediately assimilated to a schema; that is to say, generically recognized as being able to give rise to a familiar behavior pattern, even when the habitual schemata are very limited in number. In what follows it goes without saying that the more these schemata are multiplied the more the new object is subjected to various attempts.

It may be seen in what such behavior patterns consist. When confronted by new objects the child does not yet try to find out in what way they are new, he limits himself, at the outset or after a short pause, to using them as aliments for his habitual behavior patterns. He therefore generalizes, without adding anything for their use, the schemata he possesses.

But the generalizing assimilation belonging to this stage is not limited to this elementary form. It sometimes happens that

the novelty presented to the child does not consist in a particular object but in an event, in an actual spectacle on which the subject has no direct influence. What occurs then? The child, desirous of seeing the spectacle prolonged, also utilizes his habitual schemata which he generalizes without adding anything to this effect. That is what is revealed by Observation 110. When Laurent, at 0;4 (8) and 0;5 (25) cannot grasp the monkey or the newspaper which he sees from afar, he at once applies to them the schemata related to hanging objects and thus seeks to act upon them from a distance. From that to trying to act upon any phenomenon whatever, independently of any real contact, is only a step.

This step is taken as a result of the following behavior pattern: It is a transitional behavior pattern which stems from secondary circular reaction but whose higher forms foretell the combinations of the fourth stage. It is the activity by means of which the child tries to make last the interesting spectacles of which he has just been witness without himself having provoked their first appearance (for example, prolonging the swinging of a watch seen from afar, etc.). Such behavior patterns still partake of circular reaction, since it is simply a matter of conserving and reproducing, but they generalize its principle, since the schemata hitherto inserted in actual circular reactions are henceforth applied to entirely new circumstances. Here are some examples of these behavior patterns:

Observation 112.—The first example will make us understand how the secondary circular reaction is prolonged in procedures to make an interesting spectacle last. Following Observation 98 at 0;3 (20) I make the following experiment on Laurent. I give him a rubber doll, unfamiliar to him and attached to the usual rattle by a string sufficiently loose so that the doll's movements do not shake the rattle. As soon as Laurent sees the doll, he grasps it in his right hand and sucks it. This preliminary phase lasts ten minutes during which the rattle has neither moved nor made a noise. After this Laurent lets his arm drop to the side while keeping the doll in his hand. I then shake the rattle without shaking the string or Laurent's hand; moreover, he did not look at the rattle at this time. Then, as soon as he hears the rattle, he looks at it and stretches out his right arm, while holding the doll in his hand, then he shakes this doll in a perfectly adapted way.

But a moment later Laurent's right hand is in contact with the doll,

without holding it. I then shake the rattle again. He immediately moves his right arm, his hand remaining empty and not attempting to grasp the doll.

Thus it may be seen how, as soon as circumstances are changed, the schema becomes dissociated and the efficacious gesture (grasping, and shaking the arm, or simply shaking the arm) is advanced to the rank of procedure to make the interesting spectacle last, in the very absence of the usual intermediaries (of the chain).

The rest of the observation well shows, in effect, that this arm movement has become, for Laurent, a constant "procedure" and has not simply consisted in an episodic effort. At 0;3 (5) for example, Laurent practices grasping my hand when it is within his direct reach; but, when I put it at a distance of 50 cm. or more, he looks at it and then swings his arms rapidly just as he does when confronted by his usual rattle.—At 0;3 (23), I present him (at a distance of 50 cm.) with an unfamiliar doll (in costume) which I swing for a moment. As long as it moves, he looks at it, motionless, but as soon as it stops, he shakes his arm. Same reaction with respect to my watch and my wallet. The same day I saw him behave spontaneously in this way while looking at his hanging doll.

At 0;3 (29) I shake his arm as soon as I stop swinging a paper knife 100 cm. away from him.—At 0;4 (18) he shakes his arm in order to make me continue when I shake his feet. He laughs and waves his arms more and more vigorously until I resume. At 0;5 (26) he does the same as soon as a grating sound stops, a sound which I had made without his seeing me. He definitely gradates his movement according to the variations of the waiting time.

At 0;6 (27) again, he shakes his arm when he does not succeed in grasping a distant object or in order to make an object move at a distance (a sheet of paper placed on a cupboard, at a distance of 150 cm. from him, etc.). Same observation at 0;7 (5).

At 0;7 (7) he looks at a tin box placed on a cushion in front of him, too remote to be grasped. I drum on it for a moment in a rhythm which makes him laugh and then present my hand (at a distance of 2 cm. from his, in front of him). He looks at it, but only for a moment, then turns toward the box; then he shakes his arm while staring at the box (then he draws himself up, strikes his coverlets, shakes his head, etc.; that is to say, he uses all the "procedures" at his disposition). He obviously waits for the phenomenon to recur. Same reaction at 0;7 (12), at 0;7 (13), 0;7 (22), 0;7 (29) and 0;8 (1) in a variety of circumstances (see Obs. 115).

It therefore seems apparent that the movement of shaking the arm, at first inserted in a circular schema of the whole, has been removed from its context to be used, more and more frequently, as a "procedure" to make any interesting spectacle last.

Observation 112 repeated.—Another hand movement of which Laurent has made use as a "procedure" is the act of "striking"; but in contradistinction to the preceding one, this schema was utilized for the first time as a "procedure," due to a simple association of continuity.

At 0;7 (2), in effect, Laurent is in the process of striking a cushion when I snap my middle finger against the ball of my thumb. Laurent then smiles and strikes the cushion but while staring at my hand; as I no longer move, he strikes harder and harder, with a definite expression of desire and expectation and, at the moment when I resume snapping my fingers, he stops as though he had achieved his object.

A moment later, I hid behind a big curtain and reappeared every few minutes. In the interim, Laurent strikes his covers harder and harder while looking at the curtain.—Same reaction while looking at an electric light. At 0;7 (5) he strikes the side of his bassinet while looking at the hanging rattles and continues for a long time despite failure.

At 0;7 (7) he strikes his coverlets while looking at a tin box on which I have just drummed (see Obs. 112). Same reactions until about 0;8.

At 0;7 (11) he strikes the wrong end of his bottle in the hope of seeing the nipple come up (see Vol. II, Obs. 78).

Observation 113.—Jacqueline, likewise, at 0;7 (16), that is to say, after Observation 100, applies the schema of pulling the strings of the hood to new circumstances. After having moved the hood by moving a hanging doll, Jacqueline looks at my watch which I swing at a certain distance. She begins by trying to grasp my watch, then she happens to graze the string hanging from the hood; then she grasps it and shakes it violently while looking at the watch, as though her movement were going to make the object continue to swing.—That evening, same reaction with regard to a doll which I swing from a distance. At 0;7 (23) after having pulled the same string to shake the hood of the bassinet, Jacqueline looks at a book which I pass back and forth in front of her, at the level of the hood but in the open. As soon as I stop, Jacqueline, who until then was motionless, without hesitation pulls the string hanging from the hood while staring at the book. She shakes the string harder and harder, ten times, then stops. I then recommence to move the book. As soon as I stop doing it, Jacqueline pulls the string, but less energetically and less frequently. And so on, twice more. If the number of shakes she gave the string during these four attempts is counted, the following series is obtained: 8–10; 5–8; 3–4; 2. It is apparent from her countenance and from this series, that Jacqueline hoped to make the book continue to move by pulling the string and that she gave up little by little. At the fifth and sixth attempts Jacqueline limits herself to looking at the book while it moves without attempting anything afterward.

At 0;8 (8) on the other hand, after having used the string to make the hood of the bassinet move, she looks at a bottle which I swing 50

cm. away from her. As soon as I stop, she pulls the string to make it continue while staring at the bottle with a very typical expression of expectation and anxiety. When she notes the failure, she tries another procedure and imitates with her hand the movement of the bottle without trying to grasp it.

At 0;8 (16) Jacqueline looks at me while my lips imitate the mewing of a cat. She holds a little bell suspended from the hood. After some other procedures (see Vol. II, Obs. 132), in order to make me continue, she shakes the little bell she holds. I answer by miauling. As soon as I stop, she again shakes the little bell, and so forth. After a few moments I definitively stop my miaulings. She shakes the bell two or three times more and, confronted by failure, she changes means.

Observation 114.—At 0;7 (29) Jacqueline answered herself by rubbing the wicker side of her bassinet with her right hand. At a given moment, when her hand is outstretched beside her, without her knowledge I shake the hood once or twice. She does not then try to pull the string but again rubs her hand against the wicker while watching the hood as though this were going to move. Same reaction many times. It is true that when the hand movements are sufficiently violent they make the whole bassinet shake slightly, but the rest of the observation will show that this relative success is not enough to explain the use of the procedure.

The next day, at 0;7 (30) I clap my hands in front of Jacqueline. When I have finished she moves her hand against the wicker of the bassinet while looking at my hands. When I recommence she stops as though she had attained her goals and, when I stop, she begins again.

A few hours later, same reaction with regard to my béret which I hold (without showing myself) 100 cm. from his eyes. At the beginning the child's expression leaves no doubt of her desire to prolong the interesting spectacle by this means, but the child progressively slackens with failure. Finally, Jacqueline only moves her hand slightly, then not at all.

Observation 115.—Everyone knows the attitude of nurslings when freely enjoying themselves or when an unexpected sight causes them a lively feeling of pleasure. They draw themselves up by pressing on their feet and shoulder blades and let themselves fall back in a heap. Moreover it is not difficult to note that this movement is often utilized in order to make the bassinet move. It is enough that the child has observed the effect of these shakes for him to draw himself up intentionally while looking at the hood and the hanging objects. Now this schema, once acquired, is subsequently applied to anything at all, in the capacity of a "procedure to make an interesting spectacle last." In the course of Volume II (Obs. 132) we shall cite a long observation made of Jacque-

line concerning the development of causality. Here is its equivalent in Laurent.

At 0;4 (2) Laurent simply shakes the bassinet by drawing himself up. But already at 0;4 (7) he utilizes this schema as a "procedure." When I stop humming he waits for a brief moment and then draws himself up, at first very gently, then more and more vigorously, while looking at me. The intention is clear. Same reaction at 0;7 (3).

Between 0;4 and 0;6 he uses the same procedure to prolong the swinging, etc. At 0;6 (6) and 0;7 (2) he makes use of it to cause me to continue to snap my fingers (see Obs. 112 repeated). He visibly gradates his effort as a function of his impatience.

From 0;7 (7) to 0;8 (1) he draws himself up to act upon a tin box on which I drummed or upon a series of similar objects (see Obs. 112).

In short, the action of drawing himself up is promoted to the rank of a magic-phenomenalistic procedure and is used in the most varied circumstances.

Observation 116.—Lucienne revealed exactly analogous behavior patterns but with procedures of course varying as a function of her earlier circular reactions. Now, we recall (Obs. 94–95) that one of her most frequent reactions was to shake her bassinet or the rattles by means of repeated vigorous shakes of legs and feet (movements analogous to pedaling). From the sixth month this behavior pattern gave rise to procedures destined to satisfy desires or make interesting spectacles last. Already at 0;4 (14) Lucienne looks at my hand which I show her from a distance. Her fingers move, but her arms and trunk remain immobile; she has an expression of desire and makes sucking movements; she becomes red with emotion while opening and closing her mouth and she brusquely moves her legs with all possible speed. But is this simply an attitude or already an attempt at action? This remains in doubt until 0;5 (21). At 0;5 (10) she shakes her legs while holding a rattle with a handle, as she does when confronted by the hanging rattle. Then at 0;5 (21) she does it as soon as I stop swinging my hands. Instead of imitating my gesture she shakes her legs to make me continue. At 0;7 (1) she does the same when I move my fingers, wag my head, move my hands, etc., in order to study imitation. She begins by imitating, then shakes her legs, paying great attention to what I do. At 0;8 (5) same reaction in the presence of all sorts of spectacles: a doll that I swing, etc. At 0;8 (13) she watches me open and close my mouth. She begins by examining me with great interest, then tries to grasp, then, not succeeding, she moves her legs slightly; as soon as I stop, she shakes them violently, apparently to make me continue. Same reaction at 0;8 (15). Now this is not a question of a simple receptive attitude but of an active procedure, for Lucienne constantly gradates her effort according to the result. She tries at first prudently and slowly and if I respond by moving my lips, she shakes herself more and more vigorously.

Observation 117.—Here are some more procedures which Lucienne has used. From 0;7 (20) when she shook her bassinet by moving her hands (see Obs. 101) she used this procedure for quite other purposes. Thus, at 0;7 (23) she looks at me with great interest when I unfold a newspaper and rumple it. As soon as I have finished she moves her hands several times in succession. The same day I appear in her visual field, disappear, reappear, etc. She is very much puzzled and (after my disappearance, I see her through the hood) moves her hands while looking in the direction where I should have reappeared. She subsequently applies this schema to everything: at 0;7 (27) to make me continue my gestures, at 0;8 (0) to move a puppet from a distance, at 0;8 (18) to make me repeat a cry, at 0;10 (12) to make me put my index finger back into my mouth, etc.

From 0;8 (5) as we have previously seen (Obs. 101), she shakes her head in order to move the bassinet. The following days she applies this schema to the most varied situations: at 0;8 (12) she tries them in order to make me repeat a cry, at 0;10 (7) in order to make swing a placard hanging in a car and rendered immobile by the stopping of the train, etc.

At 0;9 (28) she puffs through her mouth in analogous situations (to make me repeat my gestures, etc.). At 0;10 (8) she draws herself up, like Jacqueline (Obs. 115) to prolong a gesture, a whistle, a movement of the dolls, etc. At 0;10 (24) she vigorously scratches her coverlet with the same intentions, etc.

Observation 118.—Let us finally mention the manner in which Laurent has come to utilize his head movements as "procedures" charged with efficacity. From 0;3 Laurent is able to imitate a lateral displacement of the head. Now, as early as 0;3 (23) I find him moving his head in this way when confronted by a hanging rattle, as though to give it a real movement (see Vol. II, Obs. 88).

At 0;3 (29) he shakes his head when I stop swinging a paper knife. The following weeks he reacts in the same way as soon as I interrupt a movement he has observed.

At 0;7 (1) he does it to incite me to continue to snap my middle finger against my thumb. At 0;7 (5) same reaction in the presence of a newspaper which I unfolded and which remains motionless. At 0;7 (7) he shakes his head the same as he shakes his arm or draws himself up when he sees a tin box on which I have drummed.

Until toward 0;8 he thus continues to use this schema to make any interesting sight whatever last, whether it is a visually perceived movement, regardless of the direction of this movement, or even a sound (humming, etc.).

Thus it may be seen that it is not an exaggeration to talk of generalization in characterizing such behavior patterns. In the

six observations which we have just summarized there is a question of schemata elaborated in the course of the child's circular reactions, but applied to new circumstances. These situations have in common the fact that the child has just witnessed an interesting spectacle and would like to have an influence upon it in order to make it last. This desire, not being coördinated with any adapted mechanism since precisely in such cases the subject is impatient, is radiated, quite naturally, in movements connected with the circular reactions, that is to say, to situations in which the child can prolong the desirable result at will. So it is that the schemata originally related to limited circumstances are applied at first to all analogous situations, then to any activity whatever on condition that it is a question of reproducing an interesting spectacle.

But the latter condition at the same time shows us the limits of circular reaction. On the one hand, as we have already stressed, it is only a question of repeating and not of inventing, truly to adapt oneself to new circumstances. On the other hand, to the extent that there is generalization, the procedures employed do not apply in detail to these new situations. There exists, so to speak, abstract generalization (the effective gesture being applied to empty space) and not concrete insertion of the means employed into the context of the situation. Let us remark, moreover, with regard to this last point, that the same is true at all levels. Not only the schemata due to secondary circular reactions but also those due to the most precise discoveries can subsequently be applied to empty space and give rise to magic-phenomenalistic connections (see below, Chap. V, Obs. 176). But, at the level which we are now considering, that is to say, at the beginning of the action upon things and the interrelations of things, these connections remain the only possible ones.

Fortunately, there is a second method of generalization of the secondary schemata: that which we shall study in the next chapter in trying to find out how the child will intercoördinate his schemata when it will be a question not only of repeating or prolonging, but of really adapting himself to new situations.

Before that, let us emphasize the lasting importance of secondary circular reaction during the last stages of intellectual

development. Secondary circular reaction, in so far as it is a reproduction of an interesting result fortuitously obtained, is, in effect, far from constituting a behavior pattern peculiar to the child. An adult, unacquainted with machinery, does not behave differently from the baby when, having by chance touched part of a motor, he does not understand the effect produced and repeats the action which set it in motion. Like the reflexes of the first stage and the acquired associations or habits of the second, the secondary reactions are behavior patterns whose mere appearance characterizes a given stage but which are conserved as substructures in the last stages.

The peculiarity of the circular reactions of the present stage is that, during this period, they constitute the highest intellectual manifestations of which the child is capable, whereas subsequently they will only fill an increasingly derived role. This point is of a certain importance and justifies the distinction we shall henceforth make between "typical" secondary circular reactions and "derived" secondary circular reactions. In effect, when the child at the present stage tries to reproduce an interesting result, he has never obtained this result except in an entirely fortuitous way; that is to say, without the context of his activity being one of research, of experimentation, etc. On the contrary, when the child at a final stage or an adult discovers a fortuitous result, it is almost always in a context of research or of experimentation and thereafter the action of reproducing the effect obtained only constitutes a "derived" action.

For example we shall observe such "derived" reactions during the fourth stage when, in the presence of new objects, the child applies himself to attempts at "explorations" (see Chap. IV, §5). If in the course of "exploration" the child fortuitously discovers an unforeseen result, he immediately reproduces it. Such behavior is identical to that of the secondary circular reaction, but is "derived." During the fifth stage, too, it happens that in experimenting—that is to say, in organizing what we shall call the "tertiary circular reactions"—the child comes little by little to repeat the movements which engendered an unforeseen effect. He then relapses into secondary circular reaction, but this is, in such a case, once more "derived."

The time has not yet come to study these derivations. Let us limit ourselves to citing an example of these final circular reactions in order to show their structural identity with the preceding ones:

Observation 119.—Jacqueline, at 1;1 (7) continues to reproduce all the new movements which she discovers by chance and everything she is made to do. For instance, I put a stick on her head: she immediately puts it back there. I hold her cheeks in my hands, then let them go: she puts her cheek back into my hand, or grasps my hand in order to press it against her cheek, or she applies her own hand.

At 1;3 (12) she is sitting in her play pen with one leg through the bars. When she tries to lift herself up she does not at first succeed in withdrawing her foot. She moans, almost cries, then tries again. She then painfully succeeds in disengaging herself, but as soon as she has done so she replaces her leg in the same position and begins all over again. This happens four or five times in succession until the situation is completely assimilated.

At 1;3 (13) while walking, she knocks her head on a table so that she has a very noticeable red mark. Nevertheless she immediately grasps a stick which is beside her and strikes her forehead on the same place. Then as this dangerous instrument is taken from her, she again knocks herself intentionally, but with very great prudence, against the arm of a chair.

In conclusion, the deep unity of the behavior patterns of this stage may be seen. Whether it is a question of pure "secondary circular reactions" or of movements of recognitory assimilation or again of generalization of schemata in the presence of new objects or of spectacles to be prolonged, in all these cases the behavior of the child consists in repeating what he has just done or is already used to doing. The action performed by the child therefore always consists in a global and unique action of a single advent and characterized by a single schema. One can, it is true, already distinguish the means and ends in such an action, in the sense that the child's movements are seriate and complex in their interrelations. But means and ends are inseparable from one another and, consequently, produced in the same entity. On the other hand, the behavior patterns of the following stage will furnish us an example of coördinations between separate schemata, some of which will serve as ends and others as means.

Now this need of repetition, which characterizes this stage

and which explains the global aspect of assimilation through secondary schemata, also conditions the accommodation to external environment peculiar to such behavior patterns. Whether it is a question of secondary circular reactions or of the generalization of the same schemata in the presence of new objects or sights, this accommodation always consists in rediscovering with the maximum precision possible, the movements which were effectual. The accommodation characteristic of the fourth stage will be quite different. By virtue of the coördination of the schemata, it will constitute an adjustment of their contexture to the objects themselves and thus will go beyond the simple, confused and whole application.

In short, if the elaboration of the secondary schemata of the third stage marks perceptible progress over that of the primary schemata in the sense that the child begins really to act upon things, it nevertheless prolongs the assimilation and accommodation characteristic of the primary reactions in that the child's activity remains centered on itself more than on objects.

CHAPTER IV

THE FOURTH STAGE:

The Coördination of the Secondary Schemata and Their Application to New Situations

At about 8 to 9 months a certain number of solidary transformations appear concerning at the same time the mechanism of intelligence and the elaboration of objects, of spatial groups as well as of causal and temporal series. These transformations even seem important enough to characterize the appearance of a stage: that of the first actually intelligent behavior patterns.

From the point of view of the functioning of intelligence this fourth stage marks considerable progress over the preceding one. The behavior patterns of the third stage, as we have seen, only consist in "circular reactions." No doubt these reactions are related to the external environment and no longer only to the body itself. Moreover, we have called them "secondary" to distinguish them from the "primary" reactions. No doubt either that the activity of the secondary schemata can start whenever the child wishes to prolong any interesting phenomenon and no longer only the result in connection with which the schemata in question were constituted. But, as we have stated, that is only a simple generalization of schemata without elaboration of the special relations between each of them and the new goal to be reached. In short, the reactions of the third stage therefore constitute the simple prolongation of the primary circular reactions; they owe only to their complexity the fact of drawing, after the event, a distinction between transitive and final states, between means and ends. On the other hand, the behavior patterns of the fourth stage involve such a distinction from the very outset. The

criterion of their appearance is, in effect, the intercoördination of the secondary schemata. Now, in order that two schemata, until then detached, may be coördinated with one another in a single act, the subject must aim to attain an end which is not directly within reach and to put to work, with this intention, the schemata thitherto related to other situations. Thereafter the action no longer functions by simple repetition but by subsuming under the principal schema a more or less long series of transitional schemata. Hence there exists simultaneously the distinction between the end and the means, and the intentional coördination of the schemata. The intelligent act is thus constituted, which does not limit itself merely to reproducing the interesting results, but to arriving at them due to new combinations.

From the point of view of the real categories such progress leads, as we shall see in Volume II, to an essential result. By coordinating the schemata which constitute the instruments of his intelligence, the child learns *ipso facto* to put things in relationship to each other. In effect, the concrete connections uniting objects of the external world are constructed at the same time as the formal interrelations of schemata since the latter represent actions capable of being brought to bear on objects. The parallelism of these two series, real and formal, is even so close that, during the first stages, it is very difficult to dissociate action from object. On the other hand, in proportion as the action becomes complicated through coördination of schemata, the universe becomes objectified and is detached from the self.

The phenomenon is apparent, at first, with regard to the concept of "object." It is to the extent that the child learns to coördinate two separate schemata—that is to say, two actions until then independent of each other—that he becomes capable of seeking objects which disappeared and of attributing to them a consistency independent of the self. Searching for the object which has disappeared is, in effect, to set aside the screens which mask it and to conceive it as being situated behind them; it is, in short, to think of it in its relations with things seen at the present time and not only in its relations with the action.

This progress in the formation of the object is on a par with a correlative elaboration of the spatial field. As long as the child's

activity is made manifest only in the form of isolated movements —that is to say, of schemata which are not intercoördinated, the "groups" of displacements remain dependent on the movements; in other words, space is only envisaged as a function of the self and not yet as an immobile environment connecting things to one another. With the coördination of the schemata, on the other hand, begins the putting of the bodies themselves into spatial interrelationships, that is to say, the formation of objective space. To be sure, the formation of this space, like that of the "objects" which are correlative to it, is not achieved at one time and at this stage many remnants of the preceding stages are found. But the orientation of the subject's mind is henceforth different and instead of bringing the universe to himself, the child begins to place himself in a universe which is independent of him.

The same is true in the realm of causality and of time. During the present stage the causal series go beyond the purely global relations between the activity and the external movements, to become objectified and made spatial. In other words, the cause of a certain phenomenon is no longer identified by the child with the feeling he has of acting upon this phenomenon. The subject begins to discover that a spatial contact exists between cause and effect and so any object at all can be a source of activity (and not only his own body). Consequently the temporal series begin, for their part, to be set in order as a function of the sequence of events, and not only of the sequence of actions.

Thus one catches a glimpse of how the intercoördination of secondary schemata is accompanied by a correlative progress with regard to the elaboration of the "real" categories of intelligence. But let us save the study of these transformations for Volume II and only analyze at the present time the formal elaboration of the mechanism of intelligence.

§1. THE "APPLICATION OF FAMILIAR SCHEMATA TO NEW SITUATIONS." I. THE FACTS.—The essential novelty of the situation which we shall now study is that: the child no longer merely tries to repeat or prolong an effect which he has discovered or observed by chance; he pursues an end not immediately attainable and tries to reach it by different inter-

mediate "means." As far as these means are concerned it is true that familiar schemata are always involved and not new means; but, granted that the subject no longer limits himself to reproducing that which he has just done but pursues a distant goal, he adapts the familiar schema to the particulars of this situation and so raises it to the level of a true "means." With regard to the "purpose" it goes without saying that the child does not decide about it in advance, in the sense that we manage, through reflection, to impose a plan on our conduct, independently of any external suggestion. It is always under the pressure of perceived facts, or by prolonging a recent reaction, that the child acts. His acts are still, therefore, in this sense, conservative, and have no function other than the use of his earlier schemata. That conforms, moreover, to the fundamental law of assimilation and we do not see how it could be otherwise. But—and it is in this sense that the goal is set in advance and that the situation is "new"—obstacles intervene between the act and its result. Where the child wishes to grasp, to swing, to strike, etc. (as many ends as are consistent with primary and secondary circular reactions), circumstances erect barriers he must clear. Hence it is a question of keeping in mind the "goal" to be reached and of trying different known means of surmounting the difficulty. The act of intelligence properly so called develops in that way, to the extent that it is differentiation of the secondary circular reaction and involves to a higher degree the "reversal" in the consciousness which constitutes intention and of which we have spoken before.

Let us try to analyze some samples of this behavior, beginning by describing three cases intermediate between the secondary circular reactions and the true "applications of familiar means to new situations."

Observation 120.—We believe we have observed in Laurent an elementary example of these behavior patterns as early as 0;6 (1), provided that the facts we have described were observed correctly. If this was the case, the thing has nothing extraordinary about it, for three reasons. The first is that this first manifestation of the "application of familiar schemata to new situations" is not yet typical and forms the transition between the simple "secondary circular reaction" and the freer behavior patterns, the description of which will follow. The second reason is that Laurent has always been ahead of his sisters, after the circumstances

previously noted, and that consequently, at 0;6 (1) he has utilized, for three months already, all sorts of secondary circular schemata. Hence it is natural that he arrive at coördinating them to each other in certain exceptional situations. The third reason, which we emphasize strongly, is that the behavior patterns characteristic of a stage appear so much the less at one time, in the form of simultaneous manifestations, according as this stage is more evolved and these behavior patterns are therefore more complex. Thereafter it is perfectly normal that these first behavior patterns of the fourth stage are constituted sporadically from the middle of the third stage, except that these episodic productions are only systematized and consolidated one or two months later. In the same way we shall see that the behavior patterns of the fifth stage are foreshadowed from the apex of the fourth and those of the sixth are foreshadowed at the fifth stage. Inversely, it is evident that the behavior patterns belonging to a given stage do not disappear during the following stages but conserve a role whose importance only diminishes very gradually (and relatively).

Regardless of the application of these remarks to this observation, Laurent, at 0;6 (1) tries to grasp a big piece of paper that I offer him and finally place on the hood of his bassinet (and on the string connecting the hood with the handle of the bassinet). Laurent begins by stretching out his hand; then as soon as the object is placed, he reacts as he always does in the presence of distant objectives: he shakes himself, waves his arms, etc. The desire to grasp the paper seems to inspire such reactions, as I regulated it by removing the objective from the hood for a few seconds in order to move it progressively closer and farther away.—It is when the paper seems inaccessible to the hand alone that Laurent shakes himself. After having behaved thus for a moment, he seems to look for the string hanging from the hood, then pulls it harder and harder while staring at the paper. At the moment when this is ready to fall off the hood, Laurent lets go the string and reaches toward the objective of which he immediately takes possession. Several sequential attempts have yielded the same result.—It goes without saying that it cannot be demonstrated that Laurent pulled the string in order to grasp the paper, but the whole behavior pattern gave me the impression of being performed with this end in view and of being perfectly coördinated.

If such is the case, it can be asserted that the schema of "pulling the string" has momentarily served as means to attain the end assigned by the schema "grasping the objective." This of course does not mean that Laurent has foreseen the object's fall, nor that he has conceived of the string as its extension: He has simply utilized a familiar schema with a new intention, and this is what characterizes the behavior patterns of the fourth stage. But, as the paper was placed in the same situation which habitually sets in motion the schema of "pulling the string,"

such an example still partakes of the generalizing assimilations of the "secondary circular reaction" (see Obs. 99).

Observation 121.—Here is an analogous example, but easier to interpret. At 0;8 (20) Jacqueline tries to grasp a cigarette case which I present to her. I then slide it between the crossed strings which attach her dolls to the hood. She tries to reach it directly. Not succeeding, she immediately looks for the strings which are not in her hands and of which she only saw the part in which the cigarette case is entangled. She looks in front of her, grasps the strings, pulls and shakes them, etc. The cigarette case then falls and she grasps it.

Second experiment: same reactions, but without first trying to grasp the object directly.

At 0;9 (2) Jacqueline tries directly to grasp her celluloid duck when I put its head between the strings I have just described. Not succeeding, she grasps both strings, one in each hand, and pulls. She looks at the duck who shakes when she shakes. Then she grasps both strings in one hand and pulls, then grasps them in the other hand a little higher up and pulls harder until the duck falls.

I begin over again, but attach the duck more firmly. She then at once pulls the strings, systematically, until she can touch the duck with her finger, but does not succeed in making it fall. She then gives up although I shake the duck several times, which shows that she tries to grasp the duck and not to swing it.

It may be seen that these behavior patterns differ from those in Observation 113, even though in both cases it is a question of shaking the string to exert influence on a distant object. In the case of Observation 113, the child limits himself to utilizing a procedure which he has employed just previously, and to utilizing it in order to prolong a spectacle he has just had before his eyes. On the other hand, in the present case, he tries to grasp an object and to do so he must find the appropriate means. The means to which Jacqueline has recourse is of course borrowed from the schemata of her earlier circular reactions, but the act of intelligence has precisely consisted in finding the right means without limiting herself to repeating that which has already been done.

These behavior patterns must not, however, be overestimated and one must not so soon see in them a utilization of instruments (the behavior pattern of the "stick") or even a utilization of extensions of the object (the behavior pattern of the "string"). There could be no question of instruments for several more months (concerning the "string" we shall return to it in Chap. V, §2). The following observation, which can be cited in the margin of this one, shows that Jacqueline does not yet consider the strings as extensions of the desired object.

Observation 121 repeated.—At 0;9 (8) Jacqueline tries to grasp her parrot which I placed between the entwined strings (in the same position

as the duck in the previous observation). She pulls a string from the lower end of which her doll is hanging. She sees the parrot swing and, instead of trying to grasp it, henceforth she merely tries to shake it. It is then that the behavior pattern we shall emphasize here arises and which constitutes a true act of intelligent adaptation. Jacqueline looks for the doll at the other end of the string, grasps it with one hand and hits its head with the other hand while staring at the parrot. She then does this at regular intervals while alternately looking at doll and parrot and each time controlling the result (the parrot swings at each stroke).

Now the genesis of this act is easy to grasp. Three days earlier (see Obs. 102) Jacqueline shook her parrot, while holding it, in order to hear the rattle it contains. Hence when she sees the hanging parrot, she wants to grasp it in order to shake it again. Moreover, she knows how to strike objects and, in particular, has constantly struck her parrot during the preceding weeks (see Obs. 103). Thereafter as soon as she discovers that the parrot is connected to the same string as her doll, she uses the doll as a means to shake the parrot. Here again she does not limit herself to applying a gesture which she made previously (as is the case in Obs. 112–118). She really adapts a schema known earlier to a new situation.

But furthermore, when Jacqueline hits the doll she does not at all have in mind pulling the string (which connects the doll to the parrot) in order to augment the effect. As in the previous observation, the string is not therefore an "extension of the object" and Jacqueline is not yet concerned with spatial and mechanical contacts. Hence the string has no other than a tactile and kinesthetic meaning: it is only a material for manual and muscular schemata, a procedure to obtain a certain result, and not yet a physical object as the "string" and above all the "stick" are later to become.

Let us now proceed to clear cases and begin by the simplest possible: withdrawing the material objects which intervene between the intention and the result. Among the behavior patterns corresponding to this definition the most elementary of all is that which consists in moving away someone's hand or a certain object which one places between the child and the objective at the time the act of grasping is performed. The objective must of course be left entirely visible, hiding it constituting an additional difficulty which we shall only examine at the end of these observations.

Observation 122.—Concerning Laurent, this sort of behavior (the acquiring of which we have just studied closely) was made manifest only at 0;7 (13). It is this coördination between a distinctly differentiated

action serving as means (=setting aside the obstacle) and the final action (=grasping the object) which we shall consider as the beginning of the fourth stage.

Until 0;7 (13) Laurent has never really succeeded in setting aside the obstacle; he has simply attempted to take no notice of it, or else to utilize the magic-phenomenalistic "procedures" taken up in Chapter III, §4. For instance at 0;6 (0) I present Laurent with a matchbox, extending my hand laterally to make an obstacle to his prehension. Laurent tries to pass over my hand, or to the side, but he does not attempt to displace it. As each time I prevent his passage, he ends by storming at the box while waving his hand, shaking himself, wagging his head from side to side, in short, by substituting magic-phenomenalistic "procedures" for prehension rendered impossible. Afterward I hold out the box to him while merely holding it by the edge. Laurent pulls, tries to snatch it away from me, but does not push back my hand.

Same reactions at 0;6 (8), 0;6 (10), 0;6 (21), etc. At 0;6 (17) I present a rattle to him while placing my hand in front of it, so that only half the object is visible. Laurent tries to grasp it directly, but not to set aside my hand.

At 0;7 (10) Laurent tries to grasp a new box in front of which I place my hand (at a distance of 10 cm.). He sets the obstacle aside, but not intentionally; he simply tries to reach the box by sliding next to my hand and, when he touches it, tries to take no notice of it. This behavior gives the impression that he pushes the screen away, but there does not yet exist any differentiated schema, any "means" dissociated from the final action (from the schema assigning an end to the action). The behavior pattern is the same when I use a cushion as an obstacle.

Same reactions at 0;7 (12). Finally, at 0;7 (13) Laurent reacts quite differently almost from the beginning of the experiment. I present a box of matches above my hand, but behind it, so that he cannot reach it without setting the obstacle aside. But Laurent, after trying to take no notice of it, suddenly tries to hit my hand as though to remove or lower it; I let him do it to me and he grasps the box.—I recommence to bar his passage, but using as a screen a sufficiently supple cushion to keep the impress of the child's gestures. Laurent tries to reach the box, and, bothered by the obstacle, he at once strikes it, definitely lowering it until the way is clear.

At 0;7 (17) I resume the experiment without there having been intervening attempts. First I present the object (my watch) 10 cm. behind the cushion (the object of course being visible). Laurent tries at first just to grasp the watch, then pauses to hit the cushion. Same thing with his hand: he at once strikes the obstacle.

From the outset it may be seen how this act of pushing back an obstacle by striking it constitutes a new behavior pattern in relation to the behavior patterns of 0;6 (0) to 0;7 (12). Before trying to attain

his end (grasping the objective) Laurent henceforth pauses and exerts a completed action on the obstacle (striking it to remove it), definitely differentiated from the final schema and nevertheless subordinated to it.

Moreover, one notes that the intermediate act serving as means (removing the obstacle) is borrowed from a familiar schema: the schema of striking. We recall that Laurent, from 0;4 (7) and above all from 0;4 (19) has the habit of hitting hanging objects in order to swing them and finally, from 0;5 (2) of striking the objects (see Obs. 103). Now, this is the usual schema of which Laurent makes use at the present time, no longer in the capacity of an end in itself (of a final schema) but as a means (a transitional or mobile schema) which he subordinates to a different schema. The need to remove the obstacle arouses, through generalizing assimilation, the simplest of the schemata of displacement which he knows and utilizes: the schema of striking. It is noteworthy, in this respect, that the child does not yet displace the objects from one position to another as he will do later, in order to study experimentally the groups of displacements (see Vol. II, Chap. II, §§ 3 and 4). That is why the act of removing or of displacing the obstacle is so difficult when it seems so simple. The universe of the child of 0;6–0;8 is not yet a world of permanent objects animated by independent movements in space (we shall try to demonstrate this in Volume II in studying the development of the concept of space). Besides, in order to push away the obstacle the child is obliged to call upon his circular schemata of which the most appropriate to the situation is that of "striking in order to swing."

Let us observe, finally, that the procedure thus discovered by Laurent has nothing in common with the behavior pattern which consists in removing annoying objects (pillows, etc.) from in front of him. Laurent, at 0;5 (25) for example, throws aside without hesitation a pillow which I put over his face. But such a reaction, in which there is probably a reflex element, only comes into play if the screen is placed in front of the child. The child no longer removes it if he is in front of the object (see Vol. II, Obs. 27). Therein is not a "means" related to the objective, but simply the elimination of a cause of annoyance related to the subject. Consequently it would be entirely artificial to say that the child removes a screen (= means) in order to see something (= end). The act of ridding himself of the obstructive object forms an entity in itself.—So it is not a question of searching in this behavior pattern for the origin of the one which we are now examining: proof in itself that between 0;6 (0) and 0;7 (12) Laurent has never succeeded in removing the obstacles, in the sense we give to these terms here, when he knew ever since 0;5 and doubtless long before how to push any obstructive screen away from his face or from before him.

Observation 123.—From 0;7 (28) the transitional schema of "pushing the obstacle away" is slightly differentiated in Laurent: instead of sim-

ply hitting the things which intercede between his hand and the objective, he has applied himself to pushing them away or even to displacing them.

For example at 0;7 (28) I present to him a little bell 5 cm. behind the corner of a cushion. Laurent then strikes the cushion, as previously, but then depresses it with one hand while he grasps the objective with the other. Same reaction with my hand.

At 0;7 (29) he immediately depresses the cushion with his left hand in order to reach the box with his right. He does the same at 0;8 (1): when my hand intervenes as the obstacle I definitely feel that he depresses it and pushes harder and harder to overcome my resistance.

At 0;8 (1) right after the attempt just described, Laurent swings his box in order to cause the lozenges within to make a noise. I then keep my hand on his arm to prevent him from playing. First he tries to proceed with the very arm whose hand holds the box, then he brings forward his other hand and removes mine. It is the first time he succeeds in this attempt, already made for weeks and days before.

At 0;8 (28), on the other hand, I observe that he does not know how to push my hand away when I hold the object[1] nor when I move my hand closer to the object from behind in order to take it away from him.—Both these behavior patterns appeared simultaneously at 0;9 (15). When I hold one of these rattles by its extremity he pushes my hand away with his left hand while pulling the object with his right, and when I get ready to take the rattle back, he pushes away my hand or forearm before I reach it.

It may thus be seen that these perfections of the transitional schema were constituted by gradual differentiation of the original procedure: "Hitting the object in order to push it away."

Observation 124.—At 0;8 (8) Jacqueline tries to grasp her celluloid duck but I also grasp it at the same time she does. Then she firmly holds the toy in her right hand and pushes my hand away with her left. I repeat the experiment by grasping only the end of the duck's tail: she again pushes my hand away. At 0;8 (17) after taking a first spoonful of medicine, she pushes away her mother's hand which extends to her a second one. At 0;9 (20) she tries to place her duck against the wicker of the bassinet but she is bothered by the string of the bell which hangs from the hood. Then she takes the string in her right hand and moves it to the far side of the left arm (the arm holding the duck), and consequently where the string no longer is an obstacle. Same operation shortly afterward.

Unfortunately, we have been unable to determine precisely with regard to Jacqueline, from which circular schema the action of "re-

[1] I started this experiment at 0;6 (10): he simply pulls the object without removing my hand.

moving the obstacle" was differentiated. It was probably not from the schema of "striking" since she only made the latter manifest shortly before. It must be, rather, the act of holding the object in order to shake, swing, or rub it which gave her the idea of displacing the obstacles. It goes without saying that the filiation can vary in each child between the transitional or mobile schemata of the fourth stage and the circular schemata. We merely assert that the subordination of means to ends belonging to the fourth stage begins by a simple coördination of earlier circular schemata.

Observation 125.—If one is still doubtful that this assertion is well founded, in other words, if one considers the act of removing the obstacle as being too simple not to be able to be formed by itself independently of the earlier schemata, the following example is of a nature to furnish a counterproof to the foregoing considerations.

Suppose there is an intentional act, which is performed as a means and which is more elementary than that of "removing the obstacle," this is doubtless the act which consists in relinquishing an object or putting it down, in order to grasp another. In effect, as soon as the child knows how to coördinate prehension with vision (beginning of the third stage), he sometimes lets go voluntarily the objects he was holding. Furthermore, he observes this phenomenon very early, since from the first weeks of the same stage (about 0;4, in Laurent) he seeks the lost object with his hand (see Vol. II, Chap. I, §2), and from the middle of the stage (about 0;6, in Laurent) his eyes follow it (see Vol. II, §1, Obs. 6). Now, far from furnishing, from the outset, a "means" to use in all circumstances, this falling of the grasped object remains unused for a long time. It therefore in no way constitutes a "schema"—that is to say, a positive action—but simply an accident, a failure of the act. (One cannot speak, in this connection, of a negative act, for a negative act is a complex act inasmuch as it is always subordinated to another). It is only at the end of the fourth stage and the beginning of the fifth that the act of relinquishing an object becomes an intentional act (concerning Laurent, see Obs. 140 and 141): This fact obviously constitutes proof that the transitional schema of "relinquishing the object," utilized as a "means," does not merely stem from the chance falling of the object, but is formed as a function of other schemata, such as those of "removing the obstacle." This is what we shall now prove.

At 0;6 (26) Laurent holds a rattle with which he no longer does very much (satiety after use). I then offer him a doll which he immediately tries to grasp with both hands (as he always does). He grasps it with his left hand, holding the rattle in his right, but he brings his hands closer together in the obvious wish to hold only the doll. He remains perplexed, alternately looking at both objects, and does not succeed in relinquishing the rattle.

At 0;6 (29) same reaction with two other objects. I offer him a

third toy: he tries to grasp it with his right hand, without letting go what he is holding in it.—Finally of course he fortuitously loses the object in which he is no longer interested, but he does not remove it on purpose.

At 0;7 (0) he has in his hands a small celluloid doll when I offer him a box (more interesting). He grasps the latter with his left hand and tried to grasp it with both hands. He then knocks both objects together, at once separates them (very much surprised at the result), and recommences to knock them again, wishing to surround the box with his hands. He then knocks them several times in succession for fun, which gives him the idea of rubbing the box against the wicker of the bassinet. Then he tries once more to surround the box with both hands; surprised at the resistance due to the doll, he looks for a moment at the two contiguous objects.

At 0;7 (28) I again note the same reaction: he knocks involuntarily the two objects he is holding when he wants to grasp one of them with both hands.—Of course, in order that this experiment have some meaning, it is fitting to choose two objectives of very unequal interest, for otherwise one could always wonder if the child was not trying to keep them both. This objection, moreover, is not theoretical since Ch. Bühler has demonstrated that an 8-month-old baby can very well pay attention to both toys simultaneously. In fact, the look and expression of the child are enough to show which toy he prefers and wishes to conserve.—It is necessary, moreover, to act quickly and to surprise the child by means of the second object before he has let go the first, without realizing it, due to lack of interest. In daily life, that is to say, independently of the experiment under discussion, these things always take place as follows: "When the child whose two hands hold an object, sees a second one and tries to grasp it, he lets go the first involuntarily, out of pure lack of interest, whereas he strains to reach the second one. In order that the experiment succeed it is therefore necessary to offer the object at a distance of several centimeters from the hand so that the child may have no difficulty in taking possession of it nor have time to lose the first one.

Finally at 0;7 (29) Laurent finds the solution. He holds a little lamb in his left hand and a rattle in his right. A small bell is held out to him: he lets go the rattle in order to grasp the bell. The reaction is the same several times in succession but I have difficulty in discerning whether he simply lets the rattle escape him or really discards it. While he holds the bell I offer him a big box: he grasps it with his (free) left hand and with his right (by sticking the bell against the box) but, noticing the difficulty, this time he definitely discards the bell.—Same reaction at 0;7 (30) with a big rattle.

At 0;8 (1) he has a big box in his hands and I present my watch chain to him. He then places the box on his coverlets in order to grasp

the chain. Now this gesture is new (hitherto he only performed it by chance): It apparently derives from the act of "discarding" observed for three days.—I again offer him the box while he holds the chain: he removes the box.

Henceforth Laurent knows how either to discard one object in order to grasp another or to place it, or to let it fall intentionally. This transitional schema, given the tardy and complex character of its advent, apparently derives from the preceding schemata which consist in "removing the obstacle" and not in the fortuitous escape of objects held in the hand.

Observation 126.—A final behavior pattern belonging to the present group of schemata ("removing the obstacle") consists in searching under a screen for invisible objectives. We shall closely study this behavior in connection with the development of the concept of the object. But from the point of view of the working of intelligence, it is fitting to show now how such an act is constituted by coördination of independent schemata.

For example Laurent, at 0;8 (29), plays with a box which I remove from his hands in order to place it under a pillow. Although, four days before, he did not react at all in an analogous situation, this time he at once takes possession of the pillow. Though it cannot be stated that he expects to find the box under the pillow (for the behavior is too undecided), it is nevertheless apparent that Laurent is not interested in the pillow as such and that he lifts it in order to try something. The act of lifting the pillow is therefore not a sure means for the child, but it already constitutes a "means" for the attempt, that is to say, a separate action from grasping the box.

Likewise, at 0;9 (17), Laurent lifts a cushion in order to look for a cigar case. When the object is entirely hidden, the child lifts the screen with hesitation, but when one end of the case appears, Laurent removes the cushion with one hand and with the other tries to extricate the objective. The act of lifting the screen is therefore entirely separate from that of grasping the desired object and constitutes an autonomous "means," no doubt derived from earlier analogous acts (removing the obstacle, displacing and pushing away objects which are a barrier, etc.).

This proves that in all these examples, the action of "removing the obstacle" definitely constitutes a transitional schema differentiated from the final schema. As we have seen in each case analyzed, either these transitional schemata derive from earlier circular schemata (Obs. 122), or else they derive from other transitional schemata (Obs. 123–126). The subordination of transitional

schemata to final ones—hence of "means" to ends—always takes place by coördination of the independent schemata.

Here, finally, is a third group of examples of the "application of familiar schemata to new situations." Henceforth it is no longer a question of removing obstacles, but of finding intermediates between the subject and the objective. These intermediates are not yet "instruments," as in the fifth stage, but they are more complex than the organized schemata borrowed without change from the secondary circular reactions (as is the case with our first group of examples: Obs. 120–121 repeated).

Observation 127.—If Jacqueline, at 0;8 (8) has shown herself capable of removing a hand which forms an obstacle to her desires, she has not delayed in making herself capable of the inverse behavior pattern: using the other person's hand as an intermediate in order to produce a coveted result. Thus at 0;8 (13) Jacqueline looks at her mother who is swinging a flounce of material with her hand. When this spectacle is over, Jacqueline, instead of imitating this movement, which she will do shortly thereafter, begins by searching for her mother's hand, places it in front of the flounce and pushes it to make it resume its activity.

Certainly it is not a question here of only making last a spectacle which has just been seen. In this respect these cases can be compared to those of Observations 112–118. But notable progress is made by virtue of the fact that Jacqueline mentally decomposes the spectacle she witnessed and utilizes the other person's hand as intermediary. Furthermore, two months later she applies this means to a new situation.

At 0;10 (30) Jacqueline grasps my hand, places it against a swinging doll which she was not able to set going herself, and exerts pressure on my index finger to make me do the necessary (same reaction three times in succession).

We shall later study (Vol. II, Chap. III, §3), these behavior patterns in detail, in connection with making causality spatial, but it behooved us to cite them now from the point of view of the "application of familiar schemata to new situations" to show how they arise from coördination of independent schemata. The great novelty of such a behavior pattern is the following. Until about 0;8 the child, when an interesting result is produced before him, tries to act upon this result directly. He looks at the objective and, according to cases, he draws himself up, waves his hands, etc. (see Obs. 112–118). or else, if he can reach the objective (a rattle, for instance), he strokes it, shakes it, etc. Furthermore it constantly happens that he reacts in the same way solely in the presence of the adult's hand. If I snap my middle finger against my thumb and present my hand to the child, either he draws himself up, etc., if

it is unattainable, or else he strokes it, shakes it, etc., to make me continue. Hence this reveals two kinds of independent secondary circular schemata: actions upon the objective and action upon the hand (which is conceived in this case as being an objective). Now, in the observation which we have just described, Jacqueline uses the other person's hand as intermediate, in order to act upon the objective. What does this mean if not that, as with regard to the schemata of "removing the obstacle" or the simpler schemata of the first group, the child begins to coördinate two kinds of schemata thitherto independent? He tries to act upon the other person's hand, but inasmuch as this hand can itself act upon the objective: accordingly he puts one of the two schemata in relationship with the other.

Observation 128.—Laurent too, from 0;8 (7) uses my hand as an intermediate to make me resume the activities which interest him. For example, I tap my cheek with my left middle finger, then I drum on my eyeglasses (he laughs). Afterward I put my hand halfway between his eyes and my face. He looks at my glasses, then at my hands, and ends by gently pushing my hand toward my face (see the continuation of these observations, Vol. II, Obs. 144).

As in the preceding observation, for the child it is a question of prolonging an interesting spectacle. But instead of applying the habitual procedures (Obs. 112–118) without adding anything to them, or of himself reproducing the thing by imitation, Laurent utilizes as means an element of the whole which he has just observed and an element which can be assimilated to his own activity. The other person's hand is comparable to that of the subject and the child simply prolongs its action due to an intermediate whose power he knows by analogy to his own earlier experiences.

Observation 129.—At 0;9 (24) Jacqueline is seated and tries to grasp her duck which is near her feet. Not succeeding, she draws it toward her with her right foot. I was not able to see whether there was still groping or if the reaction was immediate. At 0;11 (21), on the other hand, she lets fall a celluloid swan from its swinging nest. Unsuccessful in her attempts to pick it up, she immediately displaces it with her feet and brings it nearer to her. At 1;0 (7) same immediate reaction with her parrot. Of course Jacqueline also uses her feet to strike the objects with which they come in contact, etc.

These behavior patterns can surely be conceived not as acts of intelligence but as simple coördinations analogous to those of manual prehension. But as soon as they appeared they gave rise to a series of applications which definitely bear the seal of intelligent generalization.

Thus, at 0;11 (28), Jacqueline is seated and shakes a little bell. She then pauses abruptly in order delicately to place the bell in front of her right foot, then she kicks hard. Unable to recapture it, she grasps

a ball which she places at the same spot in order to give another kick. It is therefore apparent that before the act there was intention and utilization adapted from the schema of kicking.

Furthermore, the same day, Jacqueline, while playing with a thimble, struck a wooden box. Interested by this contact, she repeated it two or three times, after which at one leap she struck the thimble against her leather shoe. This gesture was precise and rapid, revealing all the signs of a typical intentional act.

Moreover, at 1;0 (10) she is in the act of striking a piece of wood against the wicker of her bassinet when she suddenly pauses, obviously to look for her shoe. As her feet are covered by a shawl, she immediately removes the shawl (see Obs. 124) and taps her shoe.

These last three observations correspond in the most characteristic way to the definition of the behavior patterns which we shall now analyze: First, the intention preceding the act (increasing the contact of the bell, the thimble, or the piece of wood); second, the search for a "means" capable of being subordinated to such an end; third, the application to that end, of a schema discovered earlier (using the foot to stir up, strike, etc.).

Observation 130.—Laurent, at 0;10 (3) utilizes as a "means" or a transitional schema, a behavior pattern which he discovered the previous day and whose genesis we shall describe in Observation 140.—By manipulating a tin of shaving cream he learned, at 0;10 (2) to let this object fall intentionally. Now, at 0;10 (3) I give it to him again. He at once begins to open his hand to make it fall and repeats this behavior a certain number of times. I then place, 15 cm. from Laurent, a large wash basin and strike the interior of it with the tin in order to make Laurent hear the sound of the metal against this object. It is noteworthy that Laurent, already at 0;9 (0), had, while being washed, by chance struck a small pot against such a basin and immediately played at reproducing this sound by a simple circular reaction. I therefore wanted to see if Laurent was going to use the tin to repeat the phenomenon and how he was going to go about it.

Now, at once, Laurent takes possession of the tin, holds out his arm and drops it over the basin. I moved the latter, as a check. He nevertheless succeeded, several times in succession, in making the object fall on the basin. Hence this is a fine example of the coördination of two schemata of which the first serves as "means" whereas the second assigns an end to the action: the schema of "relinquishing the object" and that of "striking one object against another."

§2. THE "APPLICATION OF FAMILIAR SCHEMATA TO NEW SITUATIONS"—II. COMMENTARY.—These kinds of behavior patterns constitute the first acts of intelligence,

properly so called, that we have hitherto encountered. It is therefore fitting to try to characterize them exactly and, to do this, to begin by distinguishing them from the different varieties of behavior patterns previously under study.

These behavior patterns at first contrast with the primary circular reactions and the sensorimotor habits which derive from them. In the case of these reactions, the contact (tactile, visual, etc.) with the object immediately sets in motion a global assimilatory act without our being able to distinguish between the end of the action and the means employed, whereas in the present case the contact with the external object only releases intention and the search for appropriate means. Intention exists, that is to say, consciousness of a desire to the extent that the assimilatory schema set in motion by contact with the object is opposed by an obstacle and thereafter only is made manifest in the form of a tendency and not of immediate realization. This same circumstance explains the search for means: It is, in effect, a question of overcoming the intervening obstacle. So it is that in Observation 122, the sight of the object merely sets in motion the schema of prehension but, when obstacles intervene, prehension is promoted to the rank of a distant end and certain means must be sought in order first to remove these difficulties. When Laurent tries to grasp the object behind my hand we see, in the simplest way, how the sensorimotor schema characteristic of the steps of primary circular reaction and the first habits is differentiated in an intentional act due to the intervention of intermediate obstacles. When he raises a screen in order to find a hidden object (Obs. 126), the same thing becomes complicated, but the principle remains the same: It is therefore the dissociation of means and ends, due to intervening obstacles, which creates intention and puts the present behavior pattern in opposition to simple habits.

Perhaps the objection will be raised that the intersensorial coördinations peculiar to some of the primary circular reactions seem very early to make us witness seriations of the same kind. When the child grasps an object in order to suck it, look at it, etc., he seems to differentiate the means from the ends and, consequently, set a goal in advance. But, for want of an obstacle capable of attracting the child's attention, nothing warrants attributing

these distinctions to the subject's consciousness. Grasping in order to suck constitutes a single act in which the means and the end are one, and this single act is formed by immediate reciprocal assimilation between the schemata present. It is therefore the observer, and not the subject, who makes divisions in the case of such schemata. It is only when the child seeks to put things in themselves into relationship that the differentiation of means from ends appears—in other words, the acquisition of consciousness characterizing intention and arising when external obstacles are produced.

How are the present behavior patterns thereafter to be distinguished from the secondary circular reactions which also involve the utilization of connections between the things themselves? With regard to the actual circular reactions (Obs. 94–104), first by the way in which they assign a purpose. The circular reaction has no other end, in effect, than that of reproducing a result obtained earlier, or which has just been discovered by chance. This kind of process can be accompanied by intention but after the event, to the extent that there is repetition and when the result to be reproduced presupposes a complex activity. The effect which is to be repeated is then posited in advance, as an end, and the child tries to rediscover the means which have just led him to it (in this, let us recall, these behavior patterns already presage intelligence). But, although intentional, an end of this sort is nevertheless the simple prolongation of an earlier effect. On the contrary, in the present behavior patterns, the end is posited without having been attained beforehand, at least, not in the same situation. When the child tries to grasp his toys by pushing an obstacle away (Obs. 122–124), tries to act upon objects through the intermediary of someone's else hand (Obs. 127–128), tries to shake a parrot from a distance (Obs. 121 repeated) or to knock objects against his shoes (Obs. 129), these are projects which arise in the course of his action, in conformity, it is true, with his earlier circular reactions (the very nature of the end consequently does not differ from one behavior pattern to another), but in an actually new situation. The novelty of this situation pertains, in effect, either to the presence of obstacles or to the unexpectedness of the combinations observed. With re-

gard to the means employed, the difference is just as clear. In the secondary circular reaction, the means utilized were discovered fortuitously and were applied just beforehand; hence it is only a question of rediscovering them. In the behavior patterns now under study, on the other hand, it is necessary to improvise means and remove obstacles which separate the intention from its final result. It goes without saying that before inventing new means (which he will do later), the child at first limits himself to applying the schemata with which he is familiar. Moreover the means which are found are borrowed, like the ends themselves, from the schemata of earlier circular reactions. But the chief point was to remember them at the right time and adapt them to the current situation.

Finally, if we compare these behavior patterns to the "procedures to make an interesting spectacle last" the differences are approximately the same, although less accentuated (these "procedures" form, in effect, the transition between the circular reaction and the true act of intelligence). On the one hand, the contrast remains, from the point of view of ends, between "making last" that which one has just seen and pursuing an end in a new situation. Besides, from the point of view of means, the following difference exists. In the case of procedures to make spectacles last, the means utilized are borrowed either from a circular reaction immediately preceding which was interrupted by the interesting spectacle (for example, when the child pulls a string in order to prolong the swinging of a watch, from a distance, when he was in the process of pulling this string in order to swing the hood of his bassinet), or else they are borrowed from schemata which have become so familiar or even automatic (for instance, drawing oneself up, etc.) that no effort is any longer necessary to rediscover them and they apply to everything. In both cases the "procedure" accordingly remains, so to speak, "empty" without a precise connection with the effect sought after. On the contrary, true acts of intelligence involve a combination *sui generis* of the means and of the situation. In effect, they strive in vain to be borrowed from earlier circular reactions, for they are adjusted to the goal by a special accommodation, and it is this adjustment which characterizes the beginning of intelligent action.

In short, compared to the earlier stage ("secondary circular reactions" and "procedures to prolong an interesting spectacle"), the behavior patterns analyzed here present two new characteristics. The first pertains, not to the very nature of the goal, but to the situation in which it is pursued, hence the way in which the subject assigns it to himself. Instead of merely "reproducing" that which he has seen or done, in the situation of the initial act, the child tries to attain the desired result in the midst of difficulties not yet observed or of unforeseen combinations, that is to say, always in a new situation. The second (behavior pattern) pertains to the means employed. These means are henceforth entirely differentiated from the end itself, the child's behavior consequently consisting in a coördination of two independent schemata—the one, final (the schema assigning an end to the action), the other transitional (the schema utilized as means)—and no longer in the application of a single more or less complex schema. It is only after the event that the means and the ends become differentiated in the midst of a "secondary circular reaction." In reality, it is always a question of a single act, of a completely constructed schema, so that the use of the means supplied always is on a par with the same end, or the same kind of ends. When the child generalizes the schema, that is to say, applies it to other objects (pulling the string in order to shake a new doll which is hanging from it, etc.), one cannot say, either, that the familiar means apply to a new end. It is simply the schema in its entirety the child generalizes to a new object, just as when he grasps one object instead of another or sucks his thumb instead of the maternal breast.—With regard to the "procedures to prolong an interesting spectacle" the same applies, in spite of appearances. The child who draws himself up when confronted by anything at all, in order to prolong a movement or a sound, in no way combines two schemata, he merely generalizes a behavior pattern which he found successful. This is why we remarked before that the use of such means is, so to speak, "empty," without precise adaptation to the goal pursued. On the contrary, the "application of familiar schemata to new situations" always presupposes the coördination of two schemata thitherto independent. Hence there exists simultaneously a clear

differentiation of means from ends and an exact adjustment of the former to the latter.

In contrast to the earlier forms, intelligent adaptation is always double, since it involves a relationship between at least two acts of assimilation. The choice and the pursuit of goals constitute the first of these adaptations, the adjustment of means to ends involves, besides, a second adaptation henceforth essential to the first. Let us try to analyze the nature of these two phases.

As far as the first is concerned it can be said that the present behavior patterns merely prolong the preceding ones. Primitive intelligent activity has no other function than that of assimilating the universe to the schemata elaborated by the primary and secondary circular reactions, while accommodating these schemata to the reality of things. In other words, elementary intelligence, like all spontaneous activity, is essentially conservative. What the child seeks, in the observations just made, is to grasp or to hold, to shake or strike, in short, exactly that to which his circular reactions have accustomed him. Now we have seen that the secondary circular reaction, which is already almost intentional, is just as conservative and assimilatory, despite appearances, as the primary reaction. Hence the difference between primitive intelligent behavior patterns and the preceding activities does not stem from the nature of the goals pursued: it simply results, as we have seen, from the fact that obstacles arise between intention and realization and necessitate the use of intermediate means.

With regard to the "means" giving rise to this second adaptation which constitutes the distinguishing characteristic of intelligence, it can be said that they are adapted to the "goal" of the action in the same way that the entire act, in its intention, is adapted to the desired object. In other words, the intermediates or obstacles which intervene between the subject and the goal of his acts are themselves assimilated to familiar schemata in the same way that the object of the action is assimilated to the schema of the goal. But these transitional schemata are not chosen for themselves but as a function of the final schema. Consequently, the intermediate objects are simultaneously assimilated to the transitional schemata and to the final schema and this is what

insures the coördination between the former and the latter due to a process of reciprocal assimilation.

In order to make clear the meaning of this formulation, let us first remark that the coördination between means and ends is of the same order, in its point of departure, as that of the sensorimotor schemata belonging to the primary circular reactions. When prehension is coördinated with sucking or with vision the phenomenon is explained, as we have seen, by simple reciprocal assimilation. The mouth tries to suck that which the hand grasps, or the hand tries to grasp what the eyes see, etc. This is what gives the illusion of a subordination of means to ends (the child seems "to grasp in order to suck," etc.) when there is simply a fusion of heterogeneous schemata in new global schemata. Now, the coördination of secondary schemata which constitutes the behavior patterns of the present stage is, in its beginnings, nothing other than a reciprocal assimilation of this kind. This is the case in the elementary and transitional schemata which we have assembled in a first group (Obs. 120–121). For example, when Laurent tries to grasp a piece of paper situated too high up and to do this, searches, then pulls a string hanging from the hood, he at first assimilates the paper to a schema of prehension (or to a more complex schema: feeling it, etc.), then, without ceasing to want to apply this first schema to it, he assimilates the same object to a very familiar schema of "pulling the string in order to shake." This second assimilation is, therefore, itself subordinated to the first; that is to say, pulling the strings in order to move the paper, he continues to desire it in the capacity of an "object to grasp" (he must at least have the impression that by shaking the objective he acquires a power over it which puts it more at his disposition.). Due to this double assimilation the schema of "pulling the string" is coördinated with the schema of "grasping" and becomes a transitional schema in relation to a final schema (Obs. 120; the same applies to Obs. 121). Such a reciprocal assimilation can lead either to a symmetrical relation (pulling in order to grasp and grasping in order to pull), or to a relation of simple inclusion (pulling in order to grasp).

But it is only in the elementary situation in which the child acts on an object only (the paper, for example, which is simul-

taneously for shaking and grasping) that the coördination of the schemata works due to so primitive a reciprocal assimilation. In such cases there is almost a fusion of the schemata present, as is the case in the coördination of the primary schemata, with the sole difference that the double assimilation is not instantaneous but is arranged in a sequence of two separate moments. On the other hand, in the majority of cases, the existence of an obstacle or the necessity for varied intermediates makes the coördination of the schemata less simple. This coördination most certainly continues at that time to operate, by reciprocal assimilation, but with a double complication. In the first place the schemata henceforth simultaneously subsume several objects whose mutual relationships it is a question of establishing. In the second place, and as a result, assimilation between the schemata ceases to work by simple fusion in order to give rise to diversified operations of inclusion or of hierarchical implication, of interference and even of negation, that is to say, to multiple dissociations and regroupings.

It must be recalled that the secondary schemata hitherto examined (the third stage) only comprehend one object at a time (a rattle for shaking, etc.) even when a complex object is involved (string and hood joined together, etc.). Moreover, when two primary schemata are reciprocally assimilated they apply to one and the same object (one person to look at and listen to simultaneously, etc.). Henceforth, on the contrary, the coördination of schemata bears upon two or several separate objects produced together (the objective and the obstacle or the objective and the intermediate, etc.) in such a way that the reciprocal assimilation of the schemata surpasses simple fusion to construct a series of more complicated relationships. In short, the generic character of the schemata is accentuated according as the relations (spatial, causal, etc.) of the objects to each other multiply, the elaboration of the "kinds" or "classes" and that of the relations or quantitative relationships always being on a par.

An example of transition will enable us to understand the thing. It is that of Observation 121 repeated: striking the doll in order to shake the parrot. In this example the child utilizes an intermediate (the doll) to act upon the objective (the parrot).

But the intermediate is still only one sort of substitute for the object. Wishing to grasp her parrot to shake or strike it and not succeeding, Jacqueline at first tries a procedure to draw it to her and pull the string. Hitherto we are still in the situation of Observations 120–121. But she remembers the doll located at the other end of the string, searches for it and applies to it exactly the schema which she wanted to apply to the parrot. She could, thereafter, have forgotten the parrot and only acted upon the doll; that is how she would have behaved at the stage of the pure secondary circular reactions. But she saves herself the trouble of shaking or striking the parrot and only uses the doll as a "means" toward this "end." This entails a putting of the two objects into relationship by assimilation of their schemata. This coördination, it is true, remains very primitive since it merely results from the fact that the intermediate has been assimilated to the objective. It is nevertheless real, since the two objects are distinguished from one another while being subsumed under an identical schema and are, thenceforth, placed in interrelation. In order that the coördination become effective, it will therefore suffice that this relation become spatialized and truly objectified.

This positive progress is achieved with Observations 127–130. How do Jacqueline and Laurent, for instance, come to utilize someone else's hand in order to act upon the object aimed at? On the one hand, the child already knows how to act upon the objective varied schemata for swinging the material, etc.). Besides, he also knows how to act by imitation (see Vol. II, Chap. III, §2) on someone else's hand and thus knows its power by analogy with that of his own hands. In order that he use this hand as a "means" when the final object is inaccessible, it will therefore suffice for him to assimilate their respective schemata to each other and through this very act to put the intermediate into physical relation with the objective. Now this reciprocal assimilation is easy: The hand of someone else being, like the objective itself, a source of activities which one can make last or reappear etc. by analogy with that of one's own hands, it is natural that, not succeeding in moving the objective at a distance, the child seeks to apply to this hand the schemata he expected to apply to the objective (on the one hand, the child has just seen

or has earlier seen the hand joined together with the objective and, moreover, the hand is situated between himself and the objective). Hence the child behaves, with respect to the hand and to the objective, as in Observation 121 repeated, with respect to the doll and the parrot. But reciprocal assimilation does not here proceed by simple fusion, as in the case in which two different schemata apply to the same object (Obs. 120–121) or yet as when interconnected objects are almost entirely assimilable to each other (Obs. 121 repeated); there only exists a more remote connection between the activity of someone else's hand and that of the object aimed at (swinging of the material, etc.). Thereafter, the two objects joined together in the same total schema (someone else's hand and the objective) are maintained distinct from each other and the reciprocal assimilation works no longer by fusion but by virtue of an inclusion of one of the schemata in the other.

Now, through the very fact that the schema of the intermediary (the other person's hand) and that of the objective are assimilated without being confused, the total schema resulting from this junction comprehends two separate objects which have to be put in relationship. Here the real novelty of the behavior patterns of this stage appears. The coördination, in a way formal, of the schemata due to their reciprocal assimilation, is accompanied by a physical connection established between the objects themselves; that is to say, by their being put in spatial, temporal and causal relationships. In other words, the fact that the hand of another person and the objective are mentally connected without being confused involves the construction of a real totality by bringing together the intermediate and the final object. This explains why the child does not limit himself to striking and shaking, etc., the hand of another person, but why this schema becomes differentiated, through accommodation, into a propulsive movement destined to bring this intermediate closer to the objective.

This last remark leads us to examine the most complex of the present behavior patterns: that in which the child comes to remove obstacles and differentiates toward this end the schemata of striking, etc., into movements of repulsion. We recall how this behavior pattern arose in Laurent: After having tried for a long

time to go over my hand, or a cushion which prevented him from grasping a visible objective, the child finally pushed away these obstacles, at first by striking them, then, little by little, by really removing them. Now, it is apparent that such a behavior pattern could not be explained as simply as the preceding coördinations. The schema of the obstacle could not be assimilated, without adding something, to that of the objective, since far from joining the object-screen to the objective, the child puts the former further away from the latter. But does not such a complete contrast of meaning conceal a true identity, the tendency to remove the obstacle constituting the negative of that which is, as positive, the utilization of intermediates? It is easy to see this. In effect, the capacity of coördinating the schemata involves that of contrasting them or of feeling them to be incompatible. Affirmation presupposes the power of negation; the latter is even often used before the former, affirmation sometimes remaining implicit before being willed. So also can the first intentional coördinations be very negative. They nevertheless presuppose a reciprocal assimilation. When the child who is trying to grasp the objective knocks against the intervening obstacle, this obstacle only acquires the meaning of an "object to be removed" relatively to this objective which it prevents the subject from reaching. Consequently, it is assimilated to the schema of the objective, but with a negative result. Just as a negation only exists as a function of a previous affirmation,[2] so also an exclusion necessarily rests on an earlier assimilation. Being on the road to the objective, the obstacle is assimilated to the schema of the latter (otherwise it would not be an obstacle), but through a negative connection (as in the judgment "this stone is not heavy," the quality "heavy" is connected to the subject "stone," but in order to be excluded from it). The obstacle is therefore conceived as a thing which takes the place of the objective (in this it is assimilated to it) and from which one must detach oneself in order to reach the latter. Moreover, the child begins by turning away from the obstacle

2 "An affirmative proposition conveys a judgment passed on an object: a negative proposition conveys a judgment passed on a judgment," said H. Bergson (*Creative Evolution,* 12th edit., p. 312) after Kant, Lotze, and Sigwart.

(he passes over it or beside it) which is the simplest form of negation. But the attempt does not succeed. The difficulty requires a special behavior pattern. It is then that the coördination of the schema of the obstacle with that of the objective takes place, but a negative coördination. It is a question of assimilating the obstacle to a schema which is suitable to it in the capacity of object and, at the same time, but negatively, to the final goal of the action, hence to the schema of the objective. In Laurent's case the transitional schema chosen was that of striking. Such a schema is suitable to the hand or the intervening cushion, and at the same time it involves the element of repulsion or of negation necessary to the pursuit of the objective. Moreover, and by virtue of this double assimilation, the object-screen is put in spatial relation to the objective, but in equally negative relation: It is to be put further away rather than brought nearer.—In the case, in short, of a screen completely masking the objective, the double assimilation is of the same nature but with this supplementary difficulty that it involves coördinating the schema related to the obstacle with the pursuit of an objective which has ceased to be directly perceived.

§3. ASSIMILATION, ACCOMMODATION AND ORGANIZATION PECULIAR TO THE MOBILE SCHEMATA. —The conclusion to which we are thus led is that the coördination of means to ends always involves a reciprocal assimilation of the present schemata as well as a correlative putting into relationships of the objects subsumed by these schemata. In the simplest cases this double assimilation is almost equivalent to a fusion and so calls to mind that which accounts for the coördination of the primary schemata. In other cases it can also remain truly reciprocal and give rise in that way to symmetrical series. For example, when the child taps his hoe with a thimble which he has first rubbed against another object, he applies the schema of striking to the shoe because inversely he made use of his feet shortly before in order to strike objects, etc. But in most cases, reciprocity leads to more complex relationships of inclusion, interference, negation, etc.

In order to understand this diversity we should emphasize

a fact to which we have already referred and which will assume great importance in the rest of our analyses: This is the functional analogy of the schemata of this stage (and of the following stages) with concepts, of their assimilations with judgments and of their coördinations with logical operations or reasonings.

From the point of view of *assimilation,* two complementary aspects characterize the schemata of which we have just spoken, when we compare them to the secondary schemata of the third stage from which they nevertheless derive: they are more mobile and consequently more generic. True, the secondary schemata encroaches upon all the characteristics of the "mobile" schemata peculiar to the present stage, but in a form to some degree more condensed (because undifferentiated) and consequently more rigid. This secondary schema is a complete totality of intercoordinated movements and functions every time the child perceives the objective in connection with which the schema was formed, or analogous objectives. For instance, the schema which consists in pulling a string in order to shake a hanging rattle presupposes a very complex coördination of movements and perceptions related to at least two objects (the string and the rattle). From this first point of view, it presages the schemata of the fourth stage which involve, as we have just stated, a putting into relationship of the objects themselves. Furthermore, the schema of "pulling the string," as has been seen, applies sequentially to a series of objects hanging from the hood (and not only to the first rattle) and even to objects presented from a distance, having no connection with the hood. From this second point of view as well, it presages "mobile" schemata which are capable of unlimited generalization. But, if one examines this closely, one notices that certain essential differences are in opposition to the simple secondary schema (that of the third stage), the same schema having become "mobile" during the present stage. At first the relations between objects, relations already utilized by the secondary schema, are given just as they are in the midst of the latter without the child's having elaborated them intentionally, whereas the relations due to the coördination of "mobile" schemata have really been constructed by the subject. Through the very fact that secondary circular reaction consists in simply reproducing a

result discovered by chance, the schema which proceeds from its use constitutes a global and indissoluble totality. It applies itself in one block and if it envelops certain relations between separate objects, these relations remain purely phenomenalistic and can only be taken out of their context to give rise to new constructions. Consequently there is no coördination between schemata and the internal coördinations of each schema remain invariable and hence rigid. The considerable progress in this respect made in the fourth stage is that the same schemata are made "mobile." They intercoördinate and consequently dissociate to regroup in a new way, the relations which they involve, each in itself, becoming capable of being extracted from their respective totalities to give rise to varied combinations. Now these various novelties have combined solidarity. In becoming "mobile"—that is to say, fit for new coördinations and syntheses—the secondary schemata become detached from their usual contents to apply themselves to a growing number of objects. From particular schemata with special or peculiar contents they accordingly become generic schemata with multiple contents.

It is in this sense that the coördination of the secondary schemata, and consequently their dissociations and regroupings, give rise to a system of "mobile" schemata whose functioning is comparable to that of the concepts or judgments of verbal or reflective intelligence. In effect, the subordination of means to ends is the equivalent, on the plane of practical intelligence, of the subordination of premises to conclusions, on the plane of logical intelligence. The mutual involvement of schemata, which the former presupposes, is therefore assimilable to that of concepts, which the latter utilizes. But, in order to understand such a comparison, we should consider separately the logic of classes and that of relations, that is to say, the two complementary systems of operations constituting every act of intelligence.

As we have emphasized, the coördination of the schemata which characterizes the behavior patterns of the present stage is always on a par with a putting into relationship of the objects themselves subsumed by these schemata. In other words, the relations which determine a given object are not only relations of appurtenance which permit it to be inserted in one or several

schemata, but all the relations which define it from spatial, temporal, causal, etc., points of view. For example, in order to remove a cushion which is an obstacle to grasping the objective, the child does not simply class the cushion in the schema of striking and assimilate, by inclusion, this schema to that of the end of the action, but he must understand that the obstacle is "in front of" the objective, that it must be removed "before" trying to grasp this objective, etc. In short, the coördination of the schemata presupposes the existence of a system of relations between objects and between schemata other than the simple inherent relations. Let us observe that the schemata themselves involve, in order to be formed, these same relations. Therefore a secondary schema is not only a sort of primitive "concept," it is a number of "relations" in the sense which we have just recalled. But it is only from the time when the schemata become "mobile" that the working of "relations" is clearly dissociated from that of "classes." It is after this fourth stage, as we shall see in Volume II, that the constitutive connections of the object and of space, of causality and of time become truly differentiated from the simple, practical and subjective connections bound to the schemata themselves.—Now such a distinction between schemata and relations recalls exactly the difference brought to light by modern logic between "classes" or "concepts" on the one hand, with their inherent connections (appurtenance and inclusion) and "relations," on the other hand, with their original operations of conversion and multiplication. In order to compare the processes of sensorimotor intelligence to those of reflective intelligence, it is fitting to respect such a classification.

First, with regard to classes or kinds, it is apparent that the "mobile schema," despite all the structural differences which separate it from these logical beings, is functionally analogous to them. Like them, the mobile schema always denotes one or several objects, by "appurtenance." Like them, the mobile schemata involve each other due to diversified connections, proceeding from pure "identification" to fitting in or "inclusion," and to intersections or "interferences." So too can the mobile schema function actively due to an assimilatory operation which constitutes the equivalent of a judgment, or be applied passively in the

manner of a concept. Moreover, it is self-evident that, to the extent that sequential assimilations condition each other (as in the subordination of means to ends), such totalities are equivalent to elementary reasonings.—These functional analogies of course do not at all imply an identity of structure between such practical schemata and the unities of reflective thought. Two essential differences contrast, from this second point of view, these two extreme states of infantile intelligence. In the first place, the sensorimotor schemata are not "reflective" but are projected into the things themselves; that is to say, the child is not conscious of the operations of his intelligence and considers the results of his own activity as being imposed merely by facts as such. In the second place, and concurrently, the involvements between the schemata are not yet regulated by a system of internal norms: The only verification of which the child is capable is of the type of *success* and not of *truth*.

Concerning the "relations" implied by the coördination of the mobile schemata, their situation is the same in comparison with that of the relations of reflective intelligence. First, a functional analogy: these relations are also capable of being arranged among themselves, of "multiplying," etc. But also a structural difference: as we shall see when studying, in Volume II, the development of the object, of spatial "groups" and causal or temporal series, the first differentiated relations which the sensorimotor intelligence uses are not "objective" but are centered on the self and dominated entirely by the same perspective.

Despite these structural differences, the "relations" of the fourth stage definitely involve, still more than those of the third, the element of quantity inherent in every system of relations. If it is a question of causal relations, the child perceives a proportionality between the intensity of the cause and that of the effect (one could cite here observations analogous to Obs. 106). If, on the contrary, it is a question of spatial, kinematic, or even temporal relations, the child, in order to put the objects into relationship to each other is obliged to distinguish them in order to arrange them, and this double factor of dissociation and of serialization presages the first rudiments of number.

It is easy to furnish a counterproof:

Observation 131.—At 0;9 (4) Laurent imitates the sounds which he knows how to make spontaneously. I say "papa" to him, he replies *papa* or *baba*. When I say "papa-papa" he replies *apapa* or *bababa*. When I say "papapapapapapapa" he replies *papapapa*, etc. There exists, consequently, a global evaluation of the number of syllables: The quantity corresponding to 2 is in any case distinguished from 3, 4 or 5, which are experienced as "many."

At 0;10 (14) Laurent repeats *pa* when I say "pa," *papa* for "papa" and *papapa* for a number of 4 or more than 4.

Such are the "mobile schemata" from the point of view of assimilation. It may thus be seen that the three aspects of assimilation which we have emphasized in connection with the primary and secondary schemata (repetition, recognition, and generalization) tend to join together or to combine more and more closely to the extent that the schemata become more supple and more complex. We shall not return to these distinctions except to study, in the following paragraphs, certain particular aspects of recognition and generalization peculiar to these schemata.

On the other hand, it is fitting to emphasize the process of *accommodation* characteristic of this stage because it is above all this mechanism of accommodation which will permit us to distinguish between the "application of familiar means to new situations" and the behavior patterns of the fifth stage and, in particular, the "discovery of new means through active experimentation."

One remembers that during the preceding stage accommodation consists simply in an effort to rediscover the conditions in which the action has discovered an interesting result. Such a form of accommodation, like that of the first two stages, is therefore, so to speak, dominated by assimilation: To the extent that the child tries to reproduce his acts he accommodates the schemata to the object which does not yet interest him in itself. During the next stage, on the contrary, the child will try to discover new properties in the objects. Moreover, and in correlation with these beginnings of experimentation, in order to attain ends which are unreachable by the simple coördination of already acquired schemata, he will elaborate new means. This elaboration presupposes, as we shall see, an accommodation which also

controls assimilation, that is to say, directs it as a function of the properties of the objects.

The accommodation characteristic of the fourth stage, whether it is made manifest in the "explorations" which we shall presently describe (§5), or in the "application of familiar schemata to new circumstances," is exactly intermediate between the two types. On the one hand, it is only to the extent that the coordination of schemata works, hence their reciprocal assimilation, that their accommodation to the objects progresses. Therein, the accommodation characteristic of the fourth stage simply prolongs that of the preceding ones. But, on the other hand, such accommodation, even subordinated to the working of assimilation, leads to the discovery of new relations between objects and so presages the accommodation of the fifth stage.

With regard to the first point, it can therefore be said that during this stage accommodation only progresses as a function of the coördination of the schemata. This is obvious in behavior patterns such as pushing back the obstacle, bringing another person's hand closer to the objective, etc. In such cases the child tries neither to reach a new goal relating to the object nor to discover a new procedure. He limits himself to intercoördinating two schemata according to the variations of that of the two which assigns an end to the action, and it is in order to operate this coordination that he is obliged to accommodate the transitional schema to the situation (pushing the object back instead of simply striking it, etc.). But, while doing this, he discovers in the course of the accommodation itself a new relation ("pushing back in order to," etc.) and in that the second point consists; that is to say, the outline of a more advanced accommodation which will develop during the fifth stage.

Hence the accommodation of this stage is more refined than that of the schemata hitherto under study, since the mobile schema applies to relations between external things and no longer only to things in their mere connection with the activity itself. Does this accommodation involve representation? If one understands representation to mean the capacity to confer upon things a meaning before the action which this meaning permits, it is apparent that representation exists. The act of looking under a

shawl for a shoe in order to strike it with a piece of wood (Obs. 129) is the prototype of this behavior pattern. But a capacity of this kind, which naturally increases as a function of the intentional nature of the acts, is already observable before this and goes back to the very beginnings of mental life. On the other hand, if one understands representation to mean the capacity to evoke by a sign or a symbolic image an absent object or an action not yet carried out, then nothing yet warrants asserting its existence. In order that he look for his shoe it is not necessary that the child picture it to himself or that he imagine the contact of the wood with the leather. It suffices that a sensorimotor schema lead him to his foot and that this schema be put to work, by virtue of the fact that the collision of the wood with an object is assimilated to the kicking of the foot.

It remains for us to draw a conclusion, clarifying the meaning of mobile schemata from the point of view of *organization*. As we have already emphasized often, organization or internal adaptation characterizes the interior of each schema as well as the interrelations of the schemata. Now the chief originality of this stage in comparison to the preceding stages is that the organization of the schemata among themselves is established for the first time in an explicit way and thereby reveals the internal organization of the schemata considered as totalities.

Moreover, it is good to distinguish, as before, between the totalities in the process of elaboration and the completed totalities. Concerning the former, what has been said in §2 with regard to the subordination of means to ends sufficiently shows the existence of categories to which we have simply alluded hitherto and which henceforth acquire a precise meaning: those of "value" and of "ideal" totality.

As long as the schemata are not intercoördinated but function each for itself, the child's judgments of value, so to speak, are almost entirely confused with his judgments of reality. More precisely, they are one with the activity inherent in each schema. When confronted by rattles, for instance, either the child shakes them and their value becomes thus identified with their property of being shaken, or else he loses interest in them, and their temporary lack of value becomes identified in the same way

with the subject's inaction. On the other hand, another person's hand, in the behavior patterns of the present stage, is no longer characterized by one value only, or by the pair "value" and "nonvalue"; it can be considered either as an obstacle, or as a useful intermediate, or else as an end in itself according as to whether it is put into relationship with an objective from which it separates the child, or with an objective upon which it can act or again, is envisaged in itself. It thus assumes a series of different values according to the way in which it is utilized as means in view of different ends. With respect to these ends, it can be said that, to the extent that they require, in order to be attained, a more complex coördination of the means for use, they are more remote and so determine more "ideal" totalities. Hence the categories of "value" and of "ideal" become much more clearly differentiated during this stage than when the means and ends were enveloped within the same schemata as was the case with respect to the schemata of the third stage not intercoördinated.

Regarding the completed organizations, these are characterized by the two complementary modes of "totality" and "relation" which also are henceforth revealed more clearly than before.

As far as "totality" is concerned, we have already emphasized that every schema of assimilation constitutes a true totality, that is to say, an ensemble of sensorimotor elements mutually dependent or unable to function without each other. It is due to the fact that the schemata present this kind of structure that mental assimilation is possible and any object whatever can be incorporated or serve as aliment to a given schema. Inversely we have seen that the existence of this "total" structure was connected with the act of assimilation, a sensorimotor coalescence only constituting a true totality if it is capable of conversation or repetition due to the action of assimilation. "Total" organization and assimilation are consequently two aspects of the same reality, one internal and the other external. How can we surpass this analysis and grasp the intimate mechanism of this organization? This is precisely what the behavior patterns of this stage make possible by showing us simultaneously how the

schemata are organized in relation to each other and in which respects this coördination corresponds to their internal organization.

The important thing that the behavior patterns of the present stage teach us is that the coördination of the schemata is correlative to their differentiation; in other words, that the organization works by complementary regroupings and dissociations. Thus pushing back the obstacle in order to attain the objective presupposes a coördination between the schema of striking and that of grasping, but such a coördination that from the schema of striking is extracted that of "pushing back" which is immanent in it. Now such correlation between external coördination and internal differentiation reveals a fundamental character of organization. That is that every schema, insomuch as it is a totality, is pregnant with a series of schemata virtually contained within it, each organized totality thus being, not composed of totalities of a lower grade, but a possible source of such formations. These virtual totalities are not encased and preformed in the combined totality but result from it precisely to the extent that the combined totalities intercoördinate and thereby become differentiated.

An organized totality consequently never merely constitutes but one unity relating to the grade under consideration. Let it be said in passing, that explains why psychological assimilation or organization are of the same nature as physiological assimilation or organization, their grade alone contrasting them to these latter. Thus every act of intellectual assimilation presupposes a series of lower-grade assimilations protracted to the plane of truly vital assimilation. Moreover, if one remains on the psychological plane, and considers this relation between the coördination or external organization of the schemata and the differentiation revealing their internal organization, one understands why, in the rest of mental development, the individual's every external conquest founded upon a new coördination will reëcho if the acquisition of consciousness functions normally (that is to say, if no obstacle looms to impede it) in a thought about the self or an analysis of the internal mechanism of organization.

That parenthesis aside, the coördination of the schemata characteristic of this stage constitutes a new organization which forms above the schemata a totality actualizing the balance existing between them since the preceding stages. Now, as we see, this external totality only extends the internal totalities considered hitherto. Furthermore, the very fact that this external totality is constructed by virtue of a reciprocal assimilation of the existing schemata brings to light the existence, until now merely anticipated, of a close relationship between the categories of "totality" and "reciprocity." In effect, the fundamental property of every "totality" is that its elements maintain among themselves the relations of reciprocity.

The category of relation (reciprocity) is as fundamental to the human mind as that of totality. If the purpose of this book did not prohibit us from making digressions in the realm of the psychology of intelligence in general, this would be the time to show that the so-called "identification" in which a famous philosophy of science envisages the characteristic process of the "progress of thought" never gives itself an aim of forming relations of identity but rather of forming systems of reciprocal relations. The ultimate fact, in the analysis of intelligence, consequently is not the static affirmation of identity, but the process by which the mind distinguishes between two terms by putting them into relationship and forms this relation while solidifying them. Reciprocity is, accordingly, a dynastic identity of which the act of coördination is on a par with that of differentiation.

Now, considered thus, reciprocity is the fundamental relation which one finds within each totality. When the totality is constructed by coördination of two or more schemata, the relations existing between these schemata are relations of reciprocity, whereas the relations established between the objects subsumed by these schemata themselves constitute reciprocal relations. With regard to the internal structure of the schemata, the same applies: The parts of an organized whole necessarily maintain among themselves relations of reciprocity. We shall examine this more closely in Volume II when studying the objective, spatial or causal structures. Concerning space, in particular, it is obvious

that every motor totality tends to form a "group" whose elements are defined precisely by reciprocity.[3]

But, to be sure, true totality and complete reciprocity are only limited cases that every schema and every ensemble of schemata tend to realize to the extent that they actually tend toward a state of equilibrium. It is this difference between the state of fact and the highest state which justifies the distinction between real totalities and ideal totalities which is characteristic of the categories of organization.

§4. THE RECOGNITION OF SIGNS AND THEIR UTILIZATION IN PREVISION.—It goes without saying that an operation as complex as that of the coördination of mobile schemata involves a use of recognitory assimilation as well as of reproductive or generalizing assimilation. Moreover, it is useless to study separately the acts of repetition of which the child becomes capable at this stage. On the other hand, it is interesting to try to describe how the recognition of signs which the "application of familiar schemata to new situations" presupposes, surpasses this behavior pattern and can give rise to previsions independent of the action in progress.

It is only natural that prevision should become independent of action at the present stage and so engender a sort of concrete prevision, precisely because the constitution of the mobile schemata and their coördination attest to the power which the child acquires of dissociating the thitherto global totalities and of again combining their elements. But it is still necessary to understand through analysis of the facts how this liberation of meanings works and in which respect the signs peculiar to this stage differ from the various types of signals which have been under study.

[3] Thus, from the logical point of view, the difficulty inherent in the concept of identification is removed. Nothing formally distinguishes false from true identification and the experimental proof necessary for this distinction therefore either remains foreign to the mechanism of reason or else bound together with internal identifications whose validity one does not know how to demonstrate. On the other hand, a system of reciprocal relations obtains its security simultaneously from its internal structure and from the facts which it has succeeded in coördinating. Its constitution is a proof of its value since in itself it comprises an element of verification.

We recall that to each of the preceding stages a particular type of signs and meanings corresponds. To the reflex stage corresponds a type of recognitions and of meanings immanent in the use of the reflex. The child recognizes whether he is sucking without an object, whether he is sucking a tegument or is really nursing. The primary circular reactions then engender a second type of signs, the "signals" acquired by insertion of a new perceptive element in the familiar schemata. Whether they are simple or derived from the coördination of heterogeneous schemata, the signals thus form part of the act which they set in motion in the manner of direct perception of the objective. So it is that a heard sound provokes the search for the corresponding image, etc. As we have seen, with the secondary reactions a third type of signs begins which are intermediate between the "signal" and the "sign," that is to say, form the transition between the sign which simply releases the action and that which permits independent prevision of the act. For instance, when a child hears a bed creak and by this sign recognizes the presence of his mother who will be able to feed him (Obs. 108) he limits himself to inserting a new perception in the complex schemata coördinated with sucking and in that respect the sign is still only a "signal" but he is on the way to attributing to his mother an activity independent of himself and, to this extent, the prevision under consideration presages the true "sign."

This decisive progress, which consists in bringing prevision to bear upon events independent of the action itself, is achieved during the fourth stage in correlation with the objectifying of the relations which characterize this stage in general. In other words, a fourth type of sign is now constituted which we shall call the actual "sign," which permits the child to foresee, not only an event connected with his action, but also any event conceived as being independent and connected with the activity of the object.

Observation 132.—At 0;8 (6) Laurent recognizes by a certain noise caused by air that he is nearing the end of his feeding and, instead of insisting on drinking to the last drop, he rejects his bottle. Such a behavior pattern still pertains to the recognition of "signals" since the perception of sound is inserted in the schemata of sucking, but the fact

that, despite his hunger, Laurent at once resigns himself and rejects his bottle seems to us to show that he foresees the events as function of the object itself as much as of the action. He knows that the bottle is emptying although a few grams of milk still remain.

At 0;9 (8) likewise, I observe that Laurent constantly follows my whereabouts in the room without either seeing me or hearing my voice. His mother's voice or his sisters' voices in the hall or the neighboring rooms set no reaction in motion, whereas the slightest creak of my table or armchair starts his search or significant sounds of his voice. Hence he knows I am there and notes my presence as well as my changes of location by all these signs. Now this interest is independent of the hour of his meals.

Observation 133.—At 0;9 (15) Jacqueline wails or cries when she sees the person seated next to her get up or move away a little (giving the impression of leaving).

At 0;9 (16) she discovers more complex signs during a meal than previously. She likes the grape juice in a glass but not the soup in a bowl. She watches her mother's activity. When the spoon comes out of the glass she opens her mouth wide, whereas when it comes from the bowl, her mouth remains closed. Her mother then tries to lead her to make a mistake by taking a spoon from the bowl and passing it by the glass before offering it to Jacqueline. But she is not fooled. At 0;9 (18) Jacqueline no longer even needs to look at the spoon. She notes by the sound whether the spoonful comes from the glass or from the bowl and obstinately closes her mouth in the latter case. At 0;10 (26) Jacqueline also refuses her soup. But her mother, before holding out the spoon to her, strikes it against a silver bowl containing stewed fruit. Jacqueline is fooled this time and opens her mouth due to not having watched the move and to having depended on the sound alone.

At 1;1 (10) she has a slight scratch which is disinfected with alcohol. She cries, chiefly from fear. Subsequently, as soon as she again sees the bottle of alcohol she recommences to cry, knowing what is in store for her. Two days later, same reaction, as soon as she sees the bottle and even before it is opened.

Observation 133 repeated.—Lucienne has revealed most of the same reactions. Thus at 0;8 (23) she also closes her mouth to the spoonsful coming from the bowl (of soup) and opens it to those coming from the glass (of fruit juice). At 0;10 (19) she wails as soon as the person with whom she is playing gives the impression of leaving. It is enough to turn oneself away from her, without rising, to worry her!

Observation 134.—At 0;10 (26) Jacqueline has watched for a long time a red balloon attached to the handle of her bassinet and floating toward the ceiling. At a given moment I detach the balloon without her seeing

me. Shortly afterward she looks at the handle at the usual place, looks for the balloon and, not finding it, examines the ceiling. At 0;11 (14) she cries when I take a mirror from her hand without her watching this operation. She therefore knows she will not see it any more.

In a general way, from 0;11 Jacqueline cries when one pretends to take an object away from her because she awaits its disappearance. This kind of understanding is related to the development of the behavior patterns of searching for the absent object (see Vol. II, Chap. I).

So it is that from 0;11 (15) Jacqueline cries as soon as her mother puts her hat on. This is not due to fear or anxiety as before but due to the certainty of the departure.

Observation 135.—It is fitting, moreover, to class among the signs of the fourth stage those which the child uses to identify his own features by those which correspond to them on the face of another person.

We shall study these signs in connection with the acquisition of imitation (see "The Genesis of Imitation") and shall see that in this regard they could not be confused with simple "signals."

For example, before 0;10 (7) Laurent has not succeeded in imitating the act of sticking out the tongue. Now he sticks it out spontaneously while accompanying this movement with a sound of saliva. I then imitate him and he in turn imitates me. But the imitation fails when I stick my tongue out in silence. From 0;10 (10), on the contrary, it suffices that I stick out my tongue, even without the accompanying sound, for him to imitate me. The sound of saliva has consequently served as a sign to permit him to identify his tongue by mine.

That is not a question of a "signal" setting the act in motion since the sound alone does not induce the child to stick his tongue out, but rather of a sign enabling the subject to put a group of factors observed in someone else into relationship with the corresponding parts of his own body. The sign again is brought to bear upon events independent of the self.

The principal novelty of these facts compared with those of the preceding levels is that they presuppose a prevision independent of the action in progress. When the child notes the presence of persons independently of his meals, wails on seeing someone rise, turns on hearing a breath, recognizes an alcohol bottle, etc., he performs an operation more difficult than connecting a signal with the schemata relating to the meal (third type), and trying to see the thing he hears or to grasp an object which has grazed his fingers (second type). In these last three behavior patterns the signal has exclusively a practical meaning; that is to say, its only effect is to set in motion the action of a

schema of assimilation to which it is connected by a constant and necessary bond. True, this bond presages prevision, particularly when the intermediates between the signal and the act are complex as in the third type, but this prevision remains connected with the immediate action and is not yet dissociated from it. On the contrary, the behavior patterns of the fourth type reveal a more advanced differentiation between prevision and action. Undoubtedly all the intermediates are given between this level and the preceding ones and some of the behavior patterns cited prolong without adding anything to those of the third type and even of the second. Thus the fact of marking by the sound of the spoon or the sight of the receptacle that which the spoon will contain merely constitutes an extension of the schemata of coördination between sight and eating. But, although the signs of the present type are sometimes derived from more or less habitual schemata they can henceforth enter into new behavior patterns in the capacity of components. If Jacqueline, for example, foresees what the content of the spoons will be, this is in order to reject her soup and only accept the fruit juice. And, above all, it is remarkable that henceforth prevision becomes possible in connection with facts rarely or very recently observed, or even in connection with the actions of other people. Thus to foresee the departure of a person when he rises or even turns away is a prevision which is already very much detached from the action in progress; and to manifest aversion for a bottle of alcohol is to utilize a rare or very recently acquired connection.

In short the novelty of these behavior patterns, although difficult to grasp precisely, is made manifest in that prevision becomes objectified and detached from simply circular reaction. Thus it may be seen that these behavior patterns are in close relation with those of Observations 120–130, that is to say, with the application of familiar schemata to new situations. This application of familiar schemata also presupposes prevision, that is to say, the utilization of signs. And, primarily, the relationship between these two groups of facts consists in that in both cases the schemata utilized are "mobile," in other words, subject to multiple combinations. In the case of Observations 120–130, this mobility of the schemata is recognizable by the fact that the

familiar schemata, habitually constituting ends in themselves, serve momentarily as means for a new end. In the case of the present observations they are, on the contrary, signs ordinarily embodied in general schemata which are henceforth understood in themselves and utilized separately in order to give rise to an independent prevision. Thus the signs consisting of creaks of the table or of chairs, of the person rising, etc., were acquired, like most of the others, as function of the schemata of the meal. They are henceforth utilized in any circumstance whatever. In both cases, consequently, whether it is a question of the utilization of familiar schemata in the pursuit of a new end or of the utilization of signs in a new independent situation, the schemata become mobile and subject to unlimited combinations. The only difference between the two behavior patterns is that in Observations 120–130 there is seeking and invention of a means, whereas in the present observations there is chiefly comprehension, but, in both cases, the process of assimilation is the same.

Finally let us remark, before proceeding to the new facts, that the term prevision which we have used must not create an illusion or evoke more than concrete expectation. Deduction does not yet exist, because there is doubtless still no "representation." When Jacqueline expects to see a person where a door is opening, or fruit juice in a spoon coming out of a certain receptacle, it is not necessary, in order that there be understanding of these signs and consequently prevision, that she picture these objects to herself in their absence. It is enough that the sign set in motion a certain attitude of expectation and a certain schema of recognition of persons or of food. Thus the sight of obstacles in a cluttered street permits us to steer a bicycle or an automobile with enough prevision to adapt ourselves to the barely outlined movements of another person without our needing to picture them to ourselves in detail. It is during the later stages that true deduction, with representation, will be superposed on these elementary meanings. But we are not yet there and have not even reached the level at which the meaning of sensorial signs is constituted by the "object" itself, with its characteristics of permanence and solidity.

§5. THE EXPLORATION OF NEW OBJECTS AND PHENOMENA AND THE "DERIVED" SECONDARY REACTIONS.—Granted what we have seen of the application of familiar schemata to new situations and of the understanding of signs independently of the action in progress, we can ask what the child will do when confronted by objects or phenomena which are entirely new to him. Such objects could not set in motion behavior patterns analogous to those in Observations 120–130—that is to say, the application of familiar means to a new end—precisely because the child, when confronted by such objects, cannot set himself any definite goal except "understanding" them. Moreover the understanding of signs, while playing a role in this assimilation, would not suffice to account for it. Then what will happen? We are now to encounter a very significant behavior pattern which, more than any other, will make us grasp the importance of assimilation through mobile schemata. The child will try, by virtue of a sort of "generalizing assimilation," to make the new object enter into each of his habitual schemata, one by one. In other words, the child will try to "understand" the nature of the new object, and as comprehension is still confused with sensorimotor or practical assimilation, he will limit himself to applying each of his schemata to the object. But in doing this he will not, as in the third stage, set the schema as goal and the object as means. On the contrary, the schema will be, so to speak, the instrument of comprehension, whereas the object will remain the goal or intention of this comprehension. To put it more simply, the child will apply himself in acts to the operation to which older subjects apply themselves in words: He will define the object by its use.

Here are examples of these behavior patterns:

Observation 136.—At 0;8 (16) Jacqueline grasps an unfamiliar cigarette case which I present to her. At first she examines it very attentively, turns it over, then holds it in both hands while making the sound *apff* (a kind of hiss which she usually makes in the presence of people). After that she rubs it against the wicker of her bassinet (habitual movement of her right hand, Obs. 104), then draws herself up while looking at it (Obs. 115), then swings it above her and finally puts it into her mouth.

A ball of wool: She looks at it, turns it over, feels it, squeezes it,

then lets it go, accidentally. I rest the ball on her stomach. Jacqueline draws herself up three or four times while looking at it, then feels its surface again, pulls the string while staring at it, shakes it in all directions and finally again goes *apff*.

A tin box: Jacqueline grasps it, examines it all over, feels it, then goes *apff*. Afterward she shakes it, then hears a sound when striking it. She then strikes indefinitely, then draws herself up while looking at it and striking it. Then she examines it at length from the side while holding it in the air and going *apff*. Afterward she emits some sounds such as *adda, bva,* etc., while brandishing it and turning it in all directions. Finally, she rubs it against the wicker of the bassinet again going *apff*.

At 0;9 (4) she looks for a long time at a straw table mat, then delicately touches the edge, grows bold enough to touch it, then grasps it, holds it in the air while slowly displacing it, shakes it and ends by tapping it with her other hand. This behavior is accompanied by an expression of expectation and then of satisfaction. Jacqueline finally expresses her feelings by going *apff*. Then she rubs the object against the edge of the bassinet, etc.

Observation 137.—At 0;8 (29) Laurent examines at length a notebook which he has just grasped. He transfers it from one hand to the other while turning it in all directions, touches the cover, then one of the corners, then the cover again, and finally the edge. Afterward he shakes himself, shakes his head while looking at it, displaces it more slowly with a wide motion and ends by rubbing it against the side of the bassinet. He then observes that in rubbing against the wicker the notebook does not produce the usual effect (sound? consistency?) and examines the contact most attentively while rubbing more gently.

At 0;9 (6) he examines a series of new objects which I present to him in sequence: a wooden figure of a man with movable feet, a wooden toucan 7 cm. high, a matchbox case, a wooden elephant (10 cm. long) and a beaded purse. I observe four quite constant reactions. (1) In the first place a long visual exploration: Laurent looks at the objects which are at first immobile, then looks at them very rapidly (while transferring them from one hand to the other). He seems to study their various surfaces or perspectives. In particular, he folds the purse in two, unfolds and refolds it in order to study the transformations; as soon as he sees the hinge, he turns the object over in order to see it full face, etc. (2) After the visual exploration a tactile exploration begins: He feels the object, especially the toucan's beak, the little man's feet, and gently passes his fingers over the unevenness of the object (the carved wood of the toucan, the beads of the purse, etc.), he scratches certain places (the case of the box, the smooth wood of the elephant, etc.). (3) He slowly moves the object in space: chiefly movements perpendicular to his glance, but already perhaps desired displacements in

depth. (4) Only at last does he try the various familiar schemata, using them each in turn with a sort of prudence, as though studying the effect produced. He shakes them, strikes them, swings them, rubs them against the bassinet, draws himself up, shakes his head, sucks them, etc.

At 0;9 (21) same reactions in the presence of a big red crayon: He touches its point with interest, then turns the crayon around many times, strikes it, rubs it, shakes it, scratches it, etc. At 0;9 (26) the same applies to a bath thermometer: He looks at it, scratches it, shakes himself in front of it, then shakes it, turns it over, feels the handle around which he finally puts his hand, sucks the end of the handle (without wishing to suck, but "in order to see"), takes it out of his mouth, follows the thermometer itself with his left palm, shakes himself again, examines the glass part, touches it and scratches it, looks at the string and feels it, etc.

At 0;9 (30) same reactions in the presence of a new plush cat: He turns it in all directions, touches its head cautiously, touches the ribbon, the feet, discovers a cardboard disc attached to its tail and scratches it with his nail. He ends by striking the cat, swinging it in space; he shakes himself while looking at it, says *papa, baba,* etc.

Before discussing these facts of "exploration of new objects," let us again see how they can give rise to new "secondary circular reactions," but "derived" ones, when the exploration by chance results in the discovery of an unknown phenomenon. As we have already emphasized (see Chap. III, §4, Obs. 119), new secondary reactions are formed at every age (and not only during the third stage), but in the midst of new contexts. This is precisely what happens in the course of the behavior patterns of "exploration." It suffices that an unforeseen result has been fortuitously set in motion for it to give rise to immediate and simple repetition which in turn results in the elaboration of a schema. Here are some examples:

Observation 138.—We have already seen (Obs. 103) how the schemata of striking hanging objects is derived from simpler schemata (grasping, etc.) and has itself given rise to more complex schemata (hitting with one hand an object held in the other, etc.). Now we shall see how, with respect to Laurent, Lucienne, and Jacqueline, at about the same time, this same schema of striking engendered a new schema of "setting swinging in motion" and how this latter schema was discovered in the course of actual "explorations."

At 0;8 (30) Laurent, for the first time, examines a wooden hen from which hangs a ball which actuates the animal's head by its movements. First he sees the hen, touches it, etc., then examines the ball, feels it

and, seeing it move, strikes it immediately; he then attentively watches it swing and then studies it for its own sake; he simply sets it in motion, more and more gently. Then his attention is brought to bear on the accompanying movement of the hen and he swings the ball while watching the hen.

Observation 138 repeated.—Lucienne, at 0;8 (10), likewise, examines a new doll which I hang from the hood of her bassinet. She looks at it for a long time, touches it, then feels it by touching its feet, clothes, head, etc. She then ventures to grasp it, which makes the hood sway. She then pulls the doll while watching the effects of this movement. Then she returns to the doll, holds it in one hand while striking it with the other, sucks it and shakes it while holding it above her and finally shakes it by moving its legs.

Afterwards she strikes it without holding it, then grasps the string by which it hangs and gently swings it with the other hand. She then becomes very much interested in this slight swinging movement which is new to her and repeats it indefinitely.

Observation 139.—At 0;8 (9) Jacqueline looks at a hanging necktie which she has never seen. Her hands move around it and touch it very warily. She grasps it and feels its surface. At a certain moment part of the necktie escapes her grasp: visible anxiety, then, when the phenomenon is repeated, satisfaction and, almost immediately after, something which resembles an experience of letting go and recapturing.

That evening, Jacqueline is lying on her back having at her right a rag which is drying on a line. She tries to grasp it, then immediately swings it; then she draws it toward her, lets it go and watches it oscillate. When it stops, she rebegins, evincing great interest in this movement.

At 0;8 (12) Jacqueline at first tries to make another hanging necktie swing. She grasps it very gingerly, lets it go, etc., in a regular and continuous oscillating movement.

At 0;8 (13) Jacqueline watches her mother who is swinging the flounce on the bassinet. As soon as she stops, Jacqueline pushes her hand to make her continue. Then she herself grasps the flounce and imitates the movement.—That evening Jacqueline swings a hanging doll in the same way, with great delicacy.

At 0;8 (26) Jacqueline feels and explores the surface of a lamp shade which then begins to swing. She waits until this movement has almost stopped (after many oscillations) in order to start the object moving again with one blow of her hand. This reaction reappears regularly the days following as soon as she is brought near the lamp in her room. I again observe it at 0;9 (5), etc.

At 0;9 (6) Jacqueline accidentally starts a sudden movement of the lamp shade by striking it from the inside. She then immediately tries

to rediscover the result by placing her hand, not with the palm against the material as usual, but with her palm in the air. She gropes in this way, very warily, becomes entangled in the fringe, then is completely successful.

Observation 140.—Here is a second example of "exploration" leading to a "derived secondary circular reaction" and to a new schema, that of "letting go." This second example is particularly instructive since it presages the most important of the "tertiary circular reactions" and thus will permit us to form the transition between the behavior patterns of the present stage and those of the fifth stage.

At 0;10 (2) Laurent examines an empty (white metal) case of shaving soap which he sees for the first time. He begins by turning it in all directions while passing it from one hand to the other as he did with the objects in Observation 137. But the object being slippery and hard to handle, slips from his hand two or three times. Then Laurent, struck by this phenomenon, applies himself to reproducing it a certain number of times. At first I had some difficulty in deciding whether it was indeed an intentional act for Laurent begins each time by holding the case for a moment and turning it over before letting it go. But then it fell more and more frequently and above all systematically as revealed by the following statements concerning the procedures employed by Laurent for relinquishing the objective.

In effect, what interests Laurent at the beginning of this behavior pattern is not the trajectory of the object—that is to say the objective phenomenon of its fall—but the very act of letting go. Sometimes Laurent delicately opens his hand (his palm upturned), and the case rolls along his fingers, sometimes Laurent turns his hand over (vertically) and the case falls backward between the thumb and index finger which are separated, sometimes Laurent simply opens his hand (palm downward) and the object falls.

It is this characteristic of Laurent's behavior pattern which permits us still to class it among the secondary circular reactions and not among the tertiary reactions. The "tertiary" reaction will begin, in effect, at the time when Laurent will study the trajectory of the object and to do this will organize a true "experiment in order to see." He will vary the conditions, relinquish the object in different situations, watch it, try to recapture it, etc. For the time being, on the contrary, he limits himself to repeating the same gestures and is only interested in his own action which certainly constitutes a "secondary" reaction.

Above all, for several days Laurent has only utilized this schema of relinquishing in connection with the same object, the case of shaving soap. At 0;10 (3), for example, that is to say, the next day, he immediately makes use of the case in order to repeat his behavior of the previous day, but he does not let go either of the little box which he manipulates for a long time or of his plush cat, etc. At 0;10 (4) same

reaction. At 0;10 (5) he twice lets fall a small bottle (new to him) which fell from his hands fortuitously the first time. It is only at 0;10 (10) that he begins to throw everything to the ground, but then at the same time he becomes interested in the trajectories of the falling objects and so inaugurates his "tertiary circular reactions."

Hence it is fitting to terminate this observation by drawing the conclusion that this secondary circular reaction is apparently "derived" from "exploration" of the case of soap and thus reveals no relationship to the transitional schema of removing or letting fall the object as studied in Observation 125. On the other hand we have seen how the present schema has subsequently been utilized as a "means," in Observation 130.

Behavior patterns of this sort are found exactly between the generalization of secondary schemata in the presence of new objects (Obs. 110–111) and the "tertiary circular reactions," hence between the behavior patterns of the third stage and those of the fifth.

Like the "generalization of secondary schemata" the present behavior patterns consist, in effect, of applying acquired schemata to new objects or phenomena. Just as, at 4 to 6 months, the child strikes, shakes, rubs, etc., the unfamiliar object which is offered to him, so also, at 8 to 10 months, he displaces it, swings it, shakes it, etc. The exploration of which we now speak therefore prolongs without adding anything to the generalization of the schemata to such a degree that all the transitions are exhibited between the two behavior patterns and it is impossible to draw a definite boundary between them. Nevertheless they do not seem to us to be identical because, however delicate the evaluation of such characteristics may be, their orientation is different. At the beginning of the third stage, in effect, the new object does not interest the child at all as a novelty. Its novelty only arrests his curiosity fleetingly and the object immediately serves as aliment to habitual schemata. Interest is consequently not centered on the object as such but on its utilization. On the other hand, when the 8-month-old child examines a cigarette case or a hanging necktie everything transpires as though such objects presented a problem to his mind, as though he were trying to "understand." Not only does he look at such objects for a much longer time than the 4- to 5-month-old child before proceeding to acts, but

furthermore, he engages in an ensemble of exploratory movements relating to the object and not to himself. He feels, explores the surface, the edges, turns over and slowly displaces, etc., and the last behavior patterns are very significant of a new attitude. The unfamiliar obviously represents to the child an external reality, to which he must adapt himself and no longer a substance which is pliable at will or a simple aliment for the activity itself. Finally comes the application of habitual schemata to this reality. But in trying out each of his schemata in turn, the child at this stage gives more the impression of making an experiment than of generalizing his behavior patterns: He tries to "understand."

In other words, it is as though the child said to himself when confronted by the new object: "What is this thing? I see it, hear it, grasp it, feel it, turn it over, without recognizing it: what more can I do with it?" And as understanding, at this age, is purely practical or sensorimotor and the only concepts which exist as yet are mobile schemata, the child tries to make the new object enter into each of his schemata in order to see in which respect they suit it. As we have already noted, such behavior patterns constitute the functional equivalent of "definitions through use" so important to a child's verbal intelligence.

Concerning the secondary circular reactions which can "derive" from this exploration, their genesis is easy to understand when a new phenomenon appears unexpectedly. In effect, when the child tries to assimilate a new object to his earlier schemata, two things can happen. Either the object comes up to expectation and so fits the schemata tried out, and then adaptation is acquired. The new doll can indeed be swung, shaken, rubbed, etc., and the child is satisfied. Or else, on the contrary, the object resists and reveals properties hitherto unknown, but then the child behaves as he has always done in a similar case. He tries to rediscover what he has just discovered by chance and merely repeats the movements which led him to this fortuitous discovery. So it is that, when trying to explore the nature of a hanging necktie or a lamp shade, Jacqueline discovers the phenomenon of the spontaneous swinging of these objects. Until then she was only familiar with the swinging of toys hanging from the hood of her bassinet, a swinging prolonged or main-

tained by her schemata of "striking" or "drawing herself up," etc. (Obs. 103). Now on the contrary, she becomes aware of a swinging which is in a way inherent in the object, hence of a new phenomenon. She immediately studies it and, to do this, applies herself to reproducing it indefinitely. The same is true of Laurent (Obs. 140) when he discovers the possibility of "letting go" of objects.

New behavior patterns of this kind obviously prepare the "tertiary circular reactions" (such as throwing and picking up, causing sliding, rolling, splashing, etc.) which will develop during the fifth stage and will constitute the first real experimentations of which the child is capable. The "tertiary circular reaction" is, in effect, an "experiment in order to see," which no longer consists in simply reproducing an interesting result but in making it vary in the very course of repetition. At this level of development the object definitely becomes independent of the action. It is the source of completely autonomous activities which the child studies from the outside, as he is henceforth oriented toward novelty as such.

But if these actions of "swinging" and "letting go" which we have seen appear in the course of the "explorations" of this stage presage this kind of "experiments in order to see" it would not yet be possible to identify them entirely by the latter. Not only, in effect, does the child always limit himself to "reproducing" what he observes and not to innovating but furthermore, as we shall state primarily in Volume II, the "object" characteristic of this fourth stage remains partly dependent on the action.

At last it may be understood why we class facts of this kind in the same stage as the "application of familiar means to new situations." Like the intelligent behavior patterns under study above (Obs. 120–130), the latter consist essentially in adapting early schemata to current circumstances. In a sense, it is true, such applications protract without adding anything to secondary circular reactions but, contrary to the "procedures to make an interesting spectacle last," the present behavior patterns have as their function not only to make the given result "last" but to adjust to novelty.

Moreover these facts bring to mind the understanding of

signs, which was treated previously. In the course of attempts at assimilation of new objects many signals and signs intervene which guide the child in his choice of schemata to apply. So it is that in Observation 139 the fact that the object is mobile or immobile, hanging or handed to him, orients his search. We can again remark, in this connection, that the fewer schemata the child has at his disposition the less useful is the sign because assimilation is immediate and global; whereas the more the schemata multiply, the more the organization of signs becomes complicated and necessary to action.

But the difference between the present facts and the preceding ones is that the orientation of the assimilatory effort is different. There is an effort to understand and no longer to invent, or even to foresee. In the case of Observations 120–130 the child, from the outset of the act has the intention of applying a given schema to the object, and the problem is to know which intermediate schemata are suitable to serve as means to this end. There is therefore an attempt to invent and comprehension only intervenes to facilitate invention. In the present case, on the contrary, the problem is to know which schemata are suited to the object. Consequently there exists an attempt to comprehend and, to the extent that invention intervenes in the form of the search for schemata, it is simply in order to facilitate comprehension. With regard to the recognition of signs of which we have spoken in connection with Observations 132–135, it constitutes an intermediate behavior pattern in this respect: It is comprehension, since it is immediate assimilation of a datum to a schema, but this comprehension is oriented toward prevision; that is to say, toward the utilization of the same schema with the intention of assimilating coming events and in this sense it is invention.

In a general way, the behavior patterns peculiar to this fourth stage manifest a real unity. Intercoördination of the schemata and their adaptation to the object are their constant characteristics and these two characteristics are complementary. The "application of familiar means to new situations" is determined by the coördination of two groups of schemata of which some serve as ends and the others as means: whence a more ac-

curate adjustment of the latter to the circumstances motivating this union. The "signs" belonging to this stage permit a prevision which begins to become detached from the action itself. Consequently there exist simultaneously the application of familiar schemata to new situations and progress in adaptation to the date of perception. The same applies to the "explorations" of which we have just spoken. Doubtless the last of these behavior patterns does not necessarily presuppose coördinations between separate schemata; it only implies the application of the schemata to new objects. But, like the first behavior patterns, the last one permits a real accommodation of the schema to the object and no longer merely a global application as in the third stage.

CHAPTER V

THE FIFTH STAGE:

The "Tertiary Circular Reaction" and the "Discovery of New Means Through Active Experimentation"

During the third of the stages which we have delineated, the child, by manipulating things, constructed a series of simple schemata due to the "secondary circular reaction," such as "shaking," "rubbing," etc. These schemata, while not yet intercoördinated, nevertheless comprise, each in itself, an organization of movements and perceptions and, consequently, a commencement of putting objects into interrelations. But this organization, remaining within each schema, does not involve a clear distinction between "means" and "ends" and this putting into relationship, for the same reason, remains entirely practical and does not lead to the elaboration of actual "objects."

In the course of the fourth stage, which immediately precedes this one, the secondary schemata become intercoördinated and so give rise to the complex actions which we have designated "applications of familiar means to new situations." This coordination of schemata, which clearly differentiates "means" from "ends" and so characterizes the first acts of intelligence properly so-called, insures a new putting into relationship of objects among themselves and hence marks the beginning of the formation of real "objects." But two circumstances limit the effectiveness of this behavior and at the same time define the difference which separates it from those of the fifth stage. In the first place, in order to become adapted to the new circumstances in which he finds himself—that is to say, in order to remove the

obstacle or discover the requisite intermediate—the child at the fourth stage limits himself to intercoördinating familiar schemata, except for differentiating them through progressive accommodation while adjusting them to each other. In the second place, and through that very fact, the relations which the child establishes between things still depend on completed schemata of which only the coördination is new; besides, they do not lead to the elaboration of objects entirely independent of the action, nor of spatial "groups" which are entirely "objective," etc. This is what we shall see, in particular, in Volume II, in studying concepts of the object, of space, of causality and of time which are characteristic of the fourth stage. In short, the fourth stage, in so far as it is defined by the commencement of the coördination of schemata, appears more as a phase of initiation or of gestation than as a period of realization or accomplishment.

The fifth stage, which we now undertake to study, is, on the contrary, primarily the stage of elaboration of the "object." It is characterized, in effect, by the formation of new schemata which are due no longer to a simple reproduction of fortuitous results but to a sort of experimentation or search for novelty as such. Moreover, in correlation with this same tendency, the fifth stage is recognizable by the appearance of a higher type of coördination of schemata: the coördination directed by the search for new "means."

Now both of these behavior patterns protract those of the preceding stages. As far as the "tertiary circular reaction" is concerned, it derives directly, as we shall see, from the secondary reactions and "explorations" to which the latter finally give rise. The only difference is that in the case of "tertiary" reactions the new effect obtained fortuitously is not only reproduced but modified to the end that its nature may be studied. With regard to the "discoveries of new means through active experimentation" they simply crown the coördination of schemata already in use during the fourth stage, but the reciprocal adjustment of the schemata which we have described in the preceding chapter, becomes accommodation for the sake of accommodation, that is to say, a search for new procedures.

But if the behavior patterns of the fifth stage protract those

of the fourth and so constitute their natural result, they nevertheless mark decisive progress and the beginning of a really characteristic phase. In effect, for the first time, the child truly adapts himself to unfamiliar situations, not only by utilizing the schemata acquired earlier but also by seeking and finding new means —whence a series of consequences concerning, on the one hand, the functioning of intelligence and, on the other, the essential categories of concrete thought.

Of the first of these points of view, the coördination of schemata henceforth being accompanied by intentional accommodation and differentiated to new circumstances, it can be said that the mechanism of empirical intelligence has been definitely formed. The child is henceforth capable of resolving new problems, even if no acquired schema is directly utilizable for this purpose, and if the solution of these problems has not yet been found by deduction or representation, it is insured in principle in all cases due to the combined working of experimental search and the coördination of schemata.

With regard to the "real categories" of thought, this kind of accommodation to things, combined with the coördination of schemata already acquired during the preceding stage, result in definitely detaching the "object" from the activity itself while inserting it in coherent spatial groups in the same way as in the causal and temporal series which are independent of the self.

§1. THE TERTIARY CIRCULAR REACTION.—The peculiarity of the behavior patterns which we shall now describe is to constitute, for the first time, an effort to grasp novelties in themselves.

To be sure, from the very beginnings of mental life, it can be said that the external environment imposes a constant enlarging of the subject's reactions and that new experience always causes the old framework to crack. That is why acquired habits are sooner or later superposed on reflex schemata and on the former are superposed the schemata of intelligence. And, of course, it can also be said that the subject accepts this necessity with pleasure since the "circular reaction" at all levels is precisely an attempt to conserve novelties and establish them by reproductive

assimilation. In the third place, one can maintain in a sense that novelty stems from assimilation itself, since the heterogeneous schemata, not very numerous, which are bestowed at the beginning, tend reciprocally to assimilate each other and so lead to multiple combinations of coördinations, either intersensorial or intelligent.

But, considered from another angle, the same facts show the resistance of mental life to novelty and the momentary victory of conservation over accommodation. So it is that the peculiarity of assimilation is to neglect what is new in things and events in order to reduce them to the state of aliments for old schemata. With regard to circular reaction, if it tends to reproduce the new result fortuitously observed, it is nevertheless necessary to state that it did not seek it but that it imposed itself by appearing by chance and in connection with familiar movements. To such a degree that circular reaction only remains, at first, pure reproductive assimilation and, if it is applied to a new datum, it is, so to speak, because this datum forced the positions by surreptitiously finding its way into the interior of an already elaborated schema. Let us recall that the new external results which characterize secondary circular reaction appear as if sprung from a differentiation of primary schemata, under pressure from the external environment, and that primary circular reaction develops, by differentiation, from the reflex schemata.

Tertiary circular reaction is quite different: if it also arises by way of differentiation, from the secondary circular schemata, this differentiation is no longer imposed by the environment but is, so to speak, accepted and even desired in itself. In effect, not succeeding in assimilating certain objects or situations to the schemata hitherto examined, the child manifests an unexpected behavior pattern: He tries, through a sort of experimentation, to find out in which respect the object or the event is new. In other words, he will not only submit to but even provoke new results instead of being satisfied merely to reproduce them once they have been revealed fortuitously. The child discovers in this way that which has been called in scientific language the "experiment in order to see." But, of course, the new result, though sought after for its own sake, demands to be reproduced and the

initial experiment is immediately accompanied by circular reaction. But, there too, a difference contrasts these "tertiary" reactions to the "secondary" reactions. When the child repeats the movements which led him to the interesting result, he no longer repeats them just as they are but gradates and varies them, in such a way as to discover fluctuations in the result. The "experiment in order to see," consequently, from the very beginning, has the tendency to extend to the conquest of the external environment.

These tertiary circular reactions will lead the child to new complete acts of intelligence which we shall call "discovery of new means through active experimentation." The acts of intelligence hitherto under study have only consisted in an application of familiar means (already acquired schemata) to new situations. But what will happen when the familiar means reveal themselves to be inadequate, in other words, when the intermediates between subject and object are not assimilable to the habitual schemata? Something will transpire which is analogous to what we have just said about tertiary circular reaction. The subject will search on the spot for new means and will discover them precisely through tertiary reaction. It cannot be said that he will apply the tertiary schemata to these situations since, by definition, the tertiary circular reaction is vicarious and only exists during the elaboration of new schemata, but he will apply the method of tertiary circular reaction.

The invention of new means through active experimentation, therefore, is to tertiary reaction what the "application of familiar means to new situations" is to secondary reaction: a combination or coördination of schemata in comparison to the simple schemata. To be more precise, we are confronted here by a distinction analogous to that which can be made on the plane of reflective or verbal intelligence between reason and judgment, reason being a combination of judgments of which some serve as means and the others as ends. In effect, a judgment is nothing other, from the functional point of view which is common to reflective intelligence and to sensorimotor intelligence, than the assimilation of a datum to a schema. From this point of view simple circular reactions, whether primary, secondary or tertiary,

are judgments. Moreover, the application of familiar means to new situations or the invention of new means constitute, from the same functional point of view, actual reasonings since, as we have already emphasized, the schema used in the capacity of means (it makes little difference whether it is familiar or invented on the spot) is subsumed under the schema characterizing the final end in the same way that judgments are put into a state of mutual implication in the framework of the conclusion. With regard to the comprehension of signs, it constitutes an intermediate term between judgment and reasoning. It is judgment inasmuch as it is immediate assimilation of the sign, and reasoning inasmuch as this assimilation is fraught with prevision, that is to say, with virtual deduction. But this intermediate state also finds its equivalent in verbal thought: Most judgments are implicit reasonings.

Having stated that, let us now try to analyze the tertiary circular reactions which thus constitute what one might call the functional and sensorimotor point of departure of experimental judgments.

Observation 141.—This first example will make us understand the transition between secondary and "tertiary" reactions: that of the well-known behavior pattern by means of which the child explores distant space and constructs his representation of movement, the behavior pattern of letting go or throwing objects in order subsequently to pick them up.

One recalls (Obs. 140) how, at 0;10 (2) Laurent discovered in "exploring" a case of soap, the possibility of throwing this object and letting it fall. Now, what interested him at first was not the objective phenomenon of the fall—that is to say, the object's trajectory—but the very act of letting go. He therefore limited himself, at the beginning, merely to reproducing the result observed fortuitously, which still constitutes a "secondary" reaction, "derived," it is true, but of typical structure.

On the other hand, at 0;10 (10) the reaction changes and becomes "tertiary." That day Laurent manipulates a small piece of bread (without any alimentary interest: he has never eaten any and has no thought of tasting it) and lets it go continually. He even breaks off fragments which he lets drop. Now, in contradistinction to what has happened on the preceding days, he pays no attention to the act of letting go whereas he watches with great interest the body in motion; in particu-

lar, he looks at it for a long time when it has fallen, and picks it up when he can.

At 0;10 (11) Laurent is lying on his back but nevertheless resumes his experiments of the day before. He grasps in succession a celluloid swan, a box, etc., stretches out his arm and lets them fall. He distinctly varies the positions of the fall. Sometimes he stretches out his arm vertically, sometimes he holds it obliquely, in front of or behind his eyes, etc. When the object falls in a new position (for example on his pillow), he lets it fall two or three times more on the same place, as though to study the spatial relation; then he modifies the situation. At a certain moment the swan falls near his mouth: now, he does not suck it (even though this object habitually serves this purpose), but drops it three times more while merely making the gesture of opening his mouth.

At 0;10 (12) likewise, Laurent lets go of a series of objects while varying the conditions in order to study their fall. He is seated in an oval basket and lets the object fall over the edge, sometimes to the right, sometimes to the left, in different positions. Each time he tries to recapture it, leaning over and twisting himself around, even when the object falls 40 or 50 cm. away from him. He especially tries to find the object again when it rolls under the edge of the basket and hence cannot be seen.

Observation 142.—At 0;10 (29) Laurent examines a watch chain hanging from his index finger. At first he touches it very lightly simply "exploring" it without grasping it. He then starts it swinging a little and at once continues this, thus rediscovering a "derived secondary reaction" already described in Observation 138 (schema of swinging). But, instead of stopping there, he grasps the chain with his right hand and swings it with his left while trying some new combinations (here the "tertiary reaction" begins); in particular he slides it along the back of his left hand and sees it fall off when it reaches the end. Then he holds the end of the chain (with his right index finger and thumb) and lets it slide slowly between the fingers of his left hand (the chain is now horizontal and no longer oblique as before). He studies it carefully at the moment when the chain falls from his left hand and repeats this ten times. Afterward, still holding the end of the chain in his right hand, he shakes it violently which makes it describe a series of varied trajectories in the air. He then slows these movements in order to see how the chain slides along the quilt when he merely pulls it. Finally he drops it from different heights and so rediscovers the schema acquired in the preceding observation.

From his twelfth month Laurent has repeated this kind of experiments with everything that his hand came upon: my notebook, "plugs," ribbons, etc. He entertains himself either by making them slide or fall or by letting them go in different positions and from different heights in order to study their trajectory. Thus, at 0;11 (20) he holds a "plug"

3 cm. above the ground, then 20 cm., etc., each time observing the fall most attentively.

Observation 143.—Here is another example of an "experiment in order to see," observed with respect to Laurent and pertaining to sound.

At 1;1 (24) Laurent finds himself for the first time in the presence of a piece of furniture of which we shall speak later in connection with the "invention of new means through active experimentation": a tiered table of which each circular tier pivots around a single axis. Laurent takes hold of one of the tiers in order to draw it to him. The tier moves, but turns around instead of being displaced in a straight line as the child was expecting. Laurent then shakes it, knocks it, then yields to a definitely "experimental" reaction in order to study the sound. He strikes it several times in succession, sometimes gently, sometimes hard, and between times hits the surface of his own table. There is no doubt whatever that he is comparing the sounds. Then he strikes the back of his chair and again the wide circular tier. Consequently this is more than "exploration" since there is a comparison of several objects to each other and a serialization of the effects produced.

Afterward he again wants to pull the tier toward him and happens to make it turn. But we shall describe the rest of this behavior in Observation 148 repeated, for it quickly became complicated.

Observation 144.—From 0;11 it seemed to me that, in the same way, Jacqueline intentionally let objects she was holding fall to the ground, to take them up subsequently or limit herself to looking at them. But at first it is difficult to distinguish between chance and intention. At 0;11 (19), on the other hand, the thing is quite clear: During her meal, while she is seated, she moves a wooden horse to the edge of her table until she lets it fall. She watches it. An hour later she is given a postcard. Jacqueline throws it to the ground many times and looks for it. At 0;11 (28) likewise, she systematically pushes a thimble to the edge of the box on which it is placed and watches it fall. But it is necessary to note, in observing such behavior patterns, that the child has not yet perceived the role of gravity. In other words, when he lets go of an object, it is without knowing what is going to happen; and, when he tries to send it to the ground, he believes himself to be obliged to push it down, without confining himself simply to letting it go. Thus, at 1;0 (26), Jacqueline pushed her ball to the ground instead of letting it fall. The same day she tries to get rid of a cushion which eclipses an object. She then simply puts this cushion against the back of the sofa, as though it would stick there, and starts over again indefinitely, not through circular reaction, but to get rid of the object which is annoying her.[1] At 1;1 (28), in the same way, Jacqueline watches me while I let go

[1] A. Szeminska pointed out to me an observation she made on a one-year-and-a-half-old boy. This child tried to draw a large object to himself

of a napkin ring 15 cm. above her table, several times in succession. She then takes it and simply places it on the table; afterward she reveals definite disappointment when she notices that nothing else happens. She then does this five or six times; then, as I repeat the experiment, she systematically places the object at the same place as I did (15 cm. above the table), then, instead of letting it go, from there she places it on the table!

Concerning the schema of throwing to the ground and picking up, it has been conserved a long time while being differentiated little by little. At 1;3 (21) and 1;3 (27) I note that Jacqueline begins to let fall instead of throwing to the ground. In particular she raises her arm by holding her hand backward and so manages to let go of objects from behind. At 1;4 (1) she throws an object under her mother's work table several times in succession and has difficulty in obtaining it again. Same reaction under the tablecloth of the lunch table. Finally there is progressive accommodation in the very manner of picking things up. At 1;5 (7) Jacqueline picks up objects without sitting down and gets up without leaning on anything.

Observation 145.—At 0;11 (20), that is to say, the day after the experiment with the wooden horse (see the foregoing observation), Jacqueline slides a series of objects along her slanting coverlet. This is an experiment and not merely repetition, because she varies the objects and the positions. At 1;0 (2) she rolls a pencil on a table by dropping it just above the surface or by pushing it. The next day she does the same with a ball.

At 1;0 (3) also, she takes her plush dog and places it on a sofa, obviously expecting a movement. As the dog remains motionless, she places it somewhere else. After several vain attempts she pushes it gently, while it is several millimeters above the material, as though it would roll better. Finally she grasps it and places it on a slanting cushion, so that the dog rolls. She then immediately begins all over again. Experimentation certainly exists. But in the last attempt definite prevision cannot be seen. We know (Obs. 144) that still at 1;0 (26), Jacqueline is not able to foresee the effects of gravity.

Same reactions at 1;1 (18) with her rabbit.

At 1;1 (19), likewise, Jacqueline places her red ball on the ground and waits to see it roll. She repeats the attempt five or six times and reveals a lively interest in the object's slightest movement. Then she puts it down and gives it a slight push with her fingers: the ball rolls better. She then repeats the experiment while pushing harder and harder.

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through the bars of a play pen. Not succeeding, he decided to pull it over the bars. To do so he raised the object to the place where his arm was held back by the horizontal bar, then let go the object to try to catch it on the other side!

At 1;3 (6) having dropped a stick parallel to the framework of her play pen, Jacqueline watches it roll several centimeters on the floor (outside the play pen). As soon as I bring the stick toward her, Jacqueline grasps it and repeats the experiment. She raises it slightly, then lets it fall, so that it may roll. The same happens ten times. Then I place a cloth under the framework to prevent the stick from rolling. Jacqueline lets it fall, then seeing that it remains motionless, passes her hand through the bars and gives it a fillip. She repeats this two or three times then, noticing her failure she gives up, without trying to throw the stick from higher up.

Same attempts at 1;4 (0). During one of these, she happens to let the stick drop from a height and it rolls superbly to the end of the room. Jacqueline is stupefied by this result but, as soon as I return the stick to her, she simply places it gently on the ground 3 or 4 cm. from the pen. She looks at it for a long moment, apparently expecting it to roll by itself.

Observation 146.—At 1;2 (8) Jacqueline holds in her hands an object which is new to her: a round, flat box which she turns all over, shakes, rubs against the bassinet, etc. She lets it go and tries to pick it up. But she only succeeds in touching it with her index finger, without grasping it. She nevertheless makes an attempt and presses on the edge. The box then tilts up and falls again. Jacqueline, very much interested in this fortuitous result, immediately applies herself to studying it.—Hitherto it is only a question of an attempt at assimilation analogous to that of Observations 136 to 137, and of the fortuitous discovery of a new result, but this discovery, instead of giving rise to a simple circular reaction, is at once extended to "experiments in order to see."

In effect, Jacqueline immediately rests the box on the ground and pushes it as far as possible (it is noteworthy that care is taken to push the box far away in order to reproduce the same conditions as in the first attempt, as though this were a necessary condition for obtaining the result). Afterward Jacqueline puts her finger on the box and presses it. But as she places her finger on the center of the box she simply displaces it and makes it slide instead of tilting it up. She amuses herself with this game and keeps it up (resumes it after intervals, etc.) for several minutes. Then, changing the point of contact, she finally again places her finger on the edge of the box, which tilts it up. She repeats this many times, varying the conditions, but keeping track of her discovery: now she only presses on the edge!

A moment later I offer my cigarette case to Jacqueline. She throws it as far as possible and presses it at different points with her index finger to tilt it up. But the problem is above her level and she tires of it.

Observation 147.—In her bath, Jacqueline engages in many experiments with celluloid toys floating on the water. At 1;1 (20) and the days following, for example, not only does she drop her toys from a height to see the water splash or displace them with her hand in order to make them swim, but she pushes them halfway down in order to see them rise to the surface.

At 1;7 (20) she notices the drops of water which fall from the thermometer when she holds it in the air and shakes it. She then tries different combinations to splash from a distance. She brandishes the thermometer and stops suddenly, or makes it catapult.

Between the ages of a year and a year and a half, she amuses herself by filling with water pails, flasks, watering cans, etc., and studying the falling of the water. She also learns to carry the water carefully without spilling it and by holding the basin horizontally.

She entertains herself by filling her sponge with water and pressing it against her chest or above the water; by filling the sponge at the faucet; by running the water from the faucet; by running the water from the faucet along her arm, etc.

The relationship between these tertiary circular reactions and the secondary or even primary reactions is evident: On the one hand, the new result is always discovered by chance since, even in searching for novelty as such, the child does not know how to find it except by groping. Moreover, the "experiment" always begins by repetition. In order to study changes in position, the trajectory of objects thrown or rolled, etc., it is always necessary to return to the same movements, with the intention of varying them little by little. The "experiment in order to see" is therefore surely a circular reaction, of a higher type undoubtedly, but conforming in principle to the preceding reactions.

But the tertiary reaction brings several innovations. At first, even in repeating his movements to seek and find an interesting result, the child varies and gradates them. So it is that, when throwing objects from a distance or rolling them (Obs. 144 and 145), tilting up a box or making it slide (Obs. 146), etc., he drops these objects from increasingly high altitudes, places his finger on a certain part of the box or on another part, etc. Doubtless this is the case since the earlier circular reactions. During the secondary reactions, in particular, the child always gradates his effects. He shakes his bassinet more or less, pulls

harder or less hard on the hanging strings, gradates the sound of the rattle he shakes, etc. But, in the latter cases, it is always in the same fixed framework that these variations are made manifest and one has the impression that the child is trying rather to reproduce a certain result while exploring all possible modalities, than to discover a new one. On the contrary, in the present case, the child does not know what will happen and he tries to ferret out new phenomena, which are unfamiliar or which he simply senses. For example, in Observations 141–144, the child constantly repeats the act of letting go, of throwing or rolling; but it is without knowing what will follow and precisely with the intention of finding out. True, in Observations 146 and 147 Jacqueline tries to reproduce an effect already observed (tilting up a box, floating, throwing, pouring water, etc.), but this effect is a theme with variations, and chiefly constitutes a phenomenon to be understood rather than a simple result to be repeated. In the case of the earliest secondary reactions, on the other hand, the child seems to try less to analyze and understand than merely to reproduce.

It is this last nuance which best characterizes tertiary reaction. As we said at the beginning of this chapter, the originality of these behavior patterns is that they constitute a search for novelty. For the child it is no longer a question of applying familiar schemata to the new object but of grasping with the mind this object as it is. In this respect, varying positions, throwing or rolling objects, tilting a box, floating objects, pouring water, etc., are active experiments which, it goes without saying, are still far removed from the verification of a preliminary deduction, as in the scientific experiment, but which already constitute the functional equivalent of the "experiment in order to see." Moreover, by virtue of the very fact that the experiments are accompanied by variations and gradations, they almost presage the right experiment. When Jacqueline discovers the necessity of pressing her box at the edge, and not in the middle, in order to tilt it up (Obs. 146), she reveals a directed and controlled effort. Doubtless, here also secondary reaction outlines all that; when the child must pull a string in order to shake the objects hanging from the hood of his bassinet, he must also dis-

cover the correct movement. But it is one thing to select almost automatically the right movements in the course of more or less diffuse groping and it is another thing to search for the condition essential to a certain result.

This search for the new raises the most interesting question which we have to discuss in connection with these behavior patterns. How does it happen, when all the behavior patterns hitherto under study are essentially conservative, that at a certain time the child comes to pursue novelties? The problem is found again, in analogous conditions, in connection with situations in which the child invents new means due to this same process of active experimentation. But let us limit ourselves, for the moment, to this precise question: How to account, through the play of assimilations and accommodations, for the interest in novelties peculiar to the "experiment in order to see"?

In the course of the primitive behavior patterns, in effect, through a paradox which we must subsequently analyze, accommodation and assimilation are simultaneously slightly differentiated and antagonistic. They are relatively undifferentiated in the sense that every attempt at assimilation is at the same time an attempt at accommodation without its yet being possible to distinguish in the child's intellectual activity a particular moment corresponding to what is deduction in reflective thought (assimilation as such) and another moment corresponding to what is experience (accommodation as such). Every schema of assimilation is therefore forthwith a schema of accommodation; primitive assimilation, whether reproductive, generalizing or recognitory, only functions to the extent that it is accommodation growing toward reality. However, although differentiated and, in this sense, closely correlative, assimilation and accommodation are, in another sense, antagonistic at first. In effect, in the beginning the child only accommodates himself to things when he is in some way forced by them, whereas at the outset he tries to assimilate the real, impelled by an invincible and vital tendency. So it is that, during the first stages, he is only interested in the external environment to the extent that objects can serve as aliments to his schemata of assimilation. This is why the child's activity begins by being essentially conservative and only

accepts novelties when they are imposed within a schema already constituted (as when, busy grasping the strings hanging from the hood of his bassinet, the child perceives that he thus shakes the hood).

But after the progress of assimilation, things change little by little. In effect, once assimilation has been organized through mobile schemata (we have seen how it protracted, by continuous differentiation, assimilation through simple schemata), the child reveals two important tendencies from the point of view which now occupies us. On the one hand, he becomes increasingly interested in the external result of acts, not only because this result is to be seen, heard, grasped, etc. (hence, because it is to be assimilated by means of "primary" schemata), but also because this result, imposed at first by the external environment, progressively differentiates the "secondary" schemata and thus concentrates on itself the subject's attention. Moreover, the child then tries to make all new objects enter into the schemata already acquired and this constant effort to assimilate leads him to discover the resistance of certain objects and the existence of certain properties irreducible to these schemata. It is then that accommodation assumes an interest in itself and that it becomes differentiated from assimilation, subsequently to become more and more complementary.

Accommodation to novelties acquires interest precisely by reason of the two tendencies which we have just called to mind. To begin with the second one, it is clear that to the extent that the child, seeking to assimilate new objects, will encounter resistance, he will become interested in the unforeseen properties which he will thus discover. This interest in novelty, therefore—however paradoxical this assertion may seem—results from assimilation itself. If the new object or phenomenon had no connection with the schemata of assimilation they would not be of interest and that is why, in fact, they rouse nothing in the child who is too young (even if he already knows how to grasp) except visual or auditory attention. Whereas, to the extent that they are almost assimilable, they rouse an interest and an attempt at accommodation still greater than if they were assimilable immediately. That is why, the more complex the system of the